Class 8
Chemistry

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1. Atoms and Molecules
Solutions

LEVEL-I

1. Correct option: (B)

J.J.Bezelius was the scientist who suggested a method of representing elements using the English letters(capital as well as small)

2. Correct option: (D)

The symbol C stands for the element Carbon. It is derived from the English name of the element and the symbol is first letter of the name.

3. Correct option: (C)

Latin name of Hg is Hydrargyrum, in Greek Hydro means water and argyros means silver.Hg is a heavy silvery d-block element.

4. Correct option: (A)

Mendelevium is the element that named after the name of the scientist Mendeleev (some elements are named after the names of the scientists.)

5. Correct option: (D)

Neptunium is the element that named after the planet name in the following elements.

6. Correct option: (A)

Germanium is the element of the following elements named after the country's name.

7. Correct option: (C)

Atomic weight of a substance can be expressed in both gram atomic weight and gram atom.

8. Correct option: (A)

We know relative atomic mass of O atom =Mass of one O atom/one twelve th of mass of C-12 isotope.

=16amu/1amu

=16(amu s got cancelled)

9. Correct option: (D)

Chlorine molecule exist as Cl_2,

Gram molecular weight of chlorine is =2*atomic mass of Cl

=2*35.5g

=71.0g

10. Correct option: (B)

The molar mass of H=H_2=2*atomic mass of H

=2*1.008g

=2.016g

11. Correct option: (A)

One mole of molecules of Oxygen means, it contains $6.023*10^{23}$ molecules of Oxygen.

12. Correct option: (B)

The weight of 1 mole of Na_2CO_3=106g

The weight of 0.1 mole of Na_2CO_3=106g/0.1

=10.6g

13. Correct option: (A)

One mole of any substance contains $6.023*10^{23}$number of elementary particles.

14. Correct option: (B)
1 mole of hydrogen atom contains $6.023*10^{23}$ atoms of Hydrogen.

15. Correct option: (D)
Applying the concept of No. of molecules=W/M.wt*N_A
No. of molecules of 23g of NO_2=23/46*N_A=0.5N_A
A)8gm of oxygen(O_2) no. of molecules=8/32N_A=0.25N_A
B)28gm of carbon,(CO)No. of Molecules=28/28N_A=N_A
c)16gmof SO_2,No.of molecules=16/64N_A=0.25N_A
d)22gm of CO_2,No.of Molecules=22/44=0.5N_A

16. Correct option: (A)
Given,
Mass of CO2=44gm
To find moles of oxygen,
Here,
Mole=mass/molar mass
=44/44=1 mole
So 1 mole of CO_2 contains
1 mole of Carbon and
2 moles of oxygen atoms.

17. Correct option: (B)
The molar volume of any gas at STP is 22.414 litre.

18. Correct option: (D)
The volume of two gram moles of oxygen at STP is 22.4*2
=44.8L

19. Correct option: (A)
The combining capacity of an element is called its valency.

20. Correct option: (A)
The electrons that present in outermost energy level are known as valence electrons.

21. Correct option: (A)
The number of electrons that present in outermost shell is one.so valency is one.

22. Correct option: (B)
Silver shows more than one valency that is known as variable valency. It shows both 1 and 2 valances and 1 is the lowest valency.

23. Correct option: (A)
Mercury also shows variable valency(1,2).the higher valency of Hg is 2

24. Correct option: (D)
Cu(Copper) also shows variable valancies(1,2).the valency in cupric chloride is 2

25. Correct option: (B)
Valency of Nitrogen is three as nitrogen has five electrons in its valence shell, it can therefore, accept three electrons to complete its octate. Hence the valency of nitrogen in NH_3 is 3.

26. Correct option: (C)
In PCl_5 P combine with 5 chlorine atoms, Hence the valence of phosphorus is 5 .

27. Correct option: (C)
In CO_2n,carbon is combined with two oxygen atoms. So,the valence of Carbon is 4(one oxygen atom has 2 valency)

LEVEL-II

1. Correct option: (D)

Given all the statements are correct.

2. Correct option: (A)

The name of the element having atomic number 16 is Sulphur and the symbol is 'S'.

3. Correct option: (C)

The element Curium named after the scientist Madam Curie.

4. Correct option: (B)

The symbol 'Po' stands for Polonium.

5. Correct option: (D)

All are correct regarding A mole.

6. Correct option: (B)

No. of Moles=weight/M.Wt

M,Wt of $CuSO_4$ is= 64+32+64=160

So number of moles=1.60/160

=0.01 mole

7. Correct option: (B)

A) Mass of one gram atom of Iron=1gm

B)5 moles of N2=weight/M.wt

5=Wt/28

Mass=140g

C) 10^{23} carbon atoms

$6.023*10^{23}$ atoms are present in 12g of carbon

So 10^{23}atoms are present in =$(10^{23}/6.023*10^{23})*12$

=12/6.023

=1.992g

D) Invalid MCQ

So 5 moles of N^2 has the highest mass.

8. Correct option: (A)

Atomic weight of H is 1u,therefore 1g hydrogen has 1 mole of hydrogen and 1 mole of hydrogen contains $6*10^{23}$ atoms.

Moles of He present in 4g of He is=mass/M.Wt

=4/4=1mole

Hence,4g of Helium also contains $6*10^{23}$ atoms.

9. Correct option: (C)

M.Wt of CaCO3 =100,Given weight=50g

No.of moles of $CaCO_3$=50/100=0.5

1 mole of $CaCO_3$ has 3 moles of Oxygen

0.5 moles of $CaCO_3$ has 1.5 moles of O_2

1 mole=$6.023*10^{23}$ particles

Therefore 50g of $CaCO_3$=$1.5*6.023*10^{23}$

=$9.035*10^{23}$

10. Correct option: (B)

A) weight of 0.1g silver=0.1g

B) 5 moles of N2=weight/M.wt

5=Wt/28

Mass=140g

C) Weight of copper=1gm

D) 10^{23} carbon atoms

$6.023*10^{23}$ atoms are present in 12g of carbon
So 10^{23}atoms are present in =$(10^{23}/6.023*10^{23})*12$
=12/6.023
=1.992g
5 moles of N_2 has more weight than the remaining.

11. Correct option: (B)

Given:
Volume = 50 mL = 0.05 L
Molarity = 0.1 M
Step 1: Calculate moles of KI
Moles = $M \times V = 0.1 \times 0.05 = 0.005$ mol
Each KI contains **1 iodine atom**.
Step 2: Number of iodine atoms
$0.005 \times 6.02 \times 10^{23} = 3.01 \times 10^{21} \approx$ **3×10^{21} atoms**

12. Correct option: (B)

Molar mass of O_2 = 32 g/mol; Moles = $\frac{8}{32} = 0.25$
Number of molecules
$0.25 \times 6.02 \times 10^{23} = 1.505 \times 10^{23}$
$\approx$ **1.51×10^{23}**

13. Correct option: (A)

$Al_2(SO_4)_3$
Three sulphate ions
Oxygen atoms = 3 × 4 = **12**
Since the question asks which molecule (compound formula) contains more than six oxygen atoms,
$Al_2(SO_4)_3$ contains 12 oxygen atoms.

14. Correct option: (B)

Formula = M_3N_2
Mass of nitrogen:$2 \times 14 = 28$
Let atomic mass of metal = M
Total mass:$3M + 28$
Percentage of nitrogen:$\frac{28}{3M+28} \times 100 = 28$
$28 = 0.28(3M + 28)$
$100 = 3M + 28$
$3M = 72$
$M = 24$

15. Correct option: (A)

A) Carbon: $\frac{24}{12} = 2$ mol
B) Iron:$\frac{56}{56} = 1$ mol
C) Aluminium:$\frac{27}{27} = 1$ mol
D) Silver:$\frac{108}{108} = 1$ mol
Maximum moles = Maximum atoms

16. Correct option: (C)

1 atomic mass unit (amu)=121 of the mass of one carbon-12 atom

17. Correct option: (D)

One mole of $Mg_3(PO_4)_2$ contains
$2 \times 4 = 8$
oxygen atoms.
Required compound moles
$= \frac{0.25}{8} = 0.03125 = 3.125 \times 10^{-2}$
Correct option: (D)
6.02×10^{24} molecules
$= \frac{6.02 \times 10^{24}}{6.02 \times 10^{23}} = 10$ mol CO
Each CO contains one oxygen atom.
Hence oxygen atoms = 10 mol = **10 gram-atoms**

18. Correct option: (B)
At STP, 1 mole gas = 22.4 L
A) 10^{22}atoms He
$\frac{10^{22}}{6.02 \times 10^{23}} = 0.0166$ mol
Volume ≈ 0.37 L
B) 11 g N_2O
Molar mass = 44; $\frac{11}{44} = 0.25$ mol
Volume $=0.25 \times 22.4 = 5.6$ L
C) 0.2 gram-atom oxygen=0.2 mol O atoms
Equivalent O_2 molecules:0.1 mol
Volume = 2.24 L
D) 10^{22}molecules H_2:0.0166 mol
Volume ≈ 0.37 L
Maximum = **5.6 L**

19. Correct option: (B)
Given:
NH_3 = 1 mol
O_2 = 1 mol
Required O_2 for 1 mol $NH_3 = \frac{5}{4} = 1.25$ mol
Only 1 mol O_2 is available.
Therefore **oxygen is the limiting reagent**.
Products formed:$NO = \frac{4}{5} \times 1 = 0.8$ mol
$H_2O = \frac{6}{5} \times 1 = 1.2$ mol
Thus **all oxygen is consumed**, while some NH_3 remains unreacted.

20. Correct option: (A)
Let the percentage of ^{16}Abe x%.
Then percentage of $^{18}A$is (100 − x)%.
Average atomic mass:
$$\frac{16x + 18(100 - x)}{100} = 16.2$$
$$16x + 1800 - 18x = 1620$$
$$-2x = -180$$
$$x = 90$$
Therefore,
$^{16}A = 90\%$

$^{18}A = 10\%$

21. Correct option: (C)

HCl

Valency of H = 1

Valency of Cl = 1

Total atoms = 2

Valencies ≠ Total atoms.

B) H_2S

Valency of H = 1

Valency of S = 2

Total atoms = 3

Not equal.

C) CaO

Valency of Ca = 2

Valency of O = 2

Total atoms = 2

Both valencies = Total atoms = 2.

Correct

D) $MgCl_2$

Valency of Mg = 2

Valency of Cl = 1

Total atoms = 3

Not equal.

22. Correct option: (B)

Let percentage of $^{35}Q = x$.

Then percentage of $^{37}Q = 100 - x$.

Average mass:

$$\frac{35x + 37(100 - x)}{100} = 35.5$$

$$35x + 3700 - 37x = 3550$$

$$-2x = -150$$

$$x = 75$$

Thus,

$^{35}Q = 75\%$

$^{37}Q = 25\%$

23. Correct option: (A)

Mass of 0.5 mole of N_2 gas > Mass of 0.5 mole of N atoms

Mass of 0.5 mol N_2:$0.5 \times 28 = 14$ g

Mass of 0.5 mol N atoms:$0.5 \times 14 = 7$ g

Thus,$14 > 7$

Statement is **True**.

24. Correct option: (A)

(I) 0.5 mole N_2

$0.5 \times 28 = 14$ g

So,$x = 14$

(II) 0.5 mole N atoms

Atomic mass =14

$0.5 \times 14 = 7$ g

$y = 7$

(III) 3.011×10^{23}N atoms

This is half of Avogadro's number.

Therefore,0.5 mole

Mass

$0.5 \times 14 = 7$ g

$z = 7$

(IV) $6.023 \times 10^{23}N_2$ molecules

This is approximately **1 mole**.

Mass:$1 \times 28 = 28$ g; $w = 28$

Now compare:$x = 14, y = 7, z = 7, w = 28$

Divide by 2:$\frac{x}{2} = 7$

Hence,$\boxed{\frac{x}{2} = y = z = \frac{w}{4}}$

25. Correct option: (A,B,D)

The **Law of Constant Composition (Law of Definite Proportions)** states that:

A pure chemical compound always contains the same elements combined in the same fixed proportion by mass.

Therefore, it applies **only to pure compounds**, not to elements or individual atoms.

26. Correct option: (A,B,D)

1 atomic mass unit (1 u) is defined as

$1u = \frac{1}{12}$ of the mass of one carbon-12 atom

28.Correct option: (A,B)

A) 100 mL 0.1 M HCl

Moles = MV

$= 0.1 \times 0.1 = 0.01$ mol

Contains 10 mmol.

B) 0.224 L CO_2 at STP

At STP,

22.4 L = 1 mol

$\frac{0.224}{22.4} = 0.01$ mol

29. Correct option: (B,C)

B) 32 g SO_2

Molar mass SO_2:$32 + 32 = 64$

$\frac{32}{64} = 0.5$ mol

Volume:$0.5 \times 22.4 = 11.2L$

C) 10 g Ne

Atomic mass =20

$\frac{10}{20} = 0.5$ mol

Ne is monoatomic gas.

Volume =11.2 L

30. Correct option: (A,B)

Molarity$M = \frac{\text{moles}}{\text{volume}}$

Volume changes with temperature.

Depends on temperature.

Normality

Also depends on solution volume.

Depends on temperature.

31. Correct option: (B)

Assertion (A):

A symbol represents one atom of an element. True

Reason (R):

A symbol also represents the atomic mass of that element.True

For example,

O represents

one oxygen atom

atomic mass =16 u

But the reason does not explain why a symbol represents one atom.

Hence,Both are true, but Reason is **not** the correct explanation.

32. **Correct option: (A)**

Assertion (A):

Molecular mass is the sum of atomic masses of all atoms in one molecule. True

Reason (R):

Each atom contributes according to its number in the molecule.

Example:$H_2O = (2 \times 1) + (1 \times 16) = 18u$

True

The reason directly explains how molecular mass is calculated.

33.

(a) 1 mole of SO_2

Molar mass:$32 + 2(16) = 64$ g

Also,1 mole gas at STP = 22.4 L

(b) 1 mole of N_2

Mass:$2 \times 14 = 28$ g

Volume =22.4 L

(c) 2 moles of O_2

Mass:$2 \times 32 = 64$ g

Volume$2 \times 22.4 = 44.8L$

(d) 1 mole of CO_2

Mass:$12 + 32 = 44g$

Volume:22.4 L

34. Correct option: (D)

Empirical formula = simplest whole-number ratio.

Divide all subscripts by 6.$C_6H_{12}O_6$

$\downarrow CH_2O$

35. Correct option: (A)

Relationship: Molecular Formula = $n \times$ Empirical Formula

Therefore,

$$\frac{\text{Molecular Mass}}{\text{Empirical Formula Mass}} = n$$

where **n** is always a **positive whole number**.

Example: Glucose:$\frac{180}{30} = 6$

36. Correct option: (C)

Molecular formula tells
Elements present
Actual number of atoms
Empirical formula tells
Elements present
Simplest ratio
Both tell which elements are present.

37. Step 1: Molar mass
Cu =64;S =32;O_4 =64
$5H_2O$ =5×18=90
Total:$64 + 32 + 64 + 90 = 250 g/mol$
Step 2: Moles
$\frac{499}{250} \approx 2$
Step 3: Oxygen atoms
Each molecule contains
SO_4 →4 O
$5H_2O$ →5 O
Total$4 + 5 = 9$
So,2 moles compound contain$2 \times 9 = 18$
moles oxygen atoms.Half of this$\frac{18}{2} = 9$

38. Atomic mass of He =4 amu
Number of He atoms in 100 amu
$\frac{100}{4} = 25$
One-fifth of 25
$\frac{25}{5} = 5$
Valency

Level-I

1. Correct option: (A)
The combining capacity of an atom, i.e., the number of electrons lost, gained, or shared during bonding, is called valency.

2. Correct option: (A)
The electrons in the outermost shell take part in chemical bonding and are called valence electrons.

3. Correct option: (A)
Sodium (Na), atomic number 11, has the electronic configuration 2, 8, 1. Therefore, it has 1 valence electron.

4. Correct option: (B)
Silver commonly exhibits a valency of +1.

5. Correct option: (A)
Mercury shows two valencies: +1 (mercurous) and +2 (mercuric). The higher valency is 2.

6. Correct option: (D)
Cupric chloride is $CuCl_2$. Since each chlorine has a valency of 1, copper must have a valency of 2.

7. Correct option: (B)

The formula is not visible in the question. In most Class 8 questions, this is usually NH_3, where nitrogen has a valency of 3.

8. Correct option: (C)
Phosphorus is bonded to five chlorine atoms, so its valency is 5.

9. Correct option: (C)
Carbon forms four bonds with two oxygen atoms (double bond with each oxygen), so its valency is 4.

10. Correct option: (B)
An ion is an atom or group of atoms that carries a positive or negative charge.

11. Correct option: (C)
Na^+ → Single atom with a charge

12. Correct option: (B)
Sulphate ion is SO_4^{2-}, which carries a –2 charge.

13. Correct option: (C)
Sodium (Na^+) combines with chloride (Cl^-) in a 1 : 1 ratio, giving NaCl.

14. Correct option: (B)
In chemical formulae of ionic compounds, the **metal (cation)** is always written first (on the left), followed by the non-metal (anion).
Example: **NaCl, CaO**

15. Correct option: (B)
Molecular (covalent) compounds are formed by **sharing electrons between non-metals**.
Examples: H_2O, CO_2, NH_3

16. Correct option: (B)
The criss-cross method is used to write the **chemical formula** of ionic compounds by exchanging the valencies of the ions.
Example:$Ca^{2+} + Cl^- \rightarrow CaCl_2$

17. Correct option: (C)
Calcium has valency **2**, oxygen has valency **2**.
Simplest ratio = **CaO**

18. Correct option: (A)
Most non-metals form anions, but hydrogen is an exception because it commonly forms the H^+ cation.

19. Correct option: (C)
An ion with a **negative charge** is called an **anion**.
Examples: Cl^-, O^{2-}, SO_4^{2-}

20. Correct option: (C)
NH_4^+ → Polyatomic ion

LEVEL – II

1. Correct option: (B)
The valence of suplhur is 6,as sulphur bonded to three oxygen atoms and we know that oxygen has the valency 2
3*2=6

2. Correct option: (B)
The atomic number 20 will combine most likely with atom having atomic number 16 as both will gain stable electronic configuration.

3. Correct option: (B)

Electronic configuration of F in a/tomic state is $1s^2 2s^2 2p5$. Seven electrons were present in outermost shell. If stability attained with 6 electrons F will lose one electron. Then F becomes F^{+}.

$F^{+}=1s^2 2s^2 2p4$

4. Correct option: (A)

The element X has 2 electrons in outermost shell and Y has 1 electron in outermost shell and valency of X is 2 and y is 1 .To get stability it forms the compound by the combination of X and Y as XY_2.

5. Correct option: (B)

Electronic configuration of $A=1s^2 2s^2 2p^6 3s^2 3p^1$

The outermost shell has 3 electrons (Valency is 3)

Electronic configuration of $B=1s^2 2s^2 2p^4$

The outermost shell has 6 electrons; hence valency is 8-6=2

Element	A	B
valency	3	2

Now the compound formed by cirss-cross method is A_2B_3.

6. Correct option: (C)

A **cation** is a positively charged ion. Among the given options, only **Na^+** has a positive charge.

7. Correct option: (B)

A water molecule consists of **2 hydrogen atoms** and **1 oxygen atom**.

8. Correct option: (B)

Aluminium ion = Al^{3+}

Sulphate ion = SO_4^{2-}

Using the criss-cross method:$Al^{3+} + SO_4^{2-} \rightarrow Al_2(SO_4)_3$

9. Correct option: (B)

CO_2 contains **two oxygen atoms**, so the prefix **di-** is used.

10. Correct option: (A,B,D)

Copper → +1, +2

Iron → +2, +3

Sodium → only +1

Tin → +2, +4

11. Correct option: (A,B,D)

Cations are positively charged ions.

Na^+

Ca^{2+}

Cl^-

NH_4^+

12. Correct option: (A,B,D)

Polyatomic ions contain more than one atom.

NH_4^+

NO_3^-

Na^+ (single atom)

SO_4^{2-}

13. Correct option: (A,B,D)

Monovalent ions have a charge of **±1**.

Na^+

Cl^-

Ca^{2+}
OH^-

14. Correct option: (A,B,C)
$Mg^{2+} + Cl^- \rightarrow MgCl_2$
$Al^{3+} + SO_4^{2-} \rightarrow Al_2(SO_4)_3$
$Na^+ + O^{2-} \rightarrow Na_2O$
$Ca^{2+} + Cl^-$ should be $CaCl_2$, not CaCl

15. Correct option: (A,B,D)
Formed between non-metals
Prefixes (mono, di, tri) are used
They are covalent, not ionic
Less electronegative element is written first

16. Correct option:(D)
Cl^- is an **anion**, not a cation.
Chlorine gains one electron to form Cl^-.
Therefore,Assertion is false but Reason is true.

17. Correct option:(A)
Sodium has one electron in its outermost shell and loses it to attain a stable configuration. Hence, its valency is **1**.
Both Assertion and Reason are true, and the Reason correctly explains the Assertion.

18) (a→R,S) (b→Q) (c→ Q) (d→ R)

19) Correct option:(B)
Metals lose electrons and become positively charged.

20) Correct option:(C)
Non-metals gain electrons to complete their outer shell.

21) Correct option:(C)
Only Na^+ has a positive charge.

22) Outermost shell has 2 electrons.
It loses two electrons to become stable.
Valency = 2

23) Sulphate ion is **SO_4^{2-}**.
Charge = **−2**
Since the question asks for the **total charge**, the integer is:-2

24) The subscript of H is **2**.

25) One sodium ion and one chloride ion are formed.
Total ions = **2**

Previous year questions:

I. Correct option:(B)
The formula unit mass of calcium sulphite is 100 u.
Formula of calcium sulphite = **$CaSO_3$**
Formula unit mass= 40 + 32 + 3(16)
= 40 + 32 + 48 = 120 u
Statement is **False**.

II. Carbon dioxide obtained by different methods consists of carbon and oxygen in the ratio 3 : 8 by mass.
CO_2 contains
Carbon =12;Oxygen =32

Mass ratio$12:32 = 3:8$
According to the **Law of Constant Composition**, CO_2 always has this ratio irrespective of its source. Statement is **True**.

III. Ozone is triatomic while sulphur is polyatomic.
Ozone = O_3 (triatomic)
Sulphur exists as S_8 (polyatomic)
Statement is **True**.

IV. Nitride is an anion while phosphate is a cation.
Nitride = N^{3-} (anion)
Phosphate = PO_4^{3-} (also anion)
Statement is **False**.

2. **Correct option:(C)**
The chemical formula of zinc sulphate is $Zn(SO_4)_2$.
Correct formula is
$ZnSO_4$
Statement is False.

II. Phosphorus and chlorine are diatomic and monatomic respectively.
`Phosphorus exists as P_4 (polyatomic)
Chlorine exists as Cl_2 (diatomic)
Statement is False.

III. In magnesium sulphide, magnesium and sulphur are present in 3 : 4 mass ratio.
Formula = MgS
Mass ratio$24:32 = 3:4$
Statement is True.

IV. Formula unit mass of ammonium sulphate is 96 u.
Formula $(NH_4)_2SO_4$
Mass$2(14 + 4) + 32 + 64 = 36 + 96 = 132u$
Statement is False.

3. **Correct option:(A)**
(P) Carbon monoxide (CO)
Mass ratio$12:16 = 3:4 \rightarrow$ **(ii)**
(Q) Hydrochloric acid (HCl)
Mass ratio$1:35.5 \rightarrow$ **(i)**
(R) Carbon dioxide (CO_2)
Mass ratio:$12:32 = 3:8 \rightarrow$ **(iv)**
(S) Nitrogen dioxide (NO_2)
Mass ratio$14:32 = 7:16 \rightarrow$ **(iii)**

4. **Correct option:(A)**

I. Carbonate ion is a polyatomic ion carrying −2 charge.
CO_3^{2-}
Correct

III. Argon and oxygen are monoatomic and diatomic gases respectively.
Ar → monoatomic
O_2 → diatomic
Correct

5. **Correct option:(B)**
Since XCl_3
Cl has valency 1
Therefore Valency of X =3

Nitride ion = N^{3-}
Combining X^{3+} and N^{3-}
Ratio1:1
Formula XN

6. Correct option:(A)

28 g He
Atomic mass =4
$\frac{28}{4} = 7$ mol

II. 46 g Na
Atomic mass =23
$\frac{46}{23} = 2$ mol

III. 60 g Ca
Atomic mass =40
$\frac{60}{40} = 1.5$ mol
Decreasing order
$7 > 2 > 1.5$
I > II > III

7. **Correct option:(C)**

A) 0.3 mol N_2
Atoms
$0.3 \times 2 = 0.6$ mol atoms
B) 0.5 mol O_2
Atoms
$0.5 \times 2 = 1.0$ mol atoms
C) 0.4 mol O_3
Atoms
$0.4 \times 3 = 1.2$ mol atoms
D) 0.2 mol CO_2
Atoms per molecule =3
$0.2 \times 3 = 0.6$ mol atoms
Maximum = 1.2 mol atoms

8. Correct option:(C)

Statement II
Gas X is a compound containing nitrogen and hydrogen.
Gas X is ammonia.
Formula:NH_3
It contains nitrogen and hydrogen.
Statement IV
The number of atoms present in 22.4 L of water vapours is 18.069×10^{23}.
Explanation:
At STP,
22.4 L of water vapour = 1 mole of H_2O molecules.
Each H_2O molecule contains:
2 H atoms
1 O atom
Total = 3 atoms

Therefore,
$3 \times 6.023 \times 10^{23} = 18.069 \times 10^{23}$
atoms.
Statement IV is True.

2. Atomic Structure
Solutions

LEVEL – I

1. Correct option: (B)

John Dalton (father of atoms and atomic theory) proposed the atomic theory in the early 19th century, which explains that matter is composed of extremely small particles called atoms.

He also suggested that atoms are indivisible and indestructible particles that combine to form molecules and compounds. Dalton's atomic theory laid the foundation for our understanding of the composition and behavior of matter. But John Dalton elaborately explained about atoms. He also explained that atoms can be rearranged, combined, or separated in chemical reactions.

2. Correct option: (A)

The term "atom" was first introduced by the ancient Greek philosopher Democritus around the 5th century BCE. Democritus proposed that all matter is composed of indivisible and indestructible particles called atoms.

He believed that atoms are the primary building blocks of the matter in the universe and they are different in size, shape, and arrangement, giving rise to the variety of substances observed in the world.

3. Correct option: (A)

Elements in periodic table made up of very small particles of matter called atoms. These are the basic building blocks of elements. Each element consists of a specific type of atom that has unique number of protons, neutrons, electrons in the nucleus of atom. Elements can be distinguished from one another based on the number of protons or neutrons in their atoms, which is known as the atomic number represented by Z.

4. Correct option: (D)

Maharishi Kanada, he is an ancient Indian philosopher and he is known for his contributions to the atomic theory in the ancient Vedic period.

He proposed that matter is composed of extremely small particles called "Anu" (meaning atom) and "Paramanu" (meaning sub-atomic particle). According to Kanada, these particles are indivisible and can’t be described.

5. Correct option: (A)

According to Dalton's atomic theory, this was proposed by John Dalton, states that atoms are indivisible and indestructible particles with which all the matter is made. According to this theory, elements are composed of atoms, and each element is characterized by a unique type of atoms. Atoms of different elements differ in their properties, including mass, size, and chemical behavior.

6. Correct option: (A)

Democritus, he is an ancient Greek philosopher around the 5th century BCE. Who proposed the concept of atoms as the fundamental building blocks of matter. He suggested that matter is composed of tiny, indivisible particles called atoms. Democritus theorized that atoms are constantly moving and combine in different ways to form different substances. His ideas were only based on philosophical reasoning.

7. Correct option: (B)

The electron is a negatively charged particle, and its charge is approximately -1.6022×10^{-19} coulombs. This value is often rounded to $-1.6\times10^{(-19)}$ C for convenience.

8. Correct option: (A)

Anode rays were discovered by the German physicist Eugen Goldstein in the late 19th century. Eugen observed while studying these rays cathode rays in gas discharge tubes. He noticed that when a perforated cathode was used, a beam of positive ions traveled in the opposite direction, towards the cathode. These positive ions became known as anode rays.

9. Correct option: (B)

Proton is a subatomic particle that is found in the nucleus of an atom. It carries a positive charge. It is commonly associated with hydrogen atoms because the nucleus of a hydrogen atom consists of a single proton.

Option A) it is a nucleus of deuterium is incorrect. Deuterium is a hydrogen isotope that consists of a proton and a neutron in its nucleus.

Option C) it is an ionized hydrogen molecule is incorrect. A hydrogen molecule (H_2) consists of two hydrogen atoms bonded together, and a proton is not a hydrogen molecule.

Option D) it is a α-particle is incorrect. A α-particle is a helium nucleus consisting of two protons and two neutrons.

10. Correct option: (B)

The e/m value describes the ratio of the electron's charge (e) to its mass (m). The value of the e/m ratio for an electron is approximately $1.76\times10^{11} C/Kg$.

11. Correct option: (A)

The correct option less than e/m value of an electron.

We know that magnitude of charge is the same for a proton and an electron but the mass of an electron is less than that of the mass of a proton.

12. Correct option: (C)

Specific charge is defined as the ratio of charge of a particle by its mass. Since the charge of an electron remains constant and its mass increases with speed, the specific charge remains the same.

13. Correct option: (B)

The specific charge is defined as the ration between charges to mass of a particle, for a cathode ray is constant.

In the case of a cathode ray, which consists of electrons, the charge and mass of an electron is constant. Regardless of the gas in the discharge tube or the atomic number of the elements involved, the charge-to-mass ratio of an electron remains the same.

14. Correct options: (A and D)

The e/m value of a proton is approximately $9.58\times10^{4} C/kg$ it represents the ratio of the charge (e) to the mass (m) of a particle. For a proton, the charge is approximately $1.6\times10^{-19} C$ and the mass is approximately $1.67\times10^{-27} Kg$. Dividing the charge by the mass we get e/m value i.e. $5.8\times10^{4} C/Kg$.

15. Correct option: (B)

The specific charge of a particle is defined as the ratio of the charge (e) to its mass (m). It is a fundamental characteristic of particles. Used to differentiate between different types of particles based on their charge-to-mass ratios.

16. Correct option: (D)

The charge of a neutron is 0.Neutrons are electrically neutral particles they reside in nucleus along with proton, meaning they have no net charge. They are composed of quarks (specifically, two down quarks and one up quark), and the combination of these quarks results in a neutral overall charge.

17. Correct option: (A)

Anode rays, also known as canal rays; they are produced when a high voltage is applied across a discharge tube containing a gas at low pressure. The nature of the gas present in the discharge tube determines the characteristics of the anode rays. Different gases exhibit different behavior when influenced to electric fields. The gas molecules or atoms in the discharge tube can get ionized and form positive ions. These positive ions, known as anode rays, are accelerated towards the cathode due to the electric field. All the properties of anode rays depend on nature of gas used in discharge tube.

18. Correct option: (B)

James Chadwick, he a British physicist, discovered the neutron 1932. He conducted experiments that involved bombarding beryllium with alpha particles, which led to the observation of a neutral particle named as neutron with a mass similar to that of a proton.

19. Correct option: (C)

The mass of a neutron is slightly more than that of a proton. Both protons and neutrons are subatomic particles found in the nucleus of an atom. The mass of a neutron is approximately $1.675 \times 10^{-27} Kg$ kilograms, which is slightly greater than the mass of a proton, which is approximately $1.673 \times 10^{-27} Kg$ kilograms.

20. Correct option: (D)

To determine the highest specific charge among the given options, we need to calculate the specific charge for each element. Specific charge is defined as the charge-to-mass ratio.

A) $Na^{+}(A = 23)$ The atomic number of sodium is 11, and given charge is +1. The specific charge is calculated as follows:

$$Specific\ charge = \frac{(Charge)}{(Mass)} = \frac{(+1)}{(23)} \approx 0.0435$$

B) $Mg^{+2}(A = 24)$ The atomic number of magnesium is 12, given charge is +2. The specific charge is calculated as follows:

$$Specific\ charge = \frac{(Charge)}{(Mass)} = \frac{(+2)}{(24)} \approx 0.0833$$

C) $Al^{+2}(A = 27)$ The atomic number of aluminum is 13, given charge is +3. The specific charge is calculated as follows:

$$Specific\ charge = \frac{(Charge)}{(Mass)} = \frac{(+3)}{(27)} \approx 0.1111$$

D) $Si^{+4}(A=28)$ The atomic number of silicon is 14, given charge is +4. The specific charge is calculated as follows:

$$Specific\ charge = \frac{(Charge)}{(Mass)} = \frac{(+4)}{(28)} \approx 0.1429$$

Comparing the calculated specific charges, we find that option D) $Si^{+4}(A=28)$ has the highest specific charge.

21. Correct option: (A)

According to Thomson's model also known as “plum pudding “model of the atom, here the atom is represented as a sphere in shape. Where the atom is considered as a positively charged sphere with electrons embedded throughout it, much like the plums in a pudding. This model was proposed by J.J. Thomson in the late 19th century before the discovery of the atomic nucleus and the subsequent development of the atomic structure.

22. Correct option: (A)

Thomson's model proposed that the atom consists of a positively charged sphere with electrons embedded in it. The distribution of positive and negative charges in the atom balanced out, resulting in an overall electrically neutral atom. Drawback of this model is it didn’t explained how atoms do not collapse due to the repulsion between like charges.

However, Thomson's model does not explain the existence of orbital’s (option C) or the concept of electron orbital’s as understood in modern atomic theory.

23. Correct option: (C)

Thomson's atomic model is also known as the Plum pudding model (option B) and watermelon model (option C). This name was given to the model because it compared the structure of the atom to a positively charged "pudding" with negatively charged electrons embedded in it, resembling plums and watermelon fruit as positive charged matter and black seeds are negatively charged electrons. The model was proposed by J.J. Thomson in the late 19th century.

24. Correct option: (A)

Ernest Rutherford conducted the famous α-particle scattering experiment, also known as the gold foil experiment, in 1909. In this experiment, he projected a beam of α-particles (positively charged particles) on a thin gold foil and observed their behavior. According to Thomson's plum pudding model, the positively charged "pudding" should cause minimal deflection of the α-particles.

However, in Rutherford's observations showed that some α-particles were deflected at large angles and a few even bounced back. So Rutherford to propose a new atomic model. Based on the experimental results, Rutherford suggested that atoms have a small, dense, and positively charged nucleus at the center, which contains most of the atom's mass. He proposed that electrons orbit around the nucleus at a distance. This became known as the Rutherford atomic model or the nuclear model of the atom.

25. Correct option: (C)

A α-particle consists of two protons and two neutrons, resembling to the nucleus of a helium atom. It carries a positive charge of +2e. Due to their composition, α-particles have a relatively large mass compared to electrons and other subatomic particles. So an $\alpha-$particle is doubly charged nuclei of Helium atom.

In certain radioactive decays, such as alpha decay, atomic nuclei emit α-particles as they undergo a transformation.

26. Correct option: (D)

In Rutherford's gold foil experiment, it was observed that the majority of the α-particles passed straight through the foil without any significant deflection. This observation led Rutherford to conclude that most of the atom is empty space.

He proposed that the positively charged α-particles were mostly encountering empty space within the atom, resulting in their relatively straight path.

Most part of the atom is empty, which allows the α-particles to pass through the foil with little deflection.

27. Correct option: (A)

The Rutherford atomic model was proposed by Ernest Rutherford in 1911 explained that the atom having a small, dense, and positively charged nucleus at the center. Electrons were thought to orbit around the nucleus in a manner similar to planets orbiting the Sun, hence the name "Planetary model."

This model brought a significant advancement from Thomson's Plum pudding model and provided a more accurate representation of the atom's structure. The Planetary model laid the foundation for further developments in atomic theory, leading to the Bohr model and eventually the quantum mechanical model.

28. Correct option: (B)

When an electron jumps from the N shell to the K shell (closer to the nucleus), energy is released. Electrons in an atom occupy specific energy levels or shells. The K shell is the innermost shell, while the N shell is further away from the nucleus. When an electron transitions from a higher energy level to a lower energy level, it releases energy in the form of electromagnetic radiation.

29. Correct option: (A)

The angular momentum of the electron (mvr) is quantized and can be represented as $nh/2\pi$, where n is a positive integer and h is Planck's constant.

Therefore, the correct answer is option $nh/2\pi$.

30. Correct option: (A)

In the Bohr model of the atom, electrons occupy discrete energy levels or shells. These energy levels are labeled with letters such as K, L, M, N, and so on, innermost shell is K.

According to the Bohr model, the energy of an electron in a specific shell is inversely proportional to the square of its principal quantum number (n). Therefore, as the value of n increases, the energy of the electron in that shell increases.

Since adjacent orbits refer to energy levels that are closest to each other, the energy difference between two adjacent orbits is minimum when considering the transition between the K and L shells.

Level-II

1. Correct option: (C)

The angular momentum of the electron in the M-shell of the hydrogen atom is (option D) $4h/2\pi$.

In the Bohr model of the atom, the angular momentum of an electron in a particular orbit is given by the product of its mass (m), velocity (v), and radius (r). Mathematically, it can be expressed as mvr.

For the hydrogen atom, the angular momentum is quantized, and it follows the relationship L = nh/2π, where n is the principal quantum number and h is Planck's constant.
In the case of the M-shell, the principal quantum number (n) would be 3. Hence, substituting n = 3 in the formula L = nh/2π, we get:
L = 3h/2π

2. Correct option: (A)

We have equation

$$\frac{m_e v^2}{r} = \frac{Zk_e e^2}{r^2} \rightarrow I \quad r = \frac{n^2 h^2}{4\prod^2 mZe^2}$$

We also have another equation $mvr = \frac{nh}{2\prod} \rightarrow II$

From solving equations I and II we get $r = \frac{n^2 h^2}{4\prod^2 mZe^2}$

3. Correct option: (D)

Bohr's atomic theory, also known as the Bohr model, according to that electrons in an atom occupy specific energy levels also called as stationary states. These stationary states are characterized by fixed energy values and specific distances from the nucleus.
According to Bohr's theory, electrons can transition between these stationary states by absorbing or emitting some amounts of energy in the form of photons. This concept of stationary states laid the foundation for understanding atomic structure and the quantization of energy levels in atoms.

4. Correct option: (A)

Formula for energy difference between two orbits is $|-13.6Z^2[\frac{1}{(n+1)^2} - \frac{1}{n^2}]|$

Greater the difference, smaller the "n" value. So $E_2 - E_1 = \triangle E$ is the correct answer.

5. Correct option: (A)

According to Bohr's atomic model $r\alpha \frac{n^2}{Z}$

So, radius of nth orbit is r= $r_1 n^2$, as Z=1 for H atom.

6. Correct option: (A)

In Bohr's atomic model, the energy of an electron at an infinite distance from the nucleus is taken as Zero. According to the model, electrons in the hydrogen atom have zero energy when they are infinitely far away from the nucleus.

7. Correct option: (B)

The ratio of the radius of the Li^{+2} ion in the 3rd energy level to that of the He^{+} ion in the 2nd energy level can be calculated using the formula for the radius of an orbit in the Bohr model.

$$r \propto \frac{n^2}{Z}$$

Where:
r = radius of the orbit,
n = principal quantum number,
Z = atomic number.

For the Li^{+2} ion in the 3rd energy level (n = 3), we have:

$\frac{r_1}{r_2} = \frac{3}{2} = 3:2 \quad r_1 = (\frac{3^2}{3}) = 3$

For the He^{+} ion in the 2nd energy level (n = 2), we have $r_1 = (\frac{2^2}{2}) = 2$

Therefore, the ratio of the radii is: $\frac{r_1}{r_2} = \frac{3}{2} = 3:2$

8. Correct option: (B)

In the Bohr model of the atom, the difference between adjacent energy levels decreases as the number of orbits increases from the nucleus.

As the electron moves to higher energy levels, the energy spacing between these levels decreases. This is due to the inverse square relationship between the radius of the orbit and the energy level, it can be written as

$E \propto \frac{1}{n^2}$

Here E represents the energy level and n represents the principal quantum number. As n Increases, the energy levels get closer together, and the difference between adjacent energy levels decreases.

9. Correct option: (B)

For an object moving in the circular orbit, the centripetal force is given by $\frac{-mv^2}{r}$ And is equal to force of attraction between electron and proton.

10. Correct option: (A)

According to Bohr's postulates, the energy of an electron in a given orbit is inversely proportional to the square of the principal quantum number (n). Therefore, the energy of an electron in the first orbit (n = 1) is higher than the energy of an electron in the second orbit (n = 2).

When an electron jumps from a higher energy level to a lower energy level, it emits energy in the form of a photon. This energy emission causes the electron to lose energy and decrease its velocity.

Therefore, the change in velocity when an electron jumps from the first orbit to the second orbit is less than its original velocity. The change in velocity would be a reduction by some fraction.

11. Correct option: (D)

Bohr's model of the atom explained about the emission spectrum of hydrogen and provided a framework for understanding the stability of electron orbits. However, it also had limitations when applied to atoms with more than one electron, leading to the drawbacks as mentioned in the options.

A) **Spectrum of atoms having more than one electron**: Bohr's model was not able to predict the emission or absorption spectra of atoms with more than one electron. It failed to explain the complex spectral lines observed in such atoms.

B) **Zeeman effect**: This effect refers to the splitting of spectral lines in the presence of a magnetic field. Bohr's model was unable to account for this effect, which demonstrated the need for a more advanced model that considered electron spin and magnetic interactions.

C) **Stark effect**: The Stark effect refers to the splitting and shifting of spectral lines in the presence of an electric field. Similarly, Bohr's model did not explain this effect, highlighting its limitations.

12. Correct option: (B)

Bohr's model was specifically developed on the hydrogen atom model, which has only one electron. It successfully explained the spectral behavior of hydrogen. However, when it comes to atoms or ions with multiple electrons, Bohr's model fails to predict their properties and spectral behavior. As a result, it is not applicable to species like Be , which has two electrons.

13. Correct option: (A)

In physics, elliptical orbits are characterized by the eccentricity (e) of the ellipse, which is defined as the ratio of the distance between the foci of the ellipse to the major axis of the ellipse. The eccentricity is given by the equation: $e = c/a$

Where c represents the distance between the foci and represents the semi-major axis (half the major axis) of the ellipse.

In the context of the given options, option C) k/n matches this definition, with k representing the distance between the foci and n representing the semi-major axis of the ellipse.

14. Correct option: (D)

Let's break down the variables in this expression:

m = mass of the electron.

h= Planck's constant.

L is the orbital angular momentum of the electron.

This expression relates the mass of the electron, its orbital angular momentum, and Planck's constant. It states that the product of the mass and the square of the orbital angular momentum is equal to π (pi) times Planck's constant.

One or more than one type:

15. Correct option: (A,B,D)

A) According to classical physics, revolving electrons should continuously lose energy and fall into the nucleus. Rutherford's model could not explain atomic stability.

B) It contradicted Maxwell's theory of electrodynamics, which predicts that accelerating charges emit radiation.

C) Rutherford's model could not explain atomic line spectra.

D) It did not explain the arrangement (distribution) of electrons around the nucleus.

16. Correct option: (A,B,D)

Hydrogen-like species contain only one electron.

$He^{+} \rightarrow 2 - 1 = 1$ electron

$Li^{2+} \rightarrow 3 - 2 = 1$ electron

$Na^{+} \rightarrow 11 - 1 = 10$ electrons

H atom → 1 electron

17. Correct option: (A,B,D)

In Bohr's model:

Energy is quantized.

Angular momentum is quantized.

Radius of permitted orbits is also quantized.

Mass of the electron is constant, not quantized.

18. **Correct option: (A,B,D)**
Sommerfeld introduced **elliptical orbits**.
Angular momentum remains quantized.
It allows both circular and elliptical orbits, so "only circular" is incorrect.
The model introduced additional quantum numbers.

19. **Correct option: (A,B,C)**
Bohr's energy equation:
$E_n = -\frac{13.6}{n^2}$ eV
Energy is negative.
As $n$increases, energy becomes less negative (i.e., increases).
At infinity ($n = \infty$), energy becomes zero.
Different orbits have different energies.

20. **Correct option: (A,B,C)**
Thomson discovered the electron.
Millikan measured the charge of the electron.
Goldstein discovered canal (anode) rays.
Chadwick discovered the neutron, not the proton.

21. **Correct option(D)**
Assertion (A):
Electrons emit energy when they jump to higher orbit.
False. Electrons **absorb** energy to move to a higher orbit.
Reason (R):
Energy increases in higher orbits.
True. Higher orbits have higher (less negative) energy.
Therefore,
Assertion is false but Reason is true.

22. **Correct option(A)**
Assertion (A):
Dalton's theory could not explain the internal structure of atoms.
True
Reason (R):
Dalton assumed atoms were indivisible.
True
This assumption is the reason why Dalton could not explain atomic structure.

23. **P → b ;Q → a ;R → c ;S → d**

24. **Correct option(C)**
James Chadwick discovered the neutron in 1932.

25. **Correct option(C)**
Neutrons have **no electric charge**.

26. **Correct option(B)**
A neutron is **slightly heavier than a proton** and much heavier than an electron.

27. Hydrogen has atomic number **1**.
A neutral atom has equal numbers of protons and electrons.

28. A neutron carries **zero charge**.

29. Neutrons=Mass Number−Atomic Number= $23 - 11 = 12$

30. Using the formula:
Maximum electrons = $2n^2$

For the K-shell,
$n = 1$
$2(1)^2 = 2$

Electron distribution rules:

1. Correct option: (B)

In **1900**, **Max Planck** proposed the **Quantum Theory of Radiation** to explain the emission of energy by hot objects (black-body radiation).
According to Planck:
Energy is **not emitted or absorbed continuously**.
It is emitted or absorbed in the form of **small discrete packets** called **quanta** (or photons).
The energy of each quantum is given by:
$E = h\nu$

2. Correct option: (A)

According to Aufbau's principle, electrons in an atom can have a maximum of three quantum numbers.
Aufbau's principle states that electrons fill the available atomic orbitals in order of increasing energy levels. Each electron in an atom is described by a set of four quantum numbers they are principal quantum number (n), the azimuthal quantum number (l), the magnetic quantum number (m_l), and the spin quantum number (m_s).

However, Aufbau's principle focuses on the filling of orbital's, and it explains that each orbital is filled with a maximum of two electrons, with opposite spins, before moving to the next orbital.
Therefore, while each electron has four quantum numbers, Aufbau's principle indicates that only one quantum number is needed to describe the order of filling orbital's.

3. Correct option: (B)

According to the rules for the filling of electron orbital's in an atom, each shell can hold a maximum number of electrons equal to $2n^2$, where n represents the principal quantum number.
The principal quantum number (n) designates the energy level of an electron. For example, the first energy level (n=1) can hold a maximum of $2(1)^2 = 2$ electrons, the second energy level (n=2) can hold a maximum of $2(2)^2 = 8$ electrons and so on.

4. Correct option: (C)

Each orbital can accommodate a maximum of two electrons.
According to the Pauli Exclusion Principle, no two electrons in an atom can have the same set of four quantum numbers. Since an orbital is described by three quantum numbers (n, l, and m_l), each orbital can hold a maximum of two electrons with opposite spins (spin quantum number, m_s).

This means that in a given orbital, you can have one electron with a spin-up (+1/2) and another electron with a spin-down (-1/2). Therefore, the maximum number of electrons that can be accommodated in an orbital is two.

5. Correct option: (C)

The statement "Electrons will fill a set of degenerate orbital's by keeping their spins parallel" is based on Hund's rule.

Hund's rule is a principle in quantum mechanics that governs the filling of degenerate orbitals, which are orbital's with the same energy level. According to Hund's rule, when multiple degenerate orbital's are available, before pairing up electrons will first occupy separate orbital's within the set, each with the same spin (parallel spins) after pairing up with opposite spins. This arrangement minimizes electron-electron repulsion and stabilizes the atom.

This rule explains why, each orbital is occupied by one electron before any pairing occurs for example, when filling the p orbital's in a sub shell (such as the 2p sub shell). The electrons distribute themselves with parallel spins to maximize the stability of the atom.

6. Correct option: (D)

Aufbau's principle states that electrons fill the atomic orbitals in order of increasing energy levels. In this electronic configuration, the numbers represent the maximum number of electrons hat can be accommodated in each sub shell:

The first sub shell is the 1s sub shell, which can hold a maximum of 2 electrons.

The second sub shell is the 2s sub shell, which can also hold a maximum of 2 electrons.

The third sub shell is the 2p sub shell, which can accommodate a maximum of 6 electrons.

So, the given option satisfies the correct order of electronic configuration follows the principle of filling orbitals in order of increasing energy levels, which is the essence of Aufbau's principle.

7. Correct option: (A)

Pauli's exclusion principle states that "it is impossible for any two electrons to have the same set of all four quantum numbers."

This principle was proposed by Wolfgang Pauli, is a fundamental concept in quantum mechanics. It states that no two electrons in an atom can have identical quantum numbers. The four quantum numbers that describe an electron are the principal quantum number (n), azimuthal quantum number (l), magnetic quantum number (m_l), and spin quantum number (m_s).

According to Pauli's exclusion principle, any given set of quantum numbers must be unique for each electron in an atom.

8. Correct option: (B)

Planck's constant is represented by the symbol h.

It appears in **Planck's quantum equation**:

$E = h\nu$

where:

E= Energy of a quantum (joule)

h= **Planck's constant**

ν(nu) = Frequency of radiation (Hz)

9. Correct option: (B)

Half-filled or completely filled sub-shells impart greater stability to the atoms.

When a sub-shell is half-filled or completely filled, it results in a more stable electronic configuration. This stability arises from the balance of electron-electron repulsion and the arrangement of electrons in such a way that minimizes energy.

The stability of an atom is crucial in determining its chemical properties and reactivity. Atoms with stable electron configurations are less likely to be inert. This stability is especially significant for elements in the transition metals and noble gas

groups, as they tend to have half-filled or completely filled sub-shells, contributing to their unique chemical behavior and inertness.

10. Correct option: (A)

The Nitrogen atom has total 7 electrons and the configuration corresponds to the ground state electronic configuration of a nitrogen atom. Let's break it down:
The first two electrons occupy the 1s orbital.
The next two electrons occupy the 2s orbital.
The remaining three electrons occupy the 2p orbitals, with one electron each in the 2px, 2py, and 2pz orbitals.
This arrangement accounts for the atomic number 7, which corresponds to nitrogen. Therefore, the given configuration represents a nitrogen atom in its ground state.

11. Correct option: (D)

Hund's rule deals with the distribution of electrons in degenerate orbitals.
Degenerate orbitals refer to orbitals that have the same energy level. According to Hund's rule, when electrons occupy degenerate orbitals, they will first fill each orbital with one electron before pairing up.
This rule is based on the principle that electrons repel each other due to their negative charges. By occupying separate orbitals, they minimize their repulsion and increase the overall stability of the atom or molecule.

12. Correct option: (A)

The systematic arrangement of electrons in orbitals in the increasing order of their energies is known as electronic configuration.
The electronic configuration of an atom or molecule explains how the electrons are distributed among the various orbitals and energy levels. It specifies the number of electrons in each shell, sub shell, and orbital. The electronic configuration provides important information about the chemical behavior and properties of an element.

13. Correct option: (C)

The atomic number of chromium is 24, which means it has 24 electrons. The electron configuration of an atom describes how these electrons are distributed among the energy levels and sub shells.
The noble gas that precedes chromium in the periodic table is argon (Ar), which has the electronic configuration $1s^2\ 2s^2\ 2p^6 3s^2\ 3p^6$.

To determine the electronic configuration of chromium, we need to add the remaining electrons after argon.
The 3d sub shell can hold up to 10 electrons, and the 4s sub shell can hold up to 2 electrons. Therefore, the electronic configuration of chromium is (Ar) $3d^5\ 4s^1$.

14. Correct option: (B)

The electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^3$ represents the element phosphorus.
Phosphorus has an atomic number of 15, meaning it has 15 electrons. The electronic configuration describes the arrangement of these electrons in different energy levels and sub shells.

The first shell (n=1) has 2 electrons in the 1s sub shell ($1s^2$).

The second shell (n=2) has 2 electrons in the 2s sub shell ($2s^2$) and 6 electrons in the 2p sub shell ($2p^6$).

The third shell (n=3) has 2 electrons in the 3s sub shell ($3s^2$) and 3 electrons in the 3p sub shell ($3p^3$).

15. Correct option: (B)

For vanadium (V), the electronic configuration is:

$1s^2 2s^2 2p^6$

The valence electrons are those in the outermost energy level, which is the 4th energy level (n=4).

The sub shells in the 4th energy level are 4s, 3d, and 4p.

16. Correct option: (C)

Given electronic configuration is $1s^2 2s^2 2p^5 3s^1$ with atomic number 10 which represents Neon atom. But original electronic configuration of Neon atom is $1s^2 2s^2 2p^6$

So here one electron is transferred from 2p to 3s orbital which is a higher energy state. So the given electronic configuration is represents exited state of **NEON** atom.

17. Correct option: (A)

The electronic configuration of different atoms can be represented in the n^{1^x} method.

In the n^{1^x} method, the electronic configuration is written as a combination of the principal quantum number (n) and the azimuthal quantum number (l). The principal quantum number describes the energy level or shell of an electron, while the azimuthal quantum number describes the sub shell or orbital.

18. Correct option: (A)

The sequence in which electrons occupy various orbitals can be easily remembered with the help of Moeller's diagram.

Moeller's diagram is a graphical representation that shows the order in which electrons fill the orbitals of an atom. It is based on the Aufbau's principle, which states that electrons fill the lowest energy orbitals first before filling higher energy orbitals.

The diagram organizes the orbitals into different energy levels or shells and indicates the order in which they are filled. Each orbital is represented by a box, and arrows are used to represent electrons. The arrows point up or down to indicate the electron's spin.

By following the sequence in Moeller's diagram, one can easily determine the order in which electrons occupy various orbitals as you move across the periodic table. This helps in understanding the electron configuration of different elements and their placement in the periodic table.

19. Correct option: (A)

The electronic configuration of nitrogen is $1s^2 2s^2 2p^3$ .Atomic number of Nitrogen is 7.

In this electronic configuration, we can break it down as follows:

$1s^2$: This represents the first energy level (n = 1) and the 1s orbital, which can hold a maximum of 2 electrons. In the case of nitrogen, it has 2 electrons in the 1s orbital.

$2s^2$: This represents the second energy level (n = 2) and the 2s orbital, which can also hold a maximum of 2 electrons. Nitrogen has 2 electrons in the 2s orbital.

$2p^3$: This represents the second energy level (n = 2) and the 2p orbital, which can hold a maximum of 6 electrons. Nitrogen has 3 electrons in the 2p orbital.

Therefore, the electronic configuration of nitrogen is $1s1s^2 2s^2 2p^3$.

20. Correct option: (B)

After the 3d sub-level is completely filled, the differentiating electron enters into the 4p sublevel. Therefore, option B) 4p is the correct answer.

In the electron configuration of atoms, the sublevels are filled in a specific order based on their increasing energy.

The order of filling is generally given by the Aufbau's principle, which states that lower energy levels are filled before higher energy levels.

The filling order for the sublevels is as follows:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, and so on.

21. Correct option: (B)

To determine the number of unpaired electrons in the Ni+2 ion, we need to know the electronic configuration of the neutral nickel atom (Ni) and then consider the loss of two electrons to form the Ni^{+2} ion.

The electronic configuration of neutral nickel (Ni) is: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$.

When nickel loses two electrons to form the Ni^{+2} ion, the electron configuration become $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$

In this configuration, we can see that there are 2 unpaired electrons in the 3d sublevel ($3d^8$).

Therefore, the number of unpaired electrons in the Ni^{+2} ion is 2.

22. Correct option: (A)

The maximum number of electrons that can be present in the M and N-shells, respectively, can be determined using the formula 2n^2, where n represents the principal quantum number.

For the M-shell:

The principal quantum number (n) for the M-shell is 3.

Using the formula $2n^2$, we have $2(3)^2 = 2(9) = 18$.

Therefore, the maximum number of electrons that can be present in the M-shell is 18.

For the N-shell:

The principal quantum number (n) for the N-shell is 4.

Using the formula $2n^2$, we have $2(4)^2 = 2(16) = 32$.

Therefore, the maximum number of electrons that can be present in the N-shell is 32.

LEVEL-II

1. Correct option: (B)

According to **Planck's Quantum Theory**,

$E = h\nu$

where:

E= Energy

h= Planck's constant

ν= Frequency

Thus, energy is **directly proportional to frequency**

2. Correct option: (B)

The equation$E = h\nu$

gives the **energy of electromagnetic radiation (photon).**

3. Correct option: (C)

For M shell,$n = 3$

Maximum electrons

$2n^2 = 2(3)^2 = 18$

4. Correct option: (B)

According to the Aufbau Principle,

$4s < 3d < 4p < 5s$

Hence 4s fills first.

5. Correct option: (B)

Correct order is

$4s < 3d$

6. Correct option: (B)

Carbon has 6 electrons.

Configuration:$1s^2 2s^2 2p^2$

7. Correct option: (B)

Copper$[Ar]3d^{10}4s^1$

The completely filled d-subshell provides extra stability.

8. Correct option: (A)

Isoelectronic species have the same number of electrons.Na^+

11−1=10 electrons

Ne:10 electrons

F^-:9+1=10 electrons

All have 10 electrons.

9. Correct option: (A,C,D)

Isotones have:

Same number of neutrons

Different atomic numbers

Usually different mass numbers

They do not have the same protons.

10. Correct option: (A,B)

Extra stability is associated with:

Half-filled subshells

Completely filled subshells

11. Correct option: (A,B,C)

Hund's rule states:

Orbitals are singly occupied first

Pairing starts only after single occupation

Spins remain parallel

12. Correct option: (A,B,D)

A) Maximum electrons = $2n^2$
(Correct)

B) K shell holds 2 electrons
(Correct)

C) L shell holds 18 electrons

L shell holds only 8.(wrong)

D) M shell holds 18 electrons (Correct)

13. **Correct option: (B,C)**

B) Energy is emitted in packets

C) Energy ∝ Frequency

14. **Correct option: (B)**

Assertion

Energy is emitted in discrete packets called quanta.(True)

Reason

Energy is directly proportional to frequency.(True)

However,

The reason gives the equation$E = h\nu$

It does not explain why energy is emitted in packets.

15. **Correct option: (C)**

Assertion

Atoms having all paired electrons are diamagnetic.

True

Reason

Diamagnetic substances are weakly attracted by magnetic field.

Actually,

Diamagnetic substances are **weakly repelled** by magnetic fields.

Reason is false.

16. P → b; Q → a; R → c; S → d

17. **Correct option: (B)**

Electrons occupy singly before pairing.

18. **Correct option: (B)**

Half-filled orbitals provide maximum stability.

19. **Correct option: (B)**

Degenerate orbitals possess same energy.

20. A p-subshell contains three orbitals.

Each orbital accommodates two electrons.

$3 \times 2 = 6$

21. Neutrons$40 - 20 = 20$

22. Electronic configuration$1s^2 2s^2 2p^4$

In the 2p subshell:

↑ ↑ ↑

Then one electron pairs:

↑↓ ↑ ↑

Thus, Unpaired electrons = **2**

23. For M shell,$n = 3$

Maximum electrons$2n^2 =$ **18**

Previous year questions

1. **Answer: B)**

Explanation:

Statement I is Correct**:** According to classical electromagnetism, an accelerating charged particle (like an electron orbiting a nucleus) should continuously radiate energy. This would cause the electron to lose energy and eventually spiral into the

nucleus, making the atom unstable. Rutherford's model could not explain why this doesn't happen.

Statement II is Incorrect**:** In the electromagnetic spectrum, X-rays have a very high frequency and a very short wavelength. Microwaves, on the other hand, have a much longer wavelength.

2. **Answer: B**

Explanation:

4s: n = 4, l = 0, so m_l must be 0.

3p: n = 3, l = 1, so m_l can be -1, 0, +1.

5d: n = 5, l = 2, so m_l can be -2, -1, 0, +1, +2.

3. **Answer: C).**

Explanation:

Assertion (A) is True**:** Like any neutral atom, isotopes have an equal number of protons and electrons, making them electrically neutral.

Reason (R) is False**:** This is actually the definition of isobars. Isotopes are defined as species with the same atomic number (same number of protons) but different mass numbers (different number of neutrons).

4. **Answer: C)**

Explanation:

II. Proton numbers of elements X and Y are 6 and 7 respectively.

Correct**.** X has 6 protons and Y has 7 protons.

III. Elements X and Y are represented by ${}^{14}_{6}X$ and ${}^{14}_{7}Y$ respectively.

Correct. These symbols correctly match our calculated atomic and mass numbers.

IV. Elements X and Y are isobars.

Correct**.** Isobars are atoms of different elements (different atomic numbers) that have the same mass number (nucleon number). Since both have a mass number of 14, they are isobars.

5. Answer: C)

Explanation:

P and Q are isotopes: Correct**.** Isotopes are atoms of the same element (same number of protons) that have different numbers of neutrons (and thus different mass numbers).

6. Answer: B

Explanation:

Element X: Shows an M-shell (3rd shell) containing 2 electrons. This means its full configuration is 2, 8, 2.

Valency of X: Since it has 2 valence electrons, it tends to lose them to become stable. Its valency is 2.

Element Y: Shows an L-shell (2nd shell) containing 6 electrons. This means its configuration is 2, 6.

Valency of Y: Since it has 6 valence electrons, it needs to gain 8 - 6 = 2 electrons to complete its octet. Its valency is 2.

7. Answer: C)

Explanation:

Element P: Has 5 electrons in the L-shell. Since the K-shell (inner) must be full with 2 electrons, the total electrons = 2 + 5 = 7. Thus, the atomic number is 7 (Nitrogen).

Element Q**:** Has 5 electrons in the M-shell. This implies the K and L shells are full (2 + 8). Total electrons = 2 + 8 + 5 = 15. Thus, the atomic number is 15 (Phosphorus).
Valency: Both have 5 valence electrons, so they gain 3 electrons to reach stability. Their valency is 3, not 5.

8. Answer: D)

Explanation:

Statement III (Incorrect): P has 16 protons (Sulfur) and 18 electrons (S^{2-}). It is an ion, not a neutral noble gas.
Statement IV (Incorrect): Q has 11 protons and 10 electrons (11-10 = +1). Q is a cation, not an anion**.** U is neutral (20p, 20e).

9. Answer: C)

Explanation:

Proton Numbers (Z): U(19), V(53), W(17), X(35), Y(18). Order: W(17) < Y(18) < U(19) < X(35) < V(53). (Statement I is incorrect).
Mass Numbers (A): W(35) < U(39) < Y(40) < X(80) < V(127). (Statement II is correct).
Neutron Check: In U, neutrons = 39 - 19 = 20. In W, neutrons = 35 - 17 = 18. Protons do not equal neutrons in all elements. (Statement IV is incorrect).

10. Answer: C)

Explanation:

R (11): Configuration 2, 8, 1. It loses 1 electron to form R^+ (univalent cation).
T (17): Configuration 2, 8, 7. It gains 1 electron to form T^- (univalent anion).
Compound: R and T will combine to form RT (like NaCl), not R_2T.
P (4): Configuration 2, 2. It has 2 electrons in its L-shell, not a complete octet (8)

3.STOICHIOMETRY

Mole Concept

LEVEL-I

1. Correct option: (C)

One atomic mass unit (amu) is defined as $^{1}/_{12}{}^{th}$ the mass of a carbon-12 atom. It is approximately equal to 1.66×10^{-24} grams.

2. Correct option: (B)

A) General Atomic Weight (GAW) = Weight of 1 gm atom of an element

B) Relative Atomic Mass (RAM) = $\frac{\text{Mass of 1 atom of that element}}{^{1}/_{12}{}^{th}\text{ mass of C}-12\text{ atom}}$

C) Molecular weight refers to the sum of the atomic weights of all the atoms in a molecule.

D) Equivalent weight = $\frac{\text{Molecular weight}}{n}$

where, n = number of equivalents of the substance involved in the reaction.

3. Correct option: (B)

A) 1 Dalton = 1 amu

B) Pascal is the SI unit of pressure, not related to the atomic mass unit.

C) 1 Avogram = 1.66×10^{-24} g = 1 amu

D) One Aston = $^{1}/_{16}{}^{th}$ the mass of a oxygen-16 atom

4. Correct option: (A)

Since a hydrogen atom contains only one proton and no neutrons, its atomic mass is approximately equal to the mass of a proton.

The mass of a proton is approximately 1.66×10^{-24} grams.

According to the definition of an atomic mass unit (amu), 1 amu is equal to 1.66×10^{-24} grams.

Therefore, the mass of one hydrogen atom is approximately equal to 1 amu.

5. Correct option: (D)

Given: 1. 1 mole of sodium (Na)

Calculation:

- Moles of Na = $\frac{\text{Mass of Na}}{\text{Molecular weight of Na}}$

Mass of Na = 23 g/mol x 1 mol = 23 g.

The calculated mass of sodium as 23 g implies that options B and C are incorrect.

- No. of atoms of Na = No. of moles of Na x Avogadro's number (N_A)

$= 1 \times N_A = 6.023 \times 10^{23}$ atoms

The fact that elemental sodium (Na) exists as individual atoms, not molecules, renders option A incorrect, making option D the correct answer.

6. Correct option: (C)

Given: 1. Mass of calcium carbonate ($CaCO_3$) = 50 g

To find: Number of oxygen atoms = ?

Calculation:

- Moles of $CaCO_3 = \frac{\text{Mass of } CaCO_3}{\text{Molecular weight of } CaCO_3}$

Moles of $CaCO_3 = \frac{50\text{ g}}{100\text{ g/mol}} = 0.5$ mol

- Since there are 3 oxygen atoms in one molecule of calcium carbonate ($CaCO_3$), the moles of oxygen will be three times the moles of calcium carbonate.

Moles of O = 3 × Moles of $CaCO_3$ = 3 × 0.5 mol = 1.5 mol

- Number of oxygen atoms = Moles of O × Avogadro's number

$= 1.5 \text{ mol} \times 6.023 \times 10^{23}$

$= 9.035 \times 10^{23}$

7. Correct option: (B)

Given: 1. Abundance of ^{54}Fe = 5%

2. Abundance of ^{56}Fe = 90%

3. Abundance of ^{57}Fe = 5%

To find: Atomic mass of Fe = ?

Calculation:

- Using weighted average formula for atomic mass calculation,

$$\text{Atomic mass} = \frac{(^{54}\text{Fe Mass} \times {}^{54}\text{Fe Abundance}) + (^{56}\text{Fe Mass} \times {}^{56}\text{Fe Abundance}) + (^{57}\text{Fe Mass} \times {}^{57}\text{Fe Abundance})}{100}$$

$$\text{Atomic mass} = \frac{(54 \times 5) + (56 \times 90) + (57 \times 5)}{100} = 55.95$$

8. Correct option: (B)

Given: 1. Standard Temperature and pressure (STP) condition

To find: Molar volume of gas = ?

Calculation:

At STP, the temperature is 0 degrees Celsius (273.15 Kelvin) and the pressure is 1 atmosphere (atm).

- Calculating the molar volume of gas using the ideal gas equation.

$PV = nRT$

$1 \times V = 1 \times 0.08206 \times 273.15$

Molar volume of gas = V = 22.414 L

9. Correct option: (D)

Given: 1. Volume of N_2 gas = 1 cc = 0.001 L

2. Normal Temperature and pressure (NTP) condition

To find: Number of nitrogen molecules = ?

Calculation:

At NTP, the temperature is 25°C and pressure is 1 atm.

- Ideal gas law

No. of moles = n = $\frac{PV}{RT}$

n = $\frac{1\ atm \times 0.001L}{0.08206\ {}^{L.atm}/_{mol.K} \times 293K}$ = 4.1 x 10^{-5}

- Number of nitrogen molecules = n × Avogadro's number

$= 4.1 \times 10^{-5} \times (6.023 \times 10^{23})$

$= 2.67 \times 10^{19}$

10. Correct option: (D)

Given: 1. Mass of nitrogen = 7g

2. Mass of oxygen = 8g

3. Standard Temperature and pressure (STP) condition

To find: Volume of mixture = ?

Calculation:

- Moles of nitrogen = $\frac{\text{Mass of nitrogen}}{\text{Molecular weight of nitrogen}} = \frac{7}{28} = \frac{1}{4}$
- Moles of oxygen = $\frac{\text{Mass of oxygen}}{\text{Molecular weight of oxygen}} = \frac{8}{32} = \frac{1}{4}$
- Total moles of mixture = moles of N_2 + moles of O_2

$= 0.25 + 0.25 = 0.5$

- Ideal gas law,

$PV = nRT$

$1 \times V = 0.5 \times 0.08206 \times 273.15$

V ~ 11220 ml

11. Correct option: (D)

Given: 1. Mass of NO_2 = 23g

Calculation:

- Calculating molecules of NO_2

Moles of NO_2 = $\frac{\text{Mass of NO2}}{\text{Molecular weight of NO2}} = \frac{23}{46} = 0.5$

No. of molecules of NO_2 = Moles of NO_2 x Avogadro's number = 0.5 x N_A

A) 8g of oxygen

- Moles of $O_2 = \frac{\text{Mass of O2}}{\text{Molecular weight of O2}} = \frac{8}{32} = 0.25$
- No. of molecules of O_2 = Moles of O_2 x Avogadro's number = 0.25 x N_A

B) 28 g of carbon

- Moles of $C = \frac{\text{Mass of C}}{\text{Molecular weight of C}} = \frac{28}{12} = 2.33$
- No. of molecules of C = Moles of C x Avogadro's number = 2.33 x N_A

C) 16g of SO_2

- Moles of $SO_2 = \frac{\text{Mass of SO2}}{\text{Molecular weight of SO2}} = \frac{16}{64} = 0.25$
- No. of molecules of SO_2 = Moles of SO_2 x Avogadro's number = 0.25 x N_A

D) 22 g of CO_2

- Moles of $CO_2 = \frac{\text{Mass of CO2}}{\text{Molecular weight of CO2}} = \frac{22}{44} = 0.5$
- No. of molecules of CO_2 = Moles of CO_2 x Avogadro's number = 0.5 x N_A

12. Correct option: (D)

Given: 1. Mass of CO_2 = 44g

To find: Moles of oxygen atom = ?

Calculation:

- Moles of $CO_2 = \frac{\text{Mass of CO2}}{\text{Molecular weight of CO2}} = \frac{44}{44} = 1$
- 1 mole of CO_2 contains 2 moles of oxygen atoms

13. Correct option: (A)

Given: 1. 20 g mass of H_2

2. Standard Temperature and pressure (STP) condition

To find: Volume of H_2 = ?

Calculation:

- Moles of $H_2 = \frac{\text{Mass of H2}}{\text{Molecular weight of H2}} = \frac{20}{2} = 10$
- Ideal gas law,

$$V = \frac{nRT}{P}$$

$$V = \frac{10 \times 0.08206 \ {}^{L.atm}/_{mol.K} \times 273K}{1 \ atm} = 224 \text{ L}$$

14. Correct option: (C)

Given: 1. Mass of He = 2g

2. Mass of N_2 = 7g

3. Standard Temperature and pressure (STP) condition

To find: Total volume of mixture = ?

Calculation:

- Moles of He = $\frac{\text{Mass of He}}{\text{Molecular weight of He}} = \frac{2}{4} = 0.5$
- Moles of N_2 = $\frac{\text{Mass of N2}}{\text{Molecular weight of N2}} = \frac{7}{28} = 0.25$
- Total moles of mixture = moles of N_2 + moles of He

$= 0.25 + 0.5 = 0.75$

- Ideal gas law,

$PV = nRT$

$1 \times V = 0.75 \times 0.08206 \times 273.15$

$V = 16.8$ L

15. Correct option: (B)

Balancing the chemical reaction, $2\,CO + 1\,O_2 \rightarrow 2CO_2$

According to the balanced equation, 2 molecules of CO react with 1 molecule of O_2 to form 2 molecules of CO_2

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16. Correct option: C) 36gm of H_2O

36 gms of H2O

- Moles of H_2O_2 = $\frac{\text{Mass of H2O}}{\text{Molecular weight of H2O}} = \frac{36}{18} = 2$
- No. of molecules of H2O = Moles of H2O x Avogadro's number = 2NA

17) Correct option: A) 280g

Given: 1. 5 g-atom of Fe To find: mass of Fe = ? Calcultion:

- 5 g-atom of Fe represents 5 mol atom of Fe
- Moles of Fe = $\frac{\text{Mass of Fe}}{\text{Molecular weight of Fe}}$

Mass of Fe = 5 mol x 56 g/mol = 280g

18. Correct option: C) 4g oxygen

4g oxygen

- Moles of O2 = $\frac{\text{Mass of O2}}{\text{Molecular weight of O2}} = \frac{4}{32} = 0.125$

- No. of molecules of O_2 = Moles of O_2 x Avogadro's number = 0.125NA

19. Correct option:B) 92g

Given: 1. Vapour density of N_2O_4 = 46To find: Molar mass of N_2O_4 = ?

Calculation:$Vapour\ density\ of\ N_2O_4 = \frac{Molar\ mass\ of\ N_2O_4}{Molar\ mass\ of\ H_2}$

Molar mass of N_2O_4 = 46 x 2 = 92g

20. Correct Answer: C) 0.012 kg of C-12

Explanation:

By official chemical definition, Avogadro's number ($N_A \approx 6.022 \times 10^{23}$) is defined based on the carbon-12 isotope. It represents the exact number of atoms present in precisely 12 grams of carbon-12 (^{12}C).

Converting 12 grams into kilograms:

$$\mathbf{12\ g = \frac{12}{1000}\ kg = 0.012\ kg}$$

LEVEL-II

1. Correct option: (A)

Given: No. of He atoms = 6.022×10^{23}

To find: Mass of He = ?

Calculation:

- No. of atoms of He = Moles of He x Avogadro's number

  6.022×10^{23} = Moles of He x 6.022×10^{23}

  Moles of He = 1

- Moles of He = $\frac{\text{Mass of He}}{\text{Molecular weight of He}}$

  Mass of He = 4g/mol x 1 mol = 4g

2. Correct option: (B)

Calculation:

A) One gram atom of iron

- 1 g-atom of Fe represents 1 mole of Fe
- Mass of Fe = No. of moles of Fe x M.W of Fe = 1 x 56 = 56g

B) 5 mole of N_2

- Mass of N_2 = No. of moles of N_2 x M.W of N_2 = 5 x 28 = 140g

C) 10^{24} carbon atoms

- No. of C-atoms = No. of moles of C x N_A

10^{24} = No. of moles of C x 6.022 x 10^{23}

No. of moles of C = 1.66

- Mass of C = moles of C x M.W of C = 1.66 x 12 = 19.93g

D) 44.8 lit of He at STP

- No. of moles = n = $\frac{PV}{RT}$

$$= \frac{1\ atm \times 44.8\ L}{0.08206\ {}^{L.atm}/_{mol.K} \times 273K} = 2$$

- Mass of He = moles of He x M.W of He = 2 x 4 = 8g

3. Correct option: (A)

Given: 1. Number of atoms of gas = 6 x 10^{24}

2. Mass of gas = 200 g

3. Homo diatomic gas (X_2)

To find: Molar mass of gas = ?

Calculation:

- Gas is homodiatomic which means that it consists of two identical atoms in a molecule.
- If 1 molecule has 2 gas atoms, then there are 3 x 10^{24} molecules in 6 x 10^{24} atoms.
- No. of molecules of gas = Moles of gas x Avogadro's number

Moles of gas = $\frac{3 \times 10^{24}}{6.023 \times 10^{23}} = 4.98$

- Moles of gas = $\frac{\text{Mass of gas}}{\text{Molecular weight of gas}}$

Molecular weight of gas = $\frac{200}{4.98} \sim 40$

4. Correct option: (D)

A mole is defined as the chemical substance contains the same number of chemical units present in ${}^{1}/_{12}{}^{th}$ mass of C-12 isotope.

5. Correct option: (B)

Given: 1. 1g of hydrogen contains 6 x 10^{23} atoms

To find: Atoms in 4g of H_2 = ?

Calculation:

- Moles of H_2 = $\frac{\text{Mass of H2}}{\text{Molecular weight of H2}} = \frac{4}{2} = 2$
- Atoms of H_2 = No. of moles of H_2 x Avogadro's number

= 2 x 6.023 x 10^{23} = 12 x 10^{23} atoms

6. Correct option: (A)

Given: 1. Mass of H_2 = 100g

2. Mass of O_2 = 100g

Reaction: $2H_2 + O_2 \rightarrow 2H_2O$

To find: Mass of H_2O formed = ?

Calculation:

- Moles of $H_2 = \frac{\text{Mass of H2}}{\text{Molecular weight of H2}} = \frac{100}{2} = 50$
- Moles of $O_2 = \frac{\text{Mass of O2}}{\text{Molecular weight of O2}} = \frac{100}{32} = 3.125$
- 1 mole of O_2 reacts with 2 moles of H_2 to form 2 moles of H_2O

  Since, O_2 is the limiting agent

  Moles of H_2O formed = 2 x moles of O_2 = 2 x 3.125 = 6.25
- Moles of $H_2O = \frac{\text{Mass of H2O}}{\text{Molecular weight of H2O}}$

  Mass of H_2O = 6.25 x 18 = 112.5g

7. Correct option: (A)

Given: 1. Atomic weight of X = 32

2. Vapour density = 32

3. Volume before and after combustion is equal

Reaction: $X + O_2 \rightarrow Y$

To find: Atoms of oxygen =?

Calculation:

- Vapour density of Y = $\frac{\text{Molar mass of Y}}{\text{Molar mass of } H_2}$

Molar mass of Y = 32 x 2 = 64 g/gmol

8. None of the options correct

Case 1: Using the values given in the question statement (2.8 kg)

1. Find moles of ethylene:

   Mass of ethylene = 2.8 kg = 2800 g

   Moles of ethylene = $\frac{2800\ \text{g}}{28\ \text{g/mol}} = 100$ moles
2. Find moles of oxygen required:

   Moles of $O_2 = 3 \times 100$ moles = 300 moles
3. Find mass of oxygen required:

   Mass of O_2 = 300 moles × 32 g/mol = 9600 g = 9.6 kg

9. Correct option: (BCD)

Calculation:

- Absolute mass of Na = Relative mass of Na x amu
  = 23 amu
  = 23 x 1.66 x 10^{-24} g (1 amu = 1.66 x 10^{-24} g)
  = 23 x 1.66 x 10^{-27} kg (1 kg = 1000 g)

10. Correct option: (A)

A) One mole of Helium gas

- No. of atoms of He = No. of moles of He x N_A = N_A

B) 22.4 ml of CO_2 at STP

- No. of moles = n = $\frac{PV}{RT}$

$$= \frac{1\ atm \times 22.4 \times 10^{-3}\ L}{0.08206\ {}^{L.atm}/_{mol.K} \times 273K} = 9.993 \times 10^{-4}$$

- No. of atoms of CO_2 = No. of moles of CO_2 x N_A = 9.993 x 10^{-4} N_A

C) 11.2 litre of Hydrogen gas

- Moles (n) = $\frac{\text{Volume of gas at STP}}{22.4\ \text{L}} = \frac{11.2\ \text{L}}{22.4\ \text{L}} = 0.5$
- No. of atoms of H_2 = No. of moles of H_2 x N_A = 0.5 N_A

D) 3.2g of methane

- Moles of CH_4 = $\frac{\text{Mass of } CH_4}{\text{Molecular weight of } CH_4} = \frac{3.2}{16} = 0.2$
- No. of atoms of CH_4 = No. of moles of CH_4 x N_A = 0.2 N_A

11. Correct option: (ABCD)

A) 1g of hydrogen atom

$$\text{Moles of H} = \frac{\text{Mass of H}}{\text{Molecular weight of H}} = \frac{1}{1} = 1$$

B) 6.023×10^{23} molecules of oxygen

No. of molecules of oxygen = No. of moles of oxygen x N_A

$$\text{Moles of oxygen} = \frac{6.023 \times 10^{23}}{6.023 \times 10^{23}} = 1$$

C) 2g of Hydrogen molecules

$$\text{Moles of } H_2 = \frac{\text{Mass of } H_2}{\text{Molecular weight of } H_2} = \frac{2}{2} = 1$$

D) 32 g of oxygen molecules

$$\text{Moles of } O_2 = \frac{\text{Mass of } O_2}{\text{Molecular weight of } O_2} = \frac{32}{32} = 1$$

12. Correct option: (AB)

Given: 1. Mass of Ammonia =
34gReaction: $N_2 + 3H_2$ □
$2NH_3$

Calculation:

- Moles of $NH_3 = \frac{\text{Mass of } NH_3}{\text{Molecular weight of } NH_3} = \frac{34}{17} = 2$
- Based on stoichiometric ratio,

  To form 2 mole NH_3, we need 1 mole of N_2 & 3 mole of H_2
- Moles of $N_2 = \frac{\text{Mass of } N_2}{\text{Molecular weight of } N_2}$

  Mass of N_2 = 1 x 28 = 28g
- Moles of $H_2 = \frac{\text{Mass of } H_2}{\text{Molecular weight of } H_2}$

  Mass of H_2 = 3 x 2 = 6g

13. Correct option: (AC)

Given: 1. Moles of $KClO_3$ = 2

Reaction: $2KClO_3 \rightarrow 2KCl + 3O_2$

Calculation:

- Based on stoichiometric ratio,

2 moles of $KClO_3$ on thermal heating gives 2 moles of KCl and 3 moles of O_2

14. Correct option: (a-R, b-S, c-Q, d-P)

a) Vapour Density of gas $= \frac{\text{Molar mass of gas}}{\text{Molar mass of } H_2}$

Molar mass of gas = 2 x Vapour Density of gas

b) Avogadro number = 6.023×10^{23}

c) 1 mole of gas occupies in volume occupies 22.4 L or 22400 cm^3 volume

1 g-atom of carbon is equal to 12g because it represents the atomic mass of carbon expressed in g

15. Correct option: (a-R,S b-P c-T,S, d-Q)

a) 32g of SO_2

- Moles of $SO_2 = \frac{\text{Mass of } SO_2}{\text{Molecular weight of } SO_2} = \frac{32}{64} = 0.5$
- 0.5 mole of SO_2 at STP has 11.2 L volume occupied.

b) 22.4 lit Hydrogen at STP

- No. of moles of hydrogen = n = $\frac{PV}{RT}$

$$= \frac{1\ atm \times 22.4\ L}{0.08206\ {}^{L.atm}/_{mol.K} \times 273K} = 1$$

- Mass of hydrogen = Moles of hydrogen x M.W of hydrogen = 1 x 2 = 2g

c) 0.5 mole of CH_4

- Mass of CH_4 = Moles of CH_4 x M.W of CH_4 = 0.5 x 16 = 8g
- 0.5 mole CH_4 of at STP has 11.2 L volume occupied.

d) 32g methane

- Moles of $CH_4 = \frac{\text{Mass of CH4}}{\text{Molecular weight of CH4}} = \frac{32}{16} = 2$
- 2 mole of CH_4 at STP has 44.8 L volume occupied.

16. Correct option: (A)

Given: 1. No. of H_2 gas = 3.011 □ 10^{23}

2. Normal temperature and pressure (NTP) conditionTo find: Volume of H_2 gas = ?

Calculation:

- No. of H_2 gas atoms = No. of moles of H_2 gas x N_A

$$\text{Moles of } H_2 \text{ gas} = \frac{3.011 \times 10^{23}}{6.023 \times 10^{23}} = 0.5$$

- Ideal gas law, PV = nRT

$$V = \frac{0.5 \text{ mol} \times 0.08206\ {}^{\text{L.atm}}/_{\text{mol.K}} \times 273\text{K}}{1 \text{ atm}} = 11.19 \text{ L}$$

17. Correct option: (A)

Given: Mass of 1 atom of an element 'X' = 6.64 x 10^{-23} g

To find: Molecular weight of X_2 = ?

Calculation:

- Atomic weight of 'X' = Mass of 1 atom of 'X' x Avogadro's number

Atomic weight of 'X'

= 6.64 x 10^{-23} g x 6.022 x 10^{23} atoms/mol

Atomic weight of 'X' = 40 g/mol

- Molecular weight of X_2 = 2 x Atomic weight of 'X' Molecular weight of X_2 = 2 x 40 g/mol

Molecular weight of X_2 = 80 g/mol

18. Correct option: (B)

Given: 1. Mass of O_2 gas = 1kg at 273 □C and 2 atm

To find: Gram-atoms of oxygen = ?

Calculation:

- Moles of $O_2 = \frac{\text{Mass of } O_2}{\text{Molecular weight of } O_2} = \frac{1000}{32} = 31.25$

1 mole of O_2 = 1 g-molecule of O_2 ….(given in the passage)

& 1 g-molecules of O_2 has 2g-atom of oxygen

Therefore, 62.5 g-atom of oxygen will be present in 1 kg O_2.

19. Correct option:B) 11.2 lit

Given: 1. Mass of O_2 = 16g

2. Standard temperature and pressure (STP) condition To find: Volume of O_2 = ?

Calculation:

$$Moles\ of\ O_2 = \frac{Mass\ of\ O_2}{Molecular\ weight\ of\ O_2}$$

$\frac{16}{32} = 0.5$

Calculate the volume occupied at STP.

Volume = Moles × 22.4 L = 0.5 × 22.4 L = 11.2 L

20. Correct Option: C) 84g

67.2 lit of Nitrogen gas at STP weighs

- Step 1: Calculate the number of moles of Nitrogen gas (N_2) from the given volume.

  Number of moles = $\frac{\text{Given volume at STP}}{22.4\ \text{L}} = \frac{67.2\ \text{L}}{22.4\ \text{L}} = 3$ moles
- Step 2: Find the molar mass of Nitrogen gas (N_2).

  Molar mass of $N_2 = 2 \times 14$ g/mol = 28 g/mol
- Step 3: Calculate the mass.

  Mass = Moles × Molar mass = 3 × 28 g = 84 g

21. Correct Option: C) 5.6 L

Step 1: Use the direct molar volume relationship.

Volume = Moles × 22.4 L

Volume = 0.25 × 22.4 L = 5.6 L

22. Answer: 1.204×10^{24} **atoms** *(or* $2N_A$*)*

Explanation: 1 mole of any substance contains Avogadro's number ($N_A = 6.022 \times 10^{23}$) of particles.

Calculation:

Number of atoms = Moles × $N_A = 2 \times 6.022 \times 10^{23} = 1.204 \times 10^{24}$ atoms

23. Answer: 98

Calculation: Sum the atomic weights of all constituent atoms (H = 1 u, S = 32 u, O = 16 u):

Molecular weight = (2 × 1) + 32 + (4 × 16)

Molecular weight = 2 + 32 + 64 = **98** u (or g/mol)

24. Answer: 35.5

Calculation: Use the weighted average formula:

$$\text{Average Atomic Mass} = \frac{(\text{Mass of Isotope 1} \times \%) + (\text{Mass of Isotope 2} \times \%)}{100}$$

$$\text{Average Atomic Mass} = \frac{(35 \times 75) + (37 \times 25)}{100} = \frac{2625 + 925}{100} = \frac{3550}{100} = \mathbf{35.5}\ \text{u}$$

25. Answer: 58.5 g

Explanation: Gram formula mass is the sum of the atomic masses of the atoms in the ionic formula unit expressed in grams (Na = 23 g/mol, Cl = 35.5 g/mol).

Gram formula mass = 23 + 35.5 = **58.5** g/mol

26. Answer: 1

Step 1: Calculate the molar mass of CO_2.

Molar mass = 12 + (2 × 16) = 12 + 32 = 44 g/mol

Step 2: Calculate the number of moles.

$$\text{Number of moles} = \frac{\text{Given mass}}{\text{Molar mass}} = \frac{44\text{ g}}{44\text{ g/mol}} = \mathbf{1}\text{ mole}$$

MOLARITY AND MOLALITY

1. Correct option: (A)

A 10% (w/v) solution means that 10 grams of solute is dissolved in 100 milliliters of the solution. Since 1 milliliter (ml) is equivalent to 1 cubic centimeter (cc), the correct option is A).

2. Correct option: (D)

A) Molar solution: A solution that contains one mole of solute dissolved in one liter of solvent.

B) Normal solution: A solution that contains one equivalent weight of a solute dissolved inone liter of solvent.

C) "Mole solution" and "molar solution" are synonymous terms, both denoting a solutionwhere the solute concentration is expressed in moles per liter

D) Standard solution: A standard solution is a solution with a precisely known concentrationof a solute.

3. Correct option: (C)

A) $\text{Volume percentage} = \frac{\text{Volume of solute}}{\text{volume of solution}} \text{ x } 100$

B) $\text{Weight percentage} = \frac{\text{mass of solute}}{\text{mass of solution}} \text{ x } 100$

C) When a solute is present in trace quantities, it is convenient to express the concentration in parts per million (ppm) or parts per billion (ppb).

D) $\text{Normality} = \frac{\text{Number of equivalents of solute}}{\text{Volume of solution in liters}}$

4. Correct option: (B)

The strength of a solution refers to the concentration or amount of solute present in a givenvolume of the solution, and it is commonly expressed in units of grams per liter (g/L)

5. Correct option: (C)

A 10% (w/w) solution means that 10 grams of solute is dissolved in 100 grams of thesolution.

6. Correct option: (B)

The strength of a solution refers to the concentration or amount of solute present in a givenvolume of the solution, and it is commonly expressed in units of grams per liter (g/L)

7. Correct option: (A)

$$\text{Strength of solution} = \frac{\text{Mass of solute in gms}}{\text{Volume of solution in litres}}$$

This formula also represents the concentration of solute in solution.

8. Correct option: (C)

Given: 1. Volume of solute = 25ml

2. Volume of solvent = 250 ml

To find: % by volume = ?

Calculation:

- Volume percentage $= \frac{\text{Volume of solute}}{\text{volume of solute+volume of solvent}} \times 100 = \frac{25}{25+250} \times 100 = 9.09$

9. Correct option: (B)

A) gL^{-1}: Unit of concentration of solution

B) $moleL^{-1}$: Unit of molarity

C) $Equivalent.L^{-1}$: Unit of normality

D) $mol.kg^{-1}$: Unit of molality

10. Correct option: (A)

Molarity provides a convenient and commonly utilized measure of concentration in various scientific disciplines.

11. Correct option: (B)

12. Correct option: (B)

Given: Dilution factor = x

Calculation:

$$\text{New Molarity} = \frac{\text{Old Molarity}}{\text{Dilution factor}} = \frac{\text{Old Molarity}}{x}$$

As evident from the equation, when a solution is diluted by a factor of x, its molarity decreases proportionally by the same factor x.

13. Correct option: (B)

Calculation:

- We define %w/v as

$$\%w/v = \frac{\text{mass of solute in g}}{\text{Volume of solution in ml}} \times 100 \quad \text{.....(1)}$$

- Converting mass of solute to number of moles of solute in eq(1)

Mass of solute = moles of solute x M.W of solute

$$\%w/v = \frac{\text{moles of solute x M.W of solute}}{\text{Volume of solution in ml}} \times 100 \quad \text{.....(2)}$$

- Converting solution volume from ml to litres in eq(2)

$$\%w/v = \frac{\text{moles of solute x M.W of solute}}{\text{Volume of solution in L x } 10^3} \times 100 \quad \text{.....(3)}$$

- We know that,

$$\text{Molarity (M)} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}}$$

Substituting the molarity formula in eq(3)

$$\%w/v = \frac{\text{Molarity x M.W of solute x 100}}{10^3} \quad \text{.....(4)}$$

- Rearranging equation (4)

$$\text{Molarity (M)} = \frac{\%w/v \text{ x } 10}{\text{M.W of solute}}$$

14. Correct option: (A)

Calculation:

- We define %w/W as

$$\%w/W = \frac{\text{mass of solute in g}}{\text{mass of solution in g}} \times 100 \quad \text{.....(1)}$$

- Converting mass of solute to number of moles of solute in eq(1)

Mass of solute = moles of solute x M.W of solute

$$\%w/W = \frac{\text{moles of solute x M.W of solute}}{\text{mass of solution in g}} \times 100 \quad \text{.....(2)}$$

- Converting solution mass to volume in eq(2)

Mass of solution (g) = density of solution (g/ml) x volume of solution (ml)

$$\%w/W = \frac{\text{moles of solute x M.W of solute}}{\rho\,(\frac{g}{ml}) \times \text{Volume of solution(ml)}} \times 100 \quad \text{.....(3)}$$

- Converting solution volume from ml to litres in eq(3)

$$\%w/W = \frac{\text{moles of solute x M.W of solute}}{\rho \text{ x Volume of solution in L x } 10^3} \times 100 \quad \text{.....(4)}$$

- We know that,

$$\text{Molarity (M)} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}}$$

Substituting the molarity formula in eq(4)

$$\%w/W = \frac{\text{Molarity x M.W of solute x 100}}{\rho \text{ x } 10^3} \quad \text{.....(5)}$$

- Rearranging equation (4)

$$\text{Molarity (M)} = \frac{\%\frac{w}{W} \text{ x } \rho \text{ x } 10}{\text{M.W of solute}}$$

15. Correct option: (A)

Calculation:

- Finding the mass of water in 1 litre

  The density of water = 1 kg/L. Therefore, mass of water in 1 litre is 1kg.

- $\text{Moles of } H_2O = \frac{\text{Mass of } H_2O}{\text{M.W of } H_2O} = \frac{1000}{18} = 55.55 \text{ mol}$

- $\text{Molarity of water} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}} = \frac{55.55 \text{ mol}}{1 \text{ L}} = 55.55\text{M}$

16. Answer:D) Moles of the solute per 1000ml of the solution

Explanation:

By definition, **Molarity (M)** is the number of moles of solute dissolved in one liter (1 L = 1000 ml) of the solution.

- Option B defines *Normality*.
- Option C is incorrect because it uses the volume of the *solvent*, not the *solution*.

17.

Answer:

B) 0·05M

Explanation:

1. **Find the molar mass of** Na_2CO_3**:**
Molar Mass = $(2 \times 23) + 12 + (3 \times 16) = 46 + 12 + 48 = 106$ g/mol

2. **Calculate the number of moles of** Na_2CO_3**:**
Moles = $\frac{\text{Given Mass}}{\text{Molar Mass}} = \frac{5.3 \text{ g}}{106 \text{ g/mol}} = 0.05$ moles

3. **Calculate Molarity:**
Since the volume is given as 1 liter,
Molarity = $\frac{\text{Moles of solute}}{\text{Volume of solution in liters}} = \frac{0.05}{1} = 0.05\text{M}$

18. **Answer:B)** 1

Explanation:
The formula relating molarity, moles, and volume is:
Molarity $(M) = \frac{\text{Number of moles } (n)}{\text{Volume in Litres } (V)}$
Rearranging the formula to find the number of moles (n):
$n = M \times V$
$n = 0.5 \text{ M} \times 2.0 \text{ L} = 1$ mole

19. **Answer:A)** 125ml

Explanation:

1. **Convert millimoles to moles:**
10 millimoles = 10×10^{-3} moles = 0.01 moles

2. **Use the Molarity formula to find Volume (V):**
$M = \frac{n}{V \text{ (in L)}} \Rightarrow V \text{ (in L)} = \frac{n}{M}$
$V = \frac{0.01}{0.08} = 0.125$ Litres

3. **Convert Litres to milliliters:**
$V = 0.125 \times 1000 = 125$ ml

20. **Answer:B)** 1.5 lit **and** 0.5 lit

Explanation:
Let the volume of 1M HCl required be V_1 litres.
Since the total final volume is 2 litres, the volume of 5M HCl required will be $V_2 = (2 - V_1)$ litres.
Using the mixing formula:
$M_1V_1 + M_2V_2 = M_fV_f$
Substitute the given values:
$(1 \times V_1) + \left(5 \times (2 - V_1)\right) = 2 \times 2$
$V_1 + 10 - 5V_1 = 4$
$-4V_1 = 4 - 10$
$-4V_1 = -6$
$V_1 = \frac{-6}{-4} = 1.5$ litres
So, the volume of 1M HCl is 1.5 lit, and the volume of 5M HCl is $2 - 1.5 = 0.5$ lit.

LEVEL-II

1. Correct option: (C)
Given: 1. %w/v = 10%
2. mole of glucose = 1
To find: volume = ?

Calculation:

- $\text{Moles of glucose} = \frac{\text{Mass of glucose}}{\text{Molecular weight of glucose}}$

$\text{Mass of glucose} = 1 \times 180 = 180 \text{ g}$

- $\%w/v = \frac{\text{Mass of glucose}}{\text{Volume of solution}} \times 100$

$10 = \frac{180}{\text{Volume of solution}} \times 100$

$\text{Volume of solution} = 1800 \text{ ml}$

2. Correct option: (D)

Given: 1. $V_1 = 750$ ml

2. $M_1 = 0.5M$

3. $V_2 = 250$ ml

4. $M_2 = 2M$

To find: M =?Calculation:

- $V = V_1 + V_2 = 750 + 250 = 1000$ ml
- Dilution formula, $MV = M_1V_1 + M_2V_2$

$M \times 1000 = 0.5 \times 750 + 2 \times 250$

$M = 0.875M$

3. Correct option: (C)

Given: 1. $V_1 = 250$ ml

2. $M_1 = 0.25M$

3. $V_2 = 500$ ml

To find: M_2 =?Calculation:

- $M_1V_1 = M_2V_2$

$0.25 \times 250 = M_2 \times 500$

$M_2 = 0.125$ M

4. Correct option: (D)

Given: 1. $V_1 = 250$ ml

2. $M_1 = 6$ M

3. $V_2 = 650$ ml

4. $M_2 = 3$ M

5. M = 3 M

To find: V_{water} = ?

Calculation:

- $V_{mix} = V_1 + V_2 = 650 + 250 = 900$ ml
- $M_{mix} V_{mix} = M_1V_1 + M_2V_2$

$M_{mix} \times 900 = 6 \times 250 + 3 \times 650$

$M_{mix} = 3.83$ M

- $M_fV_f = M_{mix} V_{mix}$

$3 \times V_f = 3450$

$V_f = 1150$ ml

- $V_f = V_{mix} + V_{water}$

Vwater = 250 ml

5. Correct Option: B) 125 ml

Explanation:

Molarity (M) is defined as the number of moles of solute per liter of solution:

$$\text{Molarity (M)} = \frac{\text{Moles of solute}}{\text{Volume of solution in Liters (L)}}$$

Rearranging the formula to solve for Volume (V):

$$V \text{ (in L)} = \frac{\text{Moles}}{\text{Molarity}} = \frac{0.1}{0.8} = 0.125 \text{ L}$$

To convert liters to milliliters, multiply by 1000:

$V = 0.125 \times 1000 = 125$ ml

6. Correct Option: B) 3.92 g

Explanation:

First, convert the volume from milliliters to liters:

$$V = \frac{400 \text{ ml}}{1000} = 0.4 \text{ L}$$

Next, use the molarity formula to find the number of moles:

Moles = Molarity $\times$ V (in L) = 0.1 M $\times$ 0.4 L = 0.04 moles

Now, calculate the mass (amount) of the acid using its formula weight (98 g/mol):

Mass = Moles $\times$ Formula Weight = $0.04 \times 98 = 3.92$ g

7. Correct Option: A) 0.5 M

Explanation:

First, find the number of moles of Urea:

$$\text{Moles} = \frac{\text{Given Mass}}{\text{Molecular Weight}} = \frac{6 \text{ g}}{60 \text{ g/mol}} = 0.1 \text{ moles}$$

Next, convert the volume of the solution to liters:

$$V = \frac{200 \text{ ml}}{1000} = 0.2 \text{ L}$$

Finally, calculate the molarity:

$$\text{Molarity} = \frac{\text{Moles}}{\text{Volume (in L)}} = \frac{0.1}{0.2} = 0.5 \text{ M}$$

8. Correct option: (AD)

Explanation:

A) A decimolar solution refers to a solution with a concentration of 0.1 M (0.1 moles per liter).

B) A molar solution is a solution with a concentration of 1 mole of solute per liter of solution, represented as 1 M.

C) A molal solution is a solution with a concentration of 1 mole of solute dissolved in 1kilogram of solvent, represented as 1 mol/kg or 1 m.

D) 0.1-gram molecule is equivalent to 0.1 mole

$$\text{Molarity of solution} = \frac{\text{Moles of urea}}{\text{Volume of solution}} = \frac{0.1}{1} = 0.1\text{M}$$

9. Correct option: (ABCD)

A) Weight percentage (w%)= $\frac{\text{mass of solute}}{\text{mass of solution}}$ x 100

B) Volume percentage (v%) = $\frac{\text{Volume of solute}}{\text{volume of solution}}$ x 100

C) %w/v = $\frac{\text{mass of solute in g}}{\text{Volume of solution in ml}}$ x 100

D) %v/W = $\frac{\text{Volume of solute in mL}}{\text{Weight of solution in g}}$ x 100

10. Correct option: (BC)

A) $\frac{v}{v+V} \times 100$ is the formula for volume percentage (v%)

D) $\frac{M}{m+M} \times 100$ is the formula for weight percentage (w%)

11. Correct option: (BC)

B) Doubling the weight of the solvent will effectively decrease the concentration of the solute in the solution, resulting in a halved molarity.

C) Doubling the volume of the solvent while keeping the amount of solute constant will dilute the solution, resulting in a lower concentration and a halved molarity.

12. Correct option: (ABC)

A) 5.85g of sodium chloride in 100ml solution

$$\text{Molarity of solution} = \frac{\text{Moles of NaCl}}{\text{Volume of solution}} = \frac{5.85}{58.44 \times 0.1} = 1\text{M}$$

B) 0.32g of methanol in 10ml of solution

$$\text{Molarity of solution} = \frac{\text{Moles of methanol}}{\text{Volume of solution}} = \frac{0.32}{32.04 \times 0.01} = 1\text{M}$$

C) 111g of calcium chloride in 1000 ml of solution

$$\text{Molarity of solution} = \frac{\text{Moles of calcium chloride}}{\text{Volume of solution}} = \frac{111}{111 \times 1} = 1\text{M}$$

D) 4gm of NaOH in 1L solution

$$\text{Molarity of solution} = \frac{\text{Moles of NaOH}}{\text{Volume of solution}} = \frac{4}{40 \times 1} = 0.1\text{M}$$

13. Correct option: (A)

Given: 1. Molarity of solution = 2 M

2. Moles of LiF = 6

To find: Volume of solution = ?

Calculation:

- $\text{Molarity of solution} = \frac{\text{Moles of LiF}}{\text{Volume of solution}}$

$$\text{Volume of solution} = \frac{6}{2} = 3\text{L}$$

14. Correct option: (B)

Given: 1. Molarity of solution = 3 M

2. Mass of NaCl = 526g

To find: Volume of solution = ?

Calculation:

- $\text{Moles of NaCl} = \frac{\text{Mass of NaCl}}{\text{Molecular weight of NaCl}} = \frac{526}{58} \sim 9 \text{ mol}$
- $\text{Molarity of solution} = \frac{\text{Moles of NaCl}}{\text{Volume of solution}}$

$\text{Volume of solution} = \frac{9}{3} = 3\text{L}$

15. Correct option: (B)

Given: 1. Volume of solution = 5.00×10^2 cm^3 = 0.5 L

2. Molarity of solution = 5 M

To find: Moles of $CaCl_2$ = ?

Calculation:

- $\text{Molarity of solution} = \frac{\text{Moles of CaCl2}}{\text{Volume of solution}}$

Moles of $CaCl_2$ = 5 x 0.5 = 2.5 moles

16. Correct option: (A-S, B-R, C-Q, D-P)

A) $\text{Weight to weight percent} = \frac{\text{Weight of solute}}{\text{Weight of solution}} \times 100$

B) $\text{Weight to volume percent} = \frac{\text{Weight of solute}}{\text{Volume of solution}} \times 100$

C) $\text{Volume to volume percent} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$

D) $\text{Volume to weight percent} = \frac{\text{Volume of solute}}{\text{Weight of solution}} \times 100$

17. Correct option: (A-Q, B-P, C-S, D-R)

A) M/2 = Semi molar. A semi-molar solution typically refers to a solution with aconcentration of 0.5 M (0.5 moles per liter).

B) M/10 = Deci molar. A decimolar solution refers to a solution with a concentration of
0.1 M (0.1 moles per liter).

C) M/100 = Centi molar. A centimolar (cM) solution refers to a solution with a concentration of 0.01 moles per liter or 10 millimoles per liter.

D) M/1000 = Milli molar. A millimolar (mM) solution refers to a solution with a concentration of 0.001 moles per liter or 1 millimole per liter.

18.Correct option: (2%)

Given: 1. Mass of solute = 2g

2. Mass of solvent = 98g

To find: Mass percent = ?

Calculation:

$$\text{Mass \%} = \frac{\text{Mass of solute}}{\text{Mass of solute+Mass of solvent}} = \frac{2}{2+98} = 2\%$$

19. Correct option: (1g/mL)

Given: 1. %w/w= 20%

2. Molarity of solution = 5 M

To find: Density of solution = ?

Calculation:

$$\text{Molarity (M)} = \frac{\%\frac{w}{W} \text{ x } \rho \text{ x } 10}{\text{M.W of NaOH}}$$

$$5 = \frac{20 \text{ x } \rho \text{ x } 10}{40}$$

$$\rho = 1 \text{ g/ml}$$

20. Step 1: Calculate the moles of $CaCl_2$

First, we need the molar mass (molecular weight) of $CaCl_2$:

- Calcium (Ca): ~ 40 g/mol
- Chlorine (Cl): ~ 35.5 g/mol × 2 = 71 g/mol
- Molar Mass of $CaCl_2$ = 40 + 71 = 111 g/mol

Now, find the number of moles:

$$\text{Moles of } CaCl_2 = \frac{\text{Mass}}{\text{Molar Mass}} = \frac{11.1 \text{ g}}{111 \text{ g/mol}} = 0.1 \text{ moles}$$

Step 2: Calculate the Molarity of the $CaCl_2$ solution

Convert the volume of the solution from milliliters to liters:

$$\text{Volume} = \frac{100 \text{ ml}}{1000} = 0.1 \text{ L}$$

Now, find the molarity (M) of the total solution:

$$\text{Molarity of } CaCl_2 = \frac{\text{Moles}}{\text{Volume (L)}} = \frac{0.1 \text{ moles}}{0.1 \text{ L}} = 1.0 \text{ M}$$

Step 3: Find the Chloride Ion (Cl^-) Concentration

When $CaCl_2$ dissolves in water, it dissociates completely into its constituent ions according to the equation:

$CaCl_2(aq) \rightarrow Ca^{2+}(aq) + 2Cl^-(aq)$

As shown by the stoichiometry of the balanced equation, 1 mole of $CaCl_2$ produces 2 moles of Cl^- ions.

Concentration of Cl^- = 2 × Molarity of $CaCl_2$

Concentration of Cl^- = 2 × 1.0 M = 2.0 M

NORMALITY AND MOLE FRACTION

1. Correct option: (A)

A) $\text{Normality} = \frac{\text{No.of.grameq.of solute}}{\text{Volume of solution(L)}}$

B) $\text{Molality} = \frac{\text{No.of.moles of solute}}{\text{Mass of the solvent in kg}}$

C) $\text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution (L)}}$

D) $\text{Mole fraction} = \frac{\text{Number of moles of the component}}{\text{Total number of moles of all components in the solution}}$

2. Correct option: (D)

A normal solution is a solution that contains 1 gram equivalent weight of solute per liter ofsolution. Since 1 litre is equivalent to 1000ml , the correct option is D).

3. Correct option: (B)

A) mole/kg: Unit of molality
B) gram equivalent/lit: Unit of normality
C) mol/lit: Unit of molarity

4. Correct option: (D) 0.001 N)

A millinormal solution is a solution with a concentration of 0.001 gram equivalent weight of
solute per liter of solution, often denoted as "mN"

5. Correct option: (A)

A) $\text{Normality} = \frac{\text{No.of.grameq.of solute}}{\text{Volume of solution(L)}}$

B) $\text{Formality} = \frac{\text{Number of formula weights of solute}}{\text{Volume of solution(L)}}$

C) $\text{Molality} = \frac{\text{Number of moles of solute}}{\text{Mass of solvent (kg)}}$

D) $\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{Volume of solution(L)}}$

6. Correct option: (A)

A) gram. eq. perlitre: Unit of normality
B) mol per lit: Unit of molarity
C) mol per kg: Unit of molality

7. Correct option: (D)

The formulas for calculating the equivalent weight of an element differ depending on the specific reactive species involved

- For acids and bases:

$$\text{Equivalent weight of acid} = \frac{\text{Mol.wt.of theacid}}{\text{Basicity}}$$

$$\text{Equivalent weight of base} = \frac{\text{Mol.wt.of the base}}{\text{Acidity}}$$

- $\text{General Equivalent weight} = \frac{\text{At.wt}}{\text{Valency}}$

8. Correct option: (C)

$$\text{Equivalent weight of base} = \frac{\text{Mol.wt.of the base}}{\text{Acidity}}$$

Where, Acidity = Number of moles of hydroxide ions (OH-) donated per molecule

9. Correct option: (D)

For ions, the formula for equivalent weight transforms slightly. In this case, valency is signified by the total ionic charges

10. Correct option: (A)

11. Correct option: ()

12. Correct option: (C)

Phosphoric acid is a triprotic acid. The basicity of H_3PO_4 is 3.

$$\text{Equivalent weight of Phosphoric acid} = \frac{\text{Mol.wt.of the phosphoric acid}}{\text{Basicity}} = \frac{\text{Mol.wt.of theacid}}{3}$$

13. Correct option: (B)

Oxalic acid is a diprotic acid. The basicity of oxalic acid is 2.

$$\text{Equivalent weight of oxalic acid} = \frac{\text{Mol.wt.of the oxalic acid}}{\text{Basicity}} = \frac{\text{Mol.wt.of theacid}}{2}$$

14. Correct option: (D)

In the context of a salt, the formula for equivalent weight undergoes a slight modification. Here, the valency is represented as the total number of either positive or negative charges.

15. Correct option: (A)

- $\text{Molarity} = \frac{\text{No.of moles of solute}}{\text{Volume of solution(L)}}$

$$= \frac{\text{Mass of solute}}{\text{Molecular mass of solute}} \text{ x } \frac{1}{\text{Volume of solution(L)}}$$

Similarly,

- $\text{Normality} = \frac{\text{No.of.grameq.of solute}}{\text{Volume of solution(L)}}$

$$= \frac{\text{Mass of solute}}{\text{Equivalent mass of solute}} \text{ x } \frac{1}{\text{Volume of solution}}$$

- $\frac{\text{Molarity}}{\text{Normality}} =$

$$\frac{\text{Mass of solute}}{\text{Molecular mass of solute}} \text{ x } \frac{1}{\text{Volume of solution}} \Big/ \frac{\text{Mass of solute}}{\text{Equivalent mass of solute}} \text{ x } \frac{1}{\text{Volume of solution}}$$

$$\frac{\text{Molarity}}{\text{Normality}} = \frac{\text{Equivalent mass of solute}}{\text{Molecular mass of solute}}$$

$$\text{Normality} = \text{Molarity x } \frac{\text{Molar mass of solute}}{\text{Equivalent mass of solute}}$$

16. Correct Option: D) 2, 3

Statement 2 is correct: The relationship between normality (N) and molarity (M) is given by Normality = Molarity × n-factor. For a dibasic acid, n-factor = 2. Therefore, $N = 2M \Longrightarrow M = \frac{N}{2}$ (molarity is half its normality).

Statement 3 is correct: Let's derive it: Normality = $\frac{\text{Mass}}{\text{GEW}\times V\text{ (in L)}}$ and Molarity = $\frac{\text{Mass}}{\text{GMW}\times V\text{ (in L)}}$. Rearranging gives Normality × GEW = Molarity × GMW. *(Note: The question text contains a minor typo using "molality" instead of "molarity" at the end of the equation, but in the context of standard chemistry relations, statement 2 and 3 represent the intended conceptual pairs for option D).*

17. Correct Option: A) 1.56N

Explanation:

1. Find Equivalent Weight of KOH:
   - Molar Mass of KOH = 39(K) + 16(O) + 1(H) = 56 g/mol.
   - Since KOH provides 1 hydroxide ion (OH^-), its n-factor = 1.
   - Gram Equivalent Weight (GEW) = $\frac{56}{1}$ = 56 g/eq.

2. Calculate Normality (N):

$$\text{Normality} = \frac{\text{Mass of solute}}{\text{GEW}} \times \frac{1000}{\text{Volume in ml}}$$

$$\text{Normality} = \frac{35}{56} \times \frac{1000}{400} = 0.625 \times 2.5 = 1.5625 \text{ N}$$

18. Correct Option: A) 0.2N

Find Equivalent Weight of Oxalic Acid ($(COOH)_2 \cdot 2H_2O$):

- Oxalic acid is a dibasic acid, so its n-factor = 2.
- GEW = $\frac{\text{Molecular Weight}}{n\text{-factor}} = \frac{126}{2} = 63$ g/eq.

Calculate Normality (N):

$$\text{Normality} = \frac{\text{Mass of solute}}{\text{GEW} \times \text{Volume in Liters}}$$

$$\text{Normality} = \frac{31.5}{63 \times 2.5} = \frac{0.5}{2.5} = 0.2 \text{ N}$$

19. Correct Option: A) $K_2Cr_2O_7$

20. Correct Option: B) $CHCl_3$

Empirical Formula: $CHCl_3$

Empirical Formula Mass: $12 + 1 + (35.5 \times 3) = 119.5$ g/mol

Since the Empirical Formula Mass equals the given Molecular Mass (119.5 amu), the molecular formula is also $CHCl_3$.

21. Correct Option: B) Fe_2O_3

22. Correct Option: C) 6

The relationship between molecular formula and empirical formula is determined by the factor n:

$$n = \frac{\text{Molecular Mass}}{\text{Empirical Formula Mass}}$$

1. Calculate Molecular Mass of $C_6H_{12}O_6$:
   Molecular Mass = $(6 \times 12) + (12 \times 1) + (6 \times 16) = 72 + 12 + 96 = 180$
2. Calculate n-value:
   $n = \frac{180}{30} = 6$

LEVEL-II

1. Correct option: (A)

Given: 1. Molarity of solution = 0.1M

M.W of water = 18

To find: Mole fraction of solvent = ?

Calculation:

Relation b/w molality and mole fraction, $m = \frac{1-X_B}{X_B} \times \frac{1000}{M_B}$

$\frac{0.1\times18}{1000} X_B = 1 - X_B$

$1.8 \times 10^{-3} X_B = 1$

$X_B = 0.998$

2. Correct option: (C)

Given: 1. Mass of urea = 6g

Mass of water = 90g

To find: Mole fraction of urea = ?

Calculation:

- Moles of urea (A) $= \frac{6}{60} = \frac{1}{10}$
- Moles of water (B) $= \frac{90}{18} = 5$
- Mole fraction of urea $= \frac{n_A}{n_A+n_B} = \frac{0.1}{5.1} = \frac{1}{51}$

3. Correct option: (A)

Given: 1. 36% water and 64% acetaldehyde

To find: Mole fraction of acetaldehyde = ?

Calculation:

- Basis = 100g solution
- Moles of water (B)$= \frac{36}{18} = 2$
- Moles of acetaldehyde (A) $= \frac{64}{44} = 1.42$
- Mole fraction of acetaldehyde $= \frac{n_A}{n_A+n_B} = \frac{1.42}{3.42} = 0.42$

4. Correct option: (C)

Given: 1. molar mass = 78

2. n = 6

Calculation:

- $n = \frac{\text{Molar mass}}{\text{Empirical formula mass}}$

Empirical formula mass $= \frac{78}{6} = 13$

Checking options to find empirical formula of mass 13, we get CH

5. Correct option: (B)

Given: 1. No. of S atoms = 3.0115×10^{23}

2.Volume of solution = 500 ml

To find: Normality = ?

Calculation:

- No. of S atoms = No. of S moles x NA

  3.0115×10^{23} = S moles x 6.023×10^{23}

  No. of S moles = 0.5
- 1 mol H_2SO_4 has 1 mol S atoms

  0.5 mol S atoms are present in 0.5 mol of H_2SO_4
  - $$\text{Molarity of } H_2SO_4 \text{ solution} = \frac{\text{No.of moles of solute}}{\text{Volume of solution(L)}}$$

    $$= \frac{0.5}{0.5} = 1 \text{ mol/L}$$
  - n-factor = 2

    Normality = Molarity x n-factor = 2 x 1 = 2N

6. Correct Option: C) Neutral

1. **Calculate** meq **of** NaOH**:**
   - Molar mass of NaOH = 23 + 16 + 1 = 40 g/mol
   - Since NaOH has one OH^- ion, its equivalent weight is equal to its molar mass (40 g/eq).
   - Number of Equivalents = $\frac{\text{Given Mass}}{\text{Equivalent Weight}} = \frac{0.4 \text{ g}}{40 \text{ g/eq}} = 0.01$ eq
   - meq of NaOH = $0.01 \times 1000 = 10$ meq
2. **Calculate** meq **of** H_2SO_4**:**
   - meq of acid = Normality × Volume (in ml)
   - meq of H_2SO_4 = 0.1 N × 100 ml = 10 meq

   Since the milliequivalents of the acid exactly equal the milliequivalents of the base (10 meq = 10 meq), complete neutralization occurs.

7. Correct Option: D) 0.75

Explanation:

1. **Find the initial normality (N_1) of the solution:**
   - Oxalic acid is a dibasic acid, so its n-factor = 2.
   - Equivalent Weight = $\frac{\text{Molar Mass}}{n\text{-factor}} = \frac{126}{2} = 63$ g/eq
   - Initial Normality (N_1) = $\frac{\text{Mass}}{\text{Equivalent Weight}} \times \frac{1000}{\text{Volume (in ml)}}$
   - $N_1 = \frac{6.3}{63} \times \frac{1000}{250} = 0.1 \times 4 = 0.4$ N
2. **Use the dilution formula to find the final volume (V_2):**

   $N_1V_1 = N_2V_2$

   0.4 N × 250 ml = 0.1 N × V_2

   $V_2 = \frac{100}{0.1} = 1000$ ml
3. **Calculate the volume of water to be added:**

   Volume of water added = $V_2 - V_1 = 1000 \text{ ml} - 250 \text{ ml} = 750$ ml
4. **Convert the added volume to liters:**

$750\ \text{ml} = \frac{750}{1000} = 0.75\ \text{Litres}$

8. Correct option: (AB)

Molarity and normality are temperature dependent because they involve volume of solutions.Volume is dependent on temperature.

Molality, mole fraction and weight percentage does not depend on temperature because theyinvolve masses of solute and solvent.

9. Correct option: (AD)

A) The solution of molality is always greater than molarity. This is because molarity is calculated as mol per unit L (volume of solution), whereas molality is calculated as moles perunit Kg (i.e. mass of solvent). So, in any case, the mass of the solvent is less than the volume of the solution, and thus the molality is greater.

B) The way we calculate formality and molarity is exactly the same but formality is used whenever the solute does not have a definite molecule; i.e., like ionic solids. For ionic solids, we have formula mass instead of molecular mass since there are no distinct molecules to talk about.

C) Mass fraction and mole fraction are different concepts and represent different ways of expressing the composition of a mixture. It is highly unlikely for them to have the same valuesimultaneously, unless there are specific circumstances or simplifications that lead to that specific scenario.

$$\text{D) Normality} = \text{Molarity x } \frac{\text{Molar mass of solute}}{\text{Equivalent mass of solute}}$$

$$= \text{Molarity x } \frac{98}{49}$$

$$\text{Normality} = \text{Molarity x } 2$$

10. Correct option: (AD)

A) 100 mL of 0.1 N HCl + 50 mL of 0.1 N NaOH: In this case, the amount of NaOH is less than the amount of HCl, so there is excess acid. The resulting solution will be acidic.

B) 75 mL of 0.1 N HCl + 75 mL of 0.2 N NaOH: Here, the amounts of HCl and NaOH are equal. However, the concentration of NoOH is higher compared to HCl, so the resulting solution will be basic in nature.

C) 50 mL of 0.1 M H2SO4 + 100 mL of 0.1 N Ca(OH)2: In this case, we have a strong acid (H2SO4) reacting with a strong base (Ca(OH)2) in a 1:2 ratio. The resulting solution will be basic because there is excess base.

D) 100 mL of 0.1 M H2SO4 + 50 mL of 0.1 N Ca(OH)2: Similar to the previous case, we have a strong acid (H2SO4) reacting with a strong base (Ca(OH)2) in a 2:1 ratio. The resulting solution will be acidic because there is excess acid.

Therefore, the solutions A) and D) contain an acidic nature.

11. Correct option: (A)

Given: 1. Volume of water = 250g

2. Molal concentration of solution = 0.1m

To find: Mass of H_2SO_4 = ?

Calculation:

- $\text{Molality} = \frac{\text{No.of } H_2SO_4 \text{ moles}}{\text{kg of water}}$

No. of H_2SO_4 moles = 0.25 x 0.1 = 0.025

- Mass of H_2SO_4 = 0.025 x 98 = 2.45g

12. Correct option: (B)

Given: 1. Density of Na_2CO_3 solution = 1.018 g/ml

2. % w/v of Na_2CO_3 solution = 8.653

To find: Molality of solution = ?

Calculation:

- $\text{Molarity (M)} = \frac{\%\text{strengthof Solutionx density of solution x10}}{\text{Molecular mass of the solute}}$

$$= \frac{8.653 \times 1.018 \times 10}{106} = 0.83\text{M}$$

- $\text{Molality} = \frac{1000 \times \text{M}}{1000\rho - \text{MM}_1}$

$$= \frac{830}{1018 - 87.98} = 0.876 \text{ mol/kg}$$

13. Correct Option: C) 2.79

Explanation:

Let's use the first relation given in the passage:

$d = \text{M}\left[\frac{1}{m} + \frac{\text{M}_{\text{solute}}}{1000}\right]$

1. Identify the given values:
   - Molarity (M) = 3 mol L^{-1}
   - Density (d) = 1.25 g mL^{-1}
   - Molar mass of solute (NaCl) = 23 (Na) + 35.5 (Cl) = 58.5 g mol^{-1}
2. Substitute into the formula:

$1.25 = 3\left[\frac{1}{m} + \frac{58.5}{1000}\right]$

$\frac{1.25}{3} = \frac{1}{m} + 0.0585$

$0.4167 = \frac{1}{m} + 0.0585$

$\frac{1}{m} = 0.4167 - 0.0585 = 0.3582$

$m = \frac{1}{0.3582} \approx 2.79$ m

14. Correction Note on Question Data/Options: Standard questions of this exact type typically have the mole fraction of ethanol as 0.05 or similar, which yields one of the choices (e.g., if X = 0.05, $m = \frac{0.05}{0.95} \times \frac{1000}{18} \approx 2.92$ m). However, evaluating the values as printed in this specific test paper (0.9 as ethanol), none of the given options match mathematically. If we reverse the components and treat water as the solute (X = 0.1) and ethanol as the solvent (M = 46), we get $m = \frac{0.1}{0.9} \times \frac{1000}{46} \approx 2.415$ m. This matches **Option B** perfectly, indicating the question paper meant the mole fraction of the *solvent* is 0.9 or calculated molality with respect to water dissolved in ethanol.

15. Correct Option: D) 1M

Explanation:

Let's use the third relation given in the passage:

$N = n \times M$

1. **Find the n-factor (basicity) of H_3PO_4:**

Phosphoric acid (H_3PO_4) is a tribasic acid because it contains 3 replaceable hydrogen ions attached to oxygen atoms. Therefore, its n-factor = 3.

2. **Calculate Molarity (M):**

$3 = 3 \times M$

$M = \frac{3}{3} = 1\ M$

16. Correct option: (a-T, b-P, c-Q, d-R)

A) Hypophosphorous acid (H3PO2) is a monobasic acid, meaning it can donate one proton (H+) per molecule. So, the equivalent weight (EW) is the same as the molar weight (MW) forhypophosphorous acid.

B) Phosphorous acid (H3PO3) is a diprotic acid, meaning it can donate 2 protons (H+) per molecule.

C) Phosphoric acid (H3PO4) is a tribasic acid, meaning it can donate three protons (H+) per molecule. So, the equivalent weight of phosphoric acid is one third of its molar weight.

D) Pyrophosphoric acid (H4P2O7) is a tetrabasic acid, meaning it can donate four protons (H+) per molecule. So, the equivalent weight of pyrophosphoric acid is one fourth of its molar weight.

17. Correct option: (a-Q, b-S, c-P, d-R)

A) $\text{Molality} = \frac{\text{No.of molesof solute}}{\text{Wt.of solvent in kg}}$

B) $\text{Mole fraction} = \frac{\text{No.of molesof one component}}{\text{No.of moles of all components}}$

C) $x_A + x_B = 1$

D) $\text{Molarity} = \frac{\text{No.of molesof solute}}{\text{Vol.of solutioninlit}}$

18. Correct option: (2)

Calculation:

$$\text{Normality} = \text{Molarity} \times \frac{\text{Molar mass of H2SO4}}{\text{Equivalent mass of H2SO4}}$$

$$= \text{Molarity} \times \frac{98}{49}$$

$$= \text{Molarity} \times 2$$

19. Correct option: (5.33)

Given: 1. Mass of nitrogen = 7g

2. Mass of oxygen = 8g

To find: Mole percentage = ?

Calculation:

- $\text{Mole fraction of oxygen} = \frac{8}{8+7} = 0.533$

- Mole percentage of oxygen = Mole fraction of oxygen x 100%

= 5.33 x 10

20. Correct option: (2)

Given: 1. % w/v of H_2SO_4 = 0.98

2. Molecular weight of H_2SO_4 = 98.08 g/mol

To Find: Normality of H_2SO_4 solution

Calculation:

- Convert % w/v to g/L 0.98 g/100 mL = 9.8 g/L
- Molarity = $\frac{mass}{Molecular\ weight \times Volume} = \frac{9.8}{98 \times 1} = 0.1$ M
- N = Molarity x n = 0.1 x 2 = 0.2 N or 2 x 10^{-1} N

Where,

n = number of H+ ions that can be donated by each molecule of H_2SO_4, which is 2 for H_2SO_4

4. Gaseous State

Solutions

LEVEL-I

1. Correct option: (D) gases

Matter has 3 states

(i) Solid (ii) Liquid (iii) Gas

Molecules of solids have least amount of space between them.

Molecules of liquids have little amount of space between them.

But molecules of gases have the largest number of space between them. This allows them to move freely. As a result if external pressure is applied, the volume of gas decreases.

Hence gases have highest compressibility.

2. Correct option: (C) They take shape of the container in which they are stored

The space between two gas molecules is large. Hence they are highly compressible. Due to this they have infinite shape and volume. They gradually take shape of the container in which they are stored.

Due to random motion of gas particles,they exert pressure on the wall.

Hence option C is correct.

3. Correct option: (B) Kelvin

Kelvin (K) is the S.I and CGS unit of temperature.

Whereas Centigrade(C) is the unit of temperature in Celsius scale and Fahrenheit (F) is the unit of temperature in Fahrenheit scale.

These two units are used widely.

$K = 273+C$

$F = (9/5)\times C + 32$

Pascal is the unit of pressure.

4. Correct option: (B) critical temperature

Critical temperature is the lowest possible temperature at which gases start to convert into liquid.

Absolute zero temperature is lowest possible temperature at which heat and energy is absent in a particle.

The value of absolute zero is 0 K or -273.15°C

Boyle's temperature is temperature at which real gases obey ideal gases equation.

Transition temperature is a temperature at which sudden change in physical properties occurs in a substance.

5. Correct option: (D) a,b c,d

Mass – Gases have have a definite mass and it can be measured in grams.

Volume – Gases do not have definite volume. But it is measurable when it occupies a container. The volume of the container is the volume of gas occupied. It can be expressed in cc or Litre.

Pressure – Force exerted by gas per unit area of a wall is called called pressure of gas. It can be measured in Pascal, atm.

Temperature – The scales which is used to measure temperature of gases are Celsius, Fehrenheit and Kelvin

6. Correct option: (A) I, II, III, IV

Matter has 3 states

(i) Solid (ii) Liquid (iii) Gas

So matter exists in solid, liquid and gas.
Molecules of gases have the largest number of space between them. This allows them to move freely. As a result if external pressure is applied, the volume of gas decreases. Hence gases are highly compressible.
Due to highly compressible, gases do not have definite shape or volume.
The gaseous state of solid and liquid is called vapour.
Example – vapour of water (liquid), vapour of iodine (solid)

7. Correct option: (B) Pressure

Torr is a unit of pressure.
It is defined as 1/760 of an atmosphere.
Unit of mass – Kg(SI), g (CGS)
Unit of volume – Litre (CGS), m^3(SI)
Unit of density – kg/m^3 (SI), g/cm^3 (CGS)

8. Correct option: (C) Measurable properties

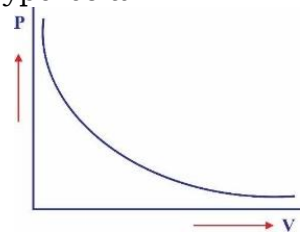
Gas laws give the relation between the different measurable properties (mass, pressure, volume, temperature) of gases.

9. Correct option: (B)

Boyle's law is a gas law that states that a gas's pressure and volume are inversely proportional.
Formula of Boyle's law
$P \propto 1/V$
Or $V \propto 1/P$
$P_1V_1 = P_2V_2$
Or $PV = K$ (K= proportionality constant)

10. Correct option: (B) Hyperbola

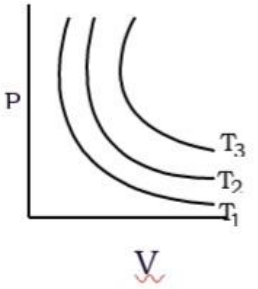
$PV = nRT$
$PV = C^2$ = hyperbola



11. Correct option: (D) Isotherms

A curve drawn at constant temperature is called an isotherm.
This shows the relationship between pressure (P) and volume (V).
The graph drawn at constant volume is known as Isochoric
The graph drawn at constant pressure is known as Isobars.

12. Correct option: (B)



According to ideal gas equation, $PV = nRT$, as n and R are constant.

$PV \propto T$

Hence $T_1 > T_2 > T_3$

13. Correct option: (C) P/3

According to Boyle's Law, volume of the gas is inversely proportional to the pressure at constant temperature.

$P_1V_1 = P_2V_2$

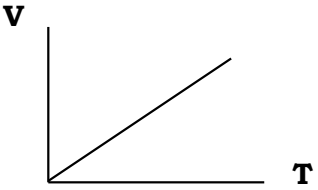
Now $V_2 = 3V_1$

$P_1V_1 = P_2 3V_1$

Hence pressure (P_2) will be = $P_1/3$

14. Correct option: (B)

B)

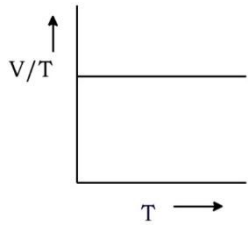


Solution : Charles' law states that at constant pressure, the volume occupied by a gas is directly proportional to the temperature.

That is $V \propto T$ at constant P

Hence option B is correct

15. Correct option: (A)



Charles' law states that at constant pressure, the volume occupied by a gas is directly proportional to the temperature.

That is $V \propto T$ at constant P

Or, V/T = Constant

So the graph will be parallel to x axis.

Hence option A is correct

16. Correct option: (D) Isobars

The graph drawn at constant pressure is known as Isobars.

A curve drawn at constant temperature is called an isotherm

The graph drawn at constant volume is known as Isochoric.

17. Correct option: (A) $V \propto T$

Charles' law states that at constant pressure, the volume occupied by a gas is directly proportional to the temperature.

That is $V \propto T$

18. Correct option: (C) 450 K

19. Correct option: (C) Isochoric

The graph drawn at constant volume is known as Isochoric

The graph drawn at constant pressure is known as Isobars.

The graph drawn at constant temperature is known as Isotherms.

20. Correct option: (C) Equal volumes of gas contain equal number of moles under similar conditions of temperature and pressure.

According to Avogadro's law, equal volumes of all gases at same temperature and pressure will have same moles. of molecules.

LEVEL-II

1. Correct option: (C) 273°C

P = constant
T_1 = 0°C = 273 K
T_2 = ?
Let V_1 = V
Hence, V_2 = 2V
We know $V_1/T_1 = V_2/T_2$
$T_2 = (V_2T_1) / V_1$
= (2V × 273) / V
= 546 K
= (546 – 273)°C
= 273°C

2. Correct option: (B) 4 atm

PV = nRT
P = nRT / V
P/(1/V) = slope
P = 60/15 = 4 atm

3. Correct option: (A) 477°C

From Charle's Law
$V_1/T_1 = V_2/T_2$
Let X be the volume of the vessel.
The volume of the air at the start (V_1) is x. After heating, 2/5 of the air fills a volume of x.
$(2/5)V_2=x$
⇨ $V_2=(5/2)x$
$=5x/2$
27 Celsius = 300 Kelvin
$x/300 = (5x/2)/T_2$
we get T_2 = 1500/2 = 750 K = 477°C

4. Correct option: (C) 400 K

PV = nRT
MP/RT = nM/V
MP/RT = d
P = dRT/M
When P, R and M is constant
Then $d_1= d_2T_2$
Or, d_1 = d
Or, d_2 = 0.75d
T_1 = 27°C = 300 K
$T_2 = d_1T_1/d_2$
= d×300/0.75d
= 400
Hence, T_2 = 400 K

5. Correct option: (B) 6 × 10^5 N/m−2

P1 = 9 × 10^5 N/m², P2 = ?

V1 = 2.5 L, V2 = 1.5L
P2 = P1V1/V2
= (9 × 10^5 × 2.5) / 1.5
= 15 × 10^5 N/m^2
Hence additional pressure required
(15 × 10^5 - 9 × 10^5) = 6×10^5 N/m^2

6. Correct option: (C) Decreased by 2 tunes

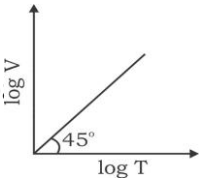
Pressure is doubled means $P_2 = 2P_1$
$P_1V_1 = P_2V_2$
$V_2 = P_1V_1/P_2$
$V_2 = P_1V_1/2P_1$
$V_2 = V_1/2$
Hence volume is decreased to half.

7. Correct option: (C) ½ atm

Initial pressure P_1 = 1atm
Initial temperature T_1 = 273°C = (273+273)K = 546 K
Final pressure P_2 = ?
Final temperature T_2 = 0°C = 273K
We know
$P_2/P_1 = T_2/T_1$
$P_2 = T_2 \times P_1/T_1$
=(273 × 1)/546
= ½ atm

8. Correct option: (A)

N = 10 moles
P = 0.821 atm
We know R = 0.0
Now PV = nRT
V = nRT/P



V = (10×0.0821×T)/0.821
V = T
Hence (log V/log T) = 1 = tan 45°
Hence option A is correct.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Correct option: (A) Pa B) Nm^{2-} C) Bar D) atm

Pressure is force applied perpendicular to the surface of an object per unit area.
Pressure = P
Force = F
Area = a
P = F/a
Unit of force in SI = N
Unit of area in SI = m^2
So unit of pressure in SI = Nm^{2-}
Other units of pressure are
Pa, Bar, atm etc

10. Correct option: (A)$V \propto T$, B) $(dV/dT)_p = k$

According to Charles's law The volume of gas is directly proportional to its absolute temperature at constant pressure

$V \propto T$ at constant pressure
$V = KT$ (K = constant)
$V/T = K$
Hence, $(dV/dT)_p = K$

11. Correct options:
(A) Volume of a gas change as it pressure and temperature change
B) Pressure has little effect on solids and liquids when compared to gases.
D) Perfumes spread because of diffusion of gases.
According to Charles's law The volume of gas is directly proportional to its absolute temperature at constant pressure.
According to Boyle's law the volume of a given amount of gas is inversely proportional to its pressure when temperature is constant.
Hence option A) is correct.
When pressure is applied on gas it becomes liquid and when pressure is applied on liquid it becomes solid.
Pressure has less effect on solid and liquid.
Hence option B) is correct.
The intermolecular attraction between gas molecules are weak. That is why they don't have definite shape or volume.
Hence option C) is incorrect.
The movement of molecules from a region of higher concentration to a region of lower concentration down the concentration gradient is known as diffusion.
As the particles of perfume are free to move, they move quickly through diffusion.
Hence option D) is correct.

12. Correct option: (A) Litre B) Millilitre C) dm^3
The amount of space occupied by any component is known as Volume.
Volume = length × width × height
Hence,
Unit of volume = unit of length × unit of width ×unit of height
= m^3(SI)
= cm^3 (CGS)
= dm^3
Dm^3 is also known as Litre.
So, Litre is also a unit of volume
1 Litre = 1000 cm^3 = 1000 millilitre
2 Milliliter is also a unit of volume
Hence option A), B and C) is correct.
Gram is a unit of mass in CGS.

13. **Correct Answer: (B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.**
Explanation:

- **Assertion (A) is True:** Gas particles are in a state of constant, rapid, and random motion. Because they move randomly in every possible direction, they collide with the walls of their container equally on all sides, resulting in equal pressure exerted in all directions.
- **Reason (R) is True:** Intermolecular spaces in gases are vast compared to solids and liquids, meaning gases contain fewer particles per unit volume. Thus, they have much lower densities.

- **Why it's not the correct explanation:** The low density of gases does not *cause* them to exert pressure equally. The equal distribution of pressure is explicitly caused by the **random, isotropic motion and collisions** of the gas molecules.

14. Hot Air Balloons

- Correct Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

  Explanation:
- Assertion (A) is True: Hot air balloons visibly and reliably rise up through the surrounding cooler atmosphere when the air inside the envelope is heated.
- Reason (R) is True: When the air inside the balloon is heated, the thermal kinetic energy of the gas molecules increases, causing them to move faster and push further apart (expansion). Because the same mass of air now occupies a much larger volume, its density decreases.
- Why it's the correct explanation: According to the principle of buoyancy (Archimedes' principle), a less dense fluid floats or rises in a denser fluid. Since the hot air inside the balloon becomes less dense than the cooler surrounding air outside, an upward buoyant force is generated that lifts the balloon.

MATRIX MATCHING TYPE

15. Correct Matching is: a → S), b → Q), c → R), d → P)

Charles' Law states that the volume of a fixed mass of a gas is directly proportional to the temperature at constant pressure.

$V \propto T$ at constant P

Gay-Lussac's Law states that the pressure of gas varies directly with the absolute temperature of the gas, when the volume is constant.

$P \propto T$ at constant V

According to Boyle's Law, volume of the gas is inversely proportional to the pressure at constant temperature.

$P \propto 1/V$(at constant T)

Avogadro's law states that equal volumes of all gases, at the same temperature and pressure, have the same number of molecules

$V \propto n$ (at constant T &P)

16. Correct Matching is: a) →Q), b) → P), c) → S), d) → R)

According to Boyle's Law, volume of the gas is inversely proportional to the pressure at constant temperature.

$P \propto 1/V$(at constant T)

That is $P_1V_1 = P_2V_2$ =constant

Curve drawn at constant temperature is called an isotherm.

This shows the relationship between pressure (P) and volume (V).

The graph drawn at constant pressure is known as Isobars.

Passage-I

17. Correct option: (C) 273°C

Let the volume of gas at 0°C = V ml

V_1 = V ml

V_2 = 2V ml

T_1 = (0+273) K = 273 K

T_2 = ?
From Charle's law,
$V_1/T_1 = V_2/T_2$
$T_2 = V_2T_1/V_1$
=(2V × 273) / V
= 546 K
= (546 – 273)°C
= 273° C
Hence option C is correct.

18. Correct option: (D) 21.38 mL

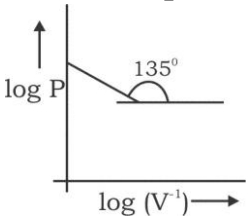
V_1 = 20mL
T_1 = (15+273) K = 288 K
T_2 = (35 + 273) K = 308 K
V_2 = ?
From Charle's law
$V_1/T_1 = V_2/T_2$
$V_2 = V_1T_2/T_1$
=(20×308)/288
= 21.38 mL
Hence option D is correct.

19. Correct option: (B) double

Let, T_1 = T
Then, T_2 = 2T
V_1 = V
V_2 = ?
Form Charle's law
$V_1/T_1 = V_2/T_2$
$V_2 = V_1T_2/T_1$
V_2 = (V×2T)/T
2V
Hence the volume will be doubled.
Option B is correct

20. Correct option: (C)

C)

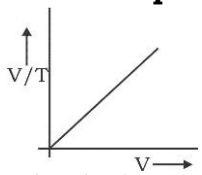


From Boyle's law P ∝ 1/V
PV = k (constant)
So it is hyper bola.
Taking log both sides
Log P = log K – log V
Log P = log K + log (V^{-1})
Comparing, y = mx + log K
For 45 ° the slope of the graph is +1
For 135 ° the slope of the graph is –1.

It doesn't obey Boyle's law.

21. Correct option: (D)

D)



Charles' Law states that the volume of a fixed mass of a gas is directly proportional to the temperature at constant pressure.

$V \propto T$ at constant P

$V = KT$

i.e. $y = mx$

So the graph V vs T is straight line.

This law applies to ideal gases held at a constant pressure, where only the volume and temperature are allowed to change.

$V/T = K$

$\log V = \log K + \log T$

Comparing, $y = mx + \log K$

Hence the slope for the graph of log V vs log T is tan 45° or +1.

$V = KT$

$VT = KT^2$

That is $y = mx$

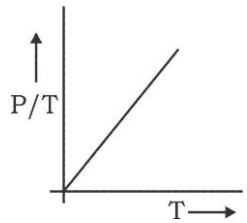
Thus, VT vs T^2 will be straight line.

$V = KT$

$V/T = K$

Thus V/T vs T will be parallel to X axis.

22. Correct option: (D)



From gay-lusacc's law

$P \propto T$

$P = KT$

Comparing $y = mx$

So it will be straight line.

$\log P = \log K + \log T$

Comparing $y = mx + \log K$

$M = \tan 45° = 1$

So it will be positive.

$P = KT$

$P/T = K$

So P/T vs P graph will be parallel to X axis

Hence option D is correct

Ideal Gas Equation
LEVEL-I

1. Correct option: (B) 2 Cal

From ideal gas equation,

PV = nRT

R = PV/nT

Here n = 1

So, PV/T = R

= 8.314 J K^{-1} mol^{-1}

= 2 Cal K^{-1} mol^{-1}

Hence option B is correct.

2. Correct option: (A) 0.5

From ideal gas equation,

PV = nRT

N = PV/RT

= (22.4 × 1) / (0.082 × 546)

= 0.5

3. Correct option: (C) 1.75 atm

2 g of hydrogen = 2/2 = 1 mole hydrogen

8 g of O_2 = 8/32 = 0.25 moles of O_2

22 g of CO_2 = 22/44 = 0.5 mole of CO_2

Total number of moles (n) = 1+0.25+0.5 = 1.75

V = 22.4 Litre, T = (0 + 273) = 273 K

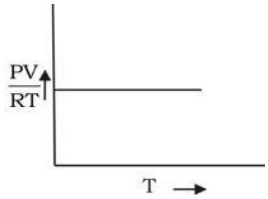
From ideal gas equation PV = nRT

P = nRT/V

= (1.75 × 0.082 × 273) /22.4

= 1.75 atm

4. Correct option: (A)



From ideal gas equation

PV = nRT

PV/RT = n, n is independent to temperature T

So the graph will be parallel to X axis.

Hence option A is correct.

5. Correct option: (C) 44.8 lit

We know that at 273 K temperature and 1 atm pressure gas occupies 22.4 Litre molar volume

Here t = (273+273) = 546 K

P = 1 atm

N = 1 mole

V_1 = 22.4 L

T_1 = 273K

V_2 = ?

$T_2 = 546K$
Volume is directly proportional to temperature at constant pressure.
$V_1/T_1 = V_2/T_2$
$V_2 = V_1T_2/T_1$
$=(22.4 \times 546) / 273$
$= 44.8$
Hence option C is correct

6. Correct option: (D) Ideal gas

According to Charles's law The volume of gas is directly proportional to its absolute temperature at constant pressure.
According to Boyle's law the volume of a given amount of gas is inversely proportional to its pressure when temperature is constant.
Gay-Lussac's Law states that the pressure of gas varies directly with the absolute temperature of the gas, when the volume is constant.
The gas which obeys all the above mentioned laws at all temperature and pressure is called ideal gas.

7. Correct option: (D) High temperature and low pressure

Real gases show ideal gas behavior at high temperature and low pressure because at high pressures and low temperature, molecules of gases are very close to each other Real gases show ideal behavior when intermolecular forces are practically negligible. This can only happen at very low pressure and very high temperature

8. Correct option: (A)Low temperature and high pressure

At Low temperature and high pressure, gas atoms or molecules get closer to each other and attracted by weak Vander Waals force. Hence a gas deviates the most from its ideal behavior.

9. Correct option: (C)273°C

1 mole oxygen occupies a volume of 22.4 L at 1 atm and 273 K
We know PV = nRT (Ideal gas equation)
If P,V and R are constant then
N is inversely proportional to T
Hence $n_1T_1 = n_2T_2$
$n_1 = 1$ mol
$T_1 = 273K$
$n_2 = 16/32 = 0.5$ mol
$T_2 = n_1T_1/n_2$
$= 273/0.5$
$= 546$ K
$= (546 – 273)°$ C
$= 273 °$ C
Hence option C is correct

10. Correct option: (C) 400k

PV = nRT
MP/RT = nM/V
MP/RT = d
P = dRT/M
When P, R and M is constant
Then $d_1T_1 = d_2T_2$
Or, $d_1 = d$

Or, $d_2 = 0.75d$
$T_1 = 27°C = 300$ K
$T_2 = d_1T_1/d_2$
$= d \times 300/0.75$ d
$= 400$
Hence, $T_2 = 400$ K

11. Correct option: (A) NO

1 mole gas occupies 22.4 L of volume at STP.
Thus 5.6 L of gas at STP contains 0.25 moles.
Number of moles = (weight / Molecular weight)
$0.25 = 7.5/M$
$M = 30$
Molecular weight of NO = (14+16) = 30
So the gas is NO

12. Correct option: (C)149.24 lit

Number of moles of hydrogen (n)= 10/2 = 5
P = 1.5 atm
T = (273+273) K = 546 K
From ideal gas equation,
PV = nRT
V = nRT/P
V = (5 × 0.082 × 546) / 1.5
V = 149.24 litre
Hence option C is correct

13. Correct option: (B) 2:1

Molar mass of $NH_3 = 14+3 = 17$
Molar mass of $PH_3 = 31+3 = 34$
The ratio of masses of,
PH_3 and NH_3
= 34: 17
= 2 : 1

14. Correct option: (C) 1.5 atm

Number of moles of Oxygen, $n_1 = 8/32 = 0.25$
Number of moles of nitrogen, $n_2 = 14/28 = 0.5$
Volume (V) = 11.2 L
Temperature = 273 K
So, $n = n_1 + n_2 = (0.25 + 0.5) = 0.75$
From ideal gas equation PV = nRT
P = nRT/V
= (0.75 × 0.082 × 273) / 11.2
= 1.5 atm

15. Correct option: (C) 8.314 $J.K^{-1}.mole^{-1}$

PV = nRT
SI unit of,
$P = N/m^2$
$V = m^3$
N = mole
Work = N.m = joule

R = PV/nT
= (N × m^3)/ (m^2 × mole × K)
= (N.m) / (mole × K)
= $J.mol^{-1}.K^{-1}$
At STP, for 1 mole of gas (n=1 mol),
The value of,
Temperature (T) = 273 K,
Pressure (P) = 1.01 x 10^5 N/m^2
Volume (V) = 22.4 x 10^{-3} m^3
From PV = nRT
R = PV/nT
= (1.01 ×10^5 × 22.4 × 10^{-3})/273
= 8.314 J.mol–1.K–1

16. Correct option: (D) All of these

From ideal gas equation
PV = nRT
R = PV/nT
Unit of
P = atm
V = litre
N = mole
T = K
So unit of R = $litre.atm.mol^{-1}.K^{-1}$
Again
R = (N × m^3)/ (m^2 × mole × K)
= (N.m) / (mole × K)
= $J.mol^{-1}.K^{-1}$
= $erg.mol^{-1}.K^{-1}$
= $Cal.mol^{-1}.K^{-1}$

17. Correct option: (B) dynes.cm^{-2} ; cm^3

$erg.K^{-1}.mol^{-1}$, this CGS unit of R
So pressure and volume will be taken in CGS unit.
CGS unit of,
Pressure = $dyne.cm^{-2}$
Volume = cm^3
Hence option B is correct.

18. Correct option: (B) Only I,II,III

At STP, for 1 mole of gas (n=1 mol),
The value of,
Temperature (T) = 273 K,
Pressure (P) = 1.01 x 10^5 N/m^2
Volume (V) = 22.4 x 10^{-3} m^3
From PV = nRT
R = PV/nT
= (1.01 ×10^5 × 22.4 × 10^{-3})/273
= 8.314 $J.mol^{-1}.K^{-1}$
= 8.314 × 10^7 ergs $K^{-1}.mole^{-1}$
Again we know

R = 0.0821 lit.atmK^{-1}.mol^{-1}

= 0.0821 × 10^3 ml. atm.K^{-1}.mol^{-1}

= 82.1 ml. atm.K^{-1}.mol^{-1}

= 0.0821 × 760 lit.mm. K^{-1}.$mole^{-1}$

= 62.4 lit.mm. K^{-1}.$mole^{-1}$

Hence I), II) & III) are correct.

19. Correct option: (A) dyne. Cm

An erg is the amount of work done by a force of one dyne exerted for a distance of one centimetre.

So

Unit of erg = Unit of force in CGS × unit of distance in CGS

= dyne.cm

20. Correct option: (D)273°C and 1 atm

From ideal gas equation

PV = nRT

MP/RT = nM/V

MP/RT = d

M & R are constant

So the density will be minimum if P/T ratio is minimum.

At STP

P = 1 atm, T = 273 K

P/T = 1/273 = 0.0037

For 0 ° C and 2 atm

P/T = 2/273 = 0.0073

For 273° C and 2 atm

P/T = 2/546 = 0.0037

For 273° C and 1 atm

P/T = 1/546 = 0.0018

LEVEL-II

1. Correct option: (B)1.25 g/1

From ideal gas equation

PV = nRT

MP/RT = nM/V

MP/RT = d

M = 28

P = 1 atm

T = 273K

Hence, d = (28×1)/(273×0.082)

= 1.25 g/l

2. Correct option: (B) 8g

From ideal gas equation

PV = nRT

MP/RT = nM/V

MP/RT = d

M = 32,

P = 380 torr = 0.5 atm,

T = 273 K,
V = 11.2 L
D = (32×0.5)/(273×0.082)
= 0.714
Mass = d × V
= 0.714 × 11.2
= 8 g

3. Correct option: (B) 89.6L

From ideal gas equation PV = nRT
MP/RT = nM/V
MP/RT = d
M = 2
P = 1.5 atm
T = (276 + 276) K = 546 K
D = (2×1.5)/(0.082×546)
= 0.067
We know mass = d × V
Hence V = 6/0.067
= 89.6 L

4. Correct option: (A) ¾

Initial pressure P_1 = 1 atm
Initial temperature T_1 = 327°C = 600 K
Final pressure P_2 = 1 atm
Final temperature T_2 = 527°C = 800 K
$P_1V_1/T_1 = P_2V_2/T_2$
$V_2 = (1\times800\times V_1)/(600\times1) = 4/3\ V_1$
Volume of air expelled, $= V_2 - V_1$
$= 4/3\ V_1 - V_1$
$= V_1/3$
Hence fraction of air repelled
$= (V_1/3)/(4V_1/3)$
= ¼
Thus fraction of air remained in the flask
= (1 – ¼)
= ¾

5. Correct option: (A) 10 %

From Charle's law
$V_1/T_1 = V_2/T_2$
Let,
$V_1 = V$
V_2 =?
$T_1 = T$
$T_2 = 11T/10$
$V_2 = (V_1T_2)/T_1$
= (V × 11T) / 10T
= 11V/10
Change in V,
= (11V/10) – V

= V/10
So percentage of change in V
= (V/10) × (100/V)
= 10%

6. Correct option: (B) 2 atm/25°C

From ideal gas equation
PV = nRT
PM = dRT
P∝d
$P_1/d_1 = P_2/d_2$
P_1 = 1 atm
d_1 = 1.458 mg/L
d_2 = 2.916
$P_2 = P_1d_2/d_1$
= 1 × 2
= 2 atm
Temperature will be same

7. Correct option: (A) 40°C

P = 0.5 atm
V = 51.4 L
T = ?
We know PV = nRT
T = PV/nR
=(0.5 × 51.5) /(0.082 × 1)
= 313 K
T = (313 – 273)° C
= 40° C

8. Correct option: (D) 68.25 mL

From ideal gas equation
PV = nRT
P = 4 atm
N = 0.5 mole
R = 2 $Cal.mol^{-1}.K^{-1}$
T = 273 K
V = ?
V = nRT/P
= (0.5 × 2 × 273) / 4
= 68.25 mL

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Correct option: (A) 0.0821L.atm.K^{-1}.$mole^{-1}$ B)1.99cal.K^{-1}.$mole^{-1}$

D)8.314J.K^{-1}.$mole^{-1}$
At STP, for 1 mole of gas (n=1 mol),
The value of,
Temperature (T) = 273 K,
Pressure (P) = 1.01 x 10^5 N/m^2
Volume (V) = 22.4 x 10^{-3} m^3
From PV = nRT
R = PV/nT

$= (1.01 \times 10^5 \times 22.4 \times 10^{-3})/273$
$= 8.314\ J.mol^{-1}.K^{-1}$
$= 1.99\ cal.\ K^{-1}.\ Mole^{-1}$
Again we know
$R = 0.0821\ lit.atmK^{-1}.mol^{-1}$
$= 8.314\ m^3.\ Pa.\ K^{-1}.mol^{-1}$
$= 8.314 \times 10^3\ Pa.\ dm^3.K^{-1}.mol^{-1}$

10. Correct option: (A) Volume increases if ideal nature is consider and 'P' & 'n' are kept constant, C) Vm = 22.4litres at 25°C assuming gas behaves ideally

PV=nRT
If P and n constant.
$V \propto T$
Volume increases with temperature if P and n kept constant and ideal nature is considered.

11. Correct option: (B) Their pressure are same if 'Y' is of half the size of 'X'
C) Temperature ratio can be made 1:1 by doubling the moles in 'Y' (PV being maintained)
D) Temperature ratio can be made 1:1 by doubling the moles in 'X' (PV being maintained).

From ideal gas equation PV = nRT
PY = nR2T/2V = nRT/V
PX = nRT/V
Option A is correct
PY = nR2T/(V/2) = 4nRT/V
PX = nRT/V
Hence option B is incorrect
2TY = PV/2nR
TY = PV /4nR
TX = PV/nR
Hence option C is incorrect
Again
TX =PV/2nR
2TY = PV/nR
TY = PV/2nR
Hence option D is correct

12. Correct option: (A) $PV = N/N_A\ RT$ B) $PV = N_KT$

$PV = N/N_A\ RT$, $N/N_A = n$
That is PV = nRT,hence it is correct.
$PV = N_KT$
Where $N_K = nR$, as $K = R/N$, hence $N_K = nR$
It is also correct.
PV = nRT
PV = W/M RT
Hence $PV = \rho/m\ RT$ is incorrect
$PV = MK_BT/m$
$PV = nK_BT = nRT$
Hence it is also incorrect

13. Correct option: (B) Velocity

The simplest known example of an equation of state is the one relating the pressure P, the volume V, and the absolute temperature T of n mole of an ideal gas—that is, the ideal gas law PV = nRT, in which R is the universal gas constant.

14. Correct Answer: (B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

Explanation:

- **Assertion (A) is True:** Ideal gas behavior assumes that there are no intermolecular forces of attraction between gas molecules and that the volume occupied by the molecules themselves is negligible compared to the total volume of the container.
  - At **low pressure**, the gas volume is huge, making the actual size of individual molecules completely negligible.
  - At **high temperature**, the kinetic energy of the molecules is so high that they rapidly overcome any attractive intermolecular forces during collisions. Thus, real gases approach ideal behavior under these conditions.
- **Reason (R) is True:** No gas is perfectly ideal in nature. At Standard Temperature and Pressure (STP:0 °C and 1 atm), intermolecular attractions and the volume of the gas molecules are significant enough that all known gases show deviations from ideal behavior. Therefore, they behave as real gases.
- **Why it's not the correct explanation:** While both statements are scientifically factual, the reason does not explain *why* high temperature and low pressure cause real gases to act ideally. The correct explanation relates to the reduction of intermolecular forces and the expansion of total volume.

15. Correct Answer: (D) Assertion is false, but Reason is true.

Explanation:

- **Assertion (A) is False:** Under standard international conventions (like IUPAC), the standard temperature at **STP** is defined as exactly 0 °C (which is 273.15 K). The value 25 °C (298.15 K) is actually used for Standard Ambient Temperature and Pressure (**SATP**) or standard thermodynamic states, not STP.
- **Reason (R) is True:** The classic standard pressure defined at STP is exactly 1 atm (or 1.01325 bar, though modern IUPAC guidelines occasionally use 1 bar). In most foundational textbook chemistry, 1 atm is standard.
- **Conclusion:** Since the assertion is completely incorrect and the reason is accurate, **(D)** is the correct choice.

16. Correct Matching is: a) → Q), b) → T), c)→P), d) → R)

R = PV/nT

= $(1.01 \times 10^5 \times 22.4 \times 10–3)/273$

= 8.314 $J.mol^{-1}.K^{-1}$ → SI unit of R

= 8.314×10^8 $ergs.k^{-1}.mol^{-1}$ = CGS unit of R

V = 22.414 lit → gram molar volume at STP

Standard pressure at STP → 1 atm = 760 mm of Hg

17. Correct Matching is: a) → Q), b) →P), c) →R), d) → T)

Standard ambient temperature = 298.15 K

Gram molar volume at STP = 22.4 L

Volume at SATP (P = 1 bar) = 24.4 L

Volume at SATP (P = 1 atm) = 24.7 L

18. Correct option: (B) 4 atm

From ideal gas equation
PV = nRT
P =(1/V)RT
Slope of P vs 1/V plot
M = 1/nRT
Again, V = 15 Litre
1/P = V/nRT
1/P = 15/60
P = 60/15
P = 4 atm

19. Correct option: (B) 4 atm

$_1V_1 = P_2V_2$
$P_2 = P_1V_1 / V_2$
= (2×6)/4
= 3 atm
Again $P_1/T_1 = P_2/T_2$
$P_2 = P_1T_2/T_1$
= (3×400)/300
= 4 atm

20. Correct option: (A) 27.5L

We know
$P_1V_1/T_1 = P_2V_2/T_2$
$V_2 = P_1V_1T_2/T_1P_2$
= (6 × 10 × 550)/(400× 3)
= 27.5

21. Weight of CO in a 22.4 Litre Flask at STP

Note: While the temperature and pressure conditions were clipped in your text, the volume of 22.4 litres *explicitly points to standard conditions, i.e., Standard Temperature and Pressure (STP).*

- Answer: 28 grams

Step-by-Step Explanation:

1. Find the number of moles: According to Avogadro's law, 1 mole of any ideal gas occupies exactly 22.4 litres of volume at STP. Therefore:
   Moles of CO = $\frac{22.4\ \text{L}}{22.4\ \text{L/mol}} = 1$ mole
2. Calculate the molar mass of Carbon Monoxide (CO):
   - Carbon (C) = 12 g/mol
   - Oxygen (O) = 16 g/mol
   Molar Mass of CO = 12 + 16 = 28 g/mol
3. Calculate the weight:
   Weight = Moles × Molar Mass = 1 mole × 28 g/mol = 28 grams

22. Correct Answer is: 6.4 g

From ideal gas equation
PV = nRT
PV = (W/M)RT
16 × 9 = (W × 300 × 0.082) / 16
W = x = 93.6 g
100-x = 100 – 93.6 = 6.4 g

Previous year questions:

1. AnswerB) Both statement-I and statement-II are not correct

Explanation

- **Statement-I Analysis:** * **Intermolecular forces** act to bind molecules together, favoring condensed phases like solids and liquids.
  - **Thermal energy** represents the kinetic energy of molecules, which causes them to move apart, favoring the gaseous state.
  - If intermolecular forces are **stronger** than thermal energy, the molecules will be tightly bound, meaning the substance will prefer to be in a **solid or liquid state**, not gaseous. Thus, Statement-I is **incorrect**.
- **Statement-II Analysis:** * There are **11 elements** that exist as gases under normal room temperature and pressure conditions:

  * **Diatomic gases (5):** Hydrogen (H_2), Nitrogen (N_2), Oxygen (O_2), Fluorine (F_2), and Chlorine (Cl_2).

  * **Noble gases (6):** Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), and Radon (Rn).
  - Since the total number is 11, Statement-II saying "10" is **incorrect**.

2. AnswerC) 1.00

Explanation

According to the ideal gas equation:

$PV = nRT$

Where concentration (C) is defined as the number of moles per unit volume ($C = \frac{n}{V}$), we can rearrange the formula:

$$C = \frac{n}{V} = \frac{P}{RT}$$

Given data:

- Pressure (P) = 16.4 atm
- Temperature (T) = 200 K
- Gas constant (R) = 0.082 L atm $mol^{-1}K^{-1}$

Substitute these values into the concentration formula:

$$C = \frac{16.4}{0.082 \times 200}$$

$$C = \frac{16.4}{16.4} = 1.00 \text{ mol } L^{-1}$$

3. Answer A) $\frac{PM}{RT}$

Explanation

We start with the ideal gas law:

$PV = nRT$

The number of moles (n) can be written as mass (m) divided by molar mass (M):

$$n = \frac{m}{M}$$

Substituting n into the ideal gas equation gives:

$$PV = \left(\frac{m}{M}\right) RT$$

Rearranging the equation to solve for pressure (P):

$$P = \left(\frac{m}{V}\right) \frac{RT}{M}$$

Since density (d) is defined as mass per unit volume ($d = \frac{m}{V}$):

$$P = \frac{dRT}{M}$$

Solving for density (d):

$$d = \frac{PM}{RT}$$

5. Metals and Non-Metals

LEVEL-I

1. Answer: C) Malleability

- *Explanation:* Malleability is the characteristic physical property of metals that allows them to be hammered, rolled, or beaten into thin sheets without breaking. Ductility is drawing into wires, and tenacity is tensile strength.

2. Answer: C) Graphite

- *Explanation:* Graphite is an allotrope of carbon (a non-metal). Due to its layered structure and the presence of free, delocalized electrons between layers, it is an excellent conductor of electricity, making it an exception among non-metals.

3. Answer: C) form acidic oxides

- *Explanation:* Metals react with oxygen to form basic oxides (e.g., Na_2O, MgO). It is non-metals that typically form acidic oxides (e.g., SO_2, CO_2).

4. Answer: A) malleability

- *Explanation:* (Repeated question concept) Malleability is the property that allows a metal to deform under compression into sheets.

5. Answer: C) Na

- *Explanation:* Na stands for Sodium, which is an alkali metal. Carbon (C), Nitrogen (N), and Oxygen (O) are all non-metals.

6. Answer: D) N

- *Explanation:* N stands for Nitrogen, which is a gas and a non-metal. Zinc (Zn), Aluminium (Al), and Iron (Fe) are all metallic elements.

7. Answer: B) Graphite

- *Explanation:* Iron, Aluminium, and Silver are metals and are highly malleable. Graphite is a non-metal crystalline form of carbon and is brittle; it shatters into powder when hammered.

8. Answer: A) Iron

- *Explanation:* Metals possess free valence electrons that allow electric current to flow easily. Plastic, wood, and glass are insulators.

9. Answer: C) Ductility

- *Explanation:* Ductility is the physical property of a material associated with the ability to be stretched or drawn into thin wires without fracturing.

10. Answer: B) sonorous

- *Explanation:* A material is said to be sonorous if it is capable of producing a deep, ringing sound when struck.

11. Answer: C) Hg

- *Explanation:* Hg stands for Mercury. It is the only metal that remains in a liquid state at standard room temperature and pressure.

12. Answer: C) Aluminium

- *Explanation:* Metals possess a characteristic metallic luster (shine) because their free electrons reflect light. Coal, sulphur, and carbon are dull non-metals.

13. Answer: C) Jewellery

- *Explanation:* Gold is highly unreactive (noble metal), does not corrode or tarnish, and is highly malleable and ductile, making it perfect for ornamental jewellery.

14. Answer: D) Basic nature

- *Explanation:* Sodium is a highly reactive metal. Its oxide (Na_2O) reacts with water to form a strong base, Sodium Hydroxide (NaOH). Hence, it is basic.

15. Answer: A) Hydrogen gas

- *Explanation:* Metals placed above hydrogen in the reactivity series displace hydrogen ions from acids, forming a metal salt and releasing H_2 gas (e.g., $Zn + 2HCl \rightarrow ZnCl_2 + H_2$).

16. Answer: A) The green solution slowly turns brown

- *Explanation:* Aluminium (Al) is more reactive than Iron (Fe). It displaces iron from $FeSO_4$ (which is light green). The iron deposits onto the strip/bottom as a brown particle mass, changing the solution profile: $2Al + 3FeSO_4 \rightarrow Al_2(SO_4)_3 + 3Fe$.

17. Answer: D) Colourless

- *Explanation:* Zinc is more reactive than copper and displaces it from blue copper sulphate solution to form colorless zinc sulphate solution: $Zn + CuSO_4$ (blue) $\rightarrow$ $ZnSO_4$ (colorless) + Cu (red-brown).

18. Answer: A) Soft and dull

- *Explanation:* The displaced metallic copper deposits onto the iron nail as a powdery, soft, and dull reddish-brown layer rather than a smooth factory finish.

19. Answer: D) Na_2CO_3

- *Explanation:* Carbonates of highly electropositive alkali metals like Sodium (Na) and Potassium (K) are extremely thermally stable and do not decompose upon heating by regular laboratory burners. $CaCO_3$, $MgCO_3$, and $ZnCO_3$ decompose easily to yield their oxides and CO_2.

20. Answer: B) Basic

- *Explanation:* Burning magnesium yields magnesium oxide (MgO) ash. Dissolving this ash in water forms magnesium hydroxide [$Mg(OH)_2$], which turns red litmus paper blue, confirming it is basic.

LEVEL - II

1. Answer: D) Sulphur dioxide

- *Explanation:* Burning sulphur in air or oxygen produces sulphur dioxide gas: $S + O_2 \rightarrow SO_2$.

2. Answer: D) Sulphurous acid is formed

- *Explanation:* SO_2 gas dissolves in water to produce sulphurous acid: $SO_2 + H_2O \rightarrow H_2SO_3$.

3. Answer: C) Sonority

- *Explanation:* Sonority refers to the ability to produce a clear metallic ringing sound upon impact.

4. Answer: B) $NaOH + HCl \rightarrow NaCl + H_2O$

- *Explanation:* A neutralization reaction is a reaction between an acid (HCl) and a base (NaOH) to produce salt (NaCl) and water (H_2O).

5. Answer: C) Mg

- *Explanation:* Sodium (Na) reacts violently with cold water. Zinc (Zn) reacts with steam. Copper (Cu) does not react with water at all. Magnesium (Mg) reacts very slowly with cold water but reacts readily with hot/boiling water to evolve hydrogen gas.

6. Answer: D) H_2SO_3

- *Explanation:* H_2SO_3 is sulphurous acid, whereas H_2SO_4 is sulphuric acid.

7. Answer: A) Red

- *Explanation:* Acids universally turn blue litmus indicators red.

8. Oxides of nonmetals are _____ in nature

- Answer: B) Acidic

- *Explanation:* Non-metal oxides (like SO_2, CO_2) react with water to yield acidic solutions.

9. Answer: C) I and III only

- *Explanation:* * (I) Iron is less reactive than Zinc, so Iron cannot displace Zinc from Zinc Sulphate (Not feasible).
  - (II) Magnesium is more reactive than Silver, so it can displace it (Feasible).
  - (III) Copper is less reactive than Hydrogen and cannot displace it from dilute acids (Not feasible).

10. Answer: C) Kerosene

- *Explanation:* Sodium is highly reactive and catches fire spontaneously upon contact with moisture/water or oxygen present in open air. It is kept submerged in inert kerosene to isolate it.

11. Answer: A) Sodium *(Note: Potassium is also correct, but out of the given typical multiple-choice setups, both alkali metals display this extreme vigor. Sodium is the standard textbook answer).*

12. Answer: C) Ca

- *Explanation:* The chemical formula is $Ca(OH)_2$. Ca stands for Calcium, which is the metal constituent.

13. Answer: B) Iron displaces Cu from $CuSO_4$

- *Explanation:* Iron is higher than copper in the reactivity series: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$.

14. Answer: B) Copper sulphate

- *Explanation:* The blue color of copper sulphate solution fades and changes to light green due to the formation of iron sulphate ($FeSO_4$).

15. Answer: B) Copper sulphate

- *Explanation:* Iron can only displace metals lower than itself in the reactivity series. Since Zinc and Aluminium are more reactive than Iron, no reaction happens in those tubes. It reacts only with $CuSO_4$, producing a brown coating of copper metal.

16. Answer: D) Zinc and aluminium

- *Explanation:* Both Zinc and Aluminium are more reactive than Iron (Fe). They will successfully displace iron out of the $FeSO_4$ solution, leading to elemental iron precipitation (seen as a dark residue). Copper is less reactive and won't react.

17. Answer: A) Copper sulphate is formed *(Note: Under high temperature/concentration copper reacts, though practically with dilute acid under standard conditions, it is extremely slow or no reaction occurs. Out of the given multiple-choice constraints, option A signifies the target copper product).*

18. Answer: C) Al > Zn > Fe > Cu

- *Explanation:* According to the activity series of metals, Aluminium is the most reactive among these, followed by Zinc, then Iron, and Copper is the least reactive.

19. Answer: C) Basic in nature

- *Explanation:* Most metallic oxides react with acids to form salt and water, showing basic properties.

20. Answer: A) SO_2

- *Explanation:* Turning blue litmus red requires an acidic solution. SO_2 is a non-metal oxide that forms sulphurous acid (H_2SO_3) in water. MgO, Fe_2O_3, and CaO are metal oxides that form basic solutions.

21. Answer: A) H_2 gas

- *Explanation:* Active metals react with acids (like HCl) as well as certain strong bases (like NaOH reacting with Zn or Al) to liberate Hydrogen (H_2) gas.

22. Answer: A) Sodium > Magnesium > Zinc > Iron > Copper

- *Explanation:* Matches the standard thermodynamic activity hierarchy of common metals perfectly.

ONE OR MORE THAN ONE CORRECT ANSWER QUESTIONS

23. Answers: A) Malleable, B) Ductile, D) conductor

- *Explanation:* Metals are malleable, ductile, and good conductors of electricity/heat. They form basic oxides, not acidic ones.

24. Answers: A) Forms anion, D) Forms acidic oxides

- *Explanation:* Non-metals readily gain electrons to form negative ions (anions) and their oxides are acidic. They generally have low tensile strength and low density.

25. Answers: A, B, C, D (All are true)

- *Explanation:* * About 80% of known elements are metals.
  - In the Earth's crust/universe, non-metals (like Oxygen, Hydrogen, Silicon) are chemically highly abundant.
  - Sodium and Potassium are exceptionally soft alkali metals that can be cut with a knife.
  - Bromine is uniquely a liquid non-metal at room temperature with a distinctive reddish-brown color.

26. Answers: A) Salt, B) Water

- *Explanation:* This is an acid-base neutralization reaction: Metal Oxide (Base) + Acid → Salt + Water.

27. Answer: C) Salt

- *Explanation:* Reactions with salts are displacement reactions where one metal replaces another in solid form ($Fe + CuSO_4 \rightarrow FeSO_4 + Cu$); no gas is liberated.

28. Answers: A) Zinc, B) Iron, C) Aluminium

- *Explanation:* Zinc, Iron, and Aluminium are all metals and therefore possess malleability. Sulphur is a brittle non-metal.

29. Answers: A) Non-metal *(Note: except graphite/diamond forms)*

- *Explanation:* Non-metals are structurally poor thermal conductors because they lack free-moving mobile electrons.

ASSERTION AND REASON TYPE

30. Answer: C) Assertion is true, but Reason is false.

- *Explanation:* Assertion is true because metals produce a ringing sound. The Reason is completely false because metals are malleable and ductile, not brittle (non-metals are brittle).

MATCHING TYPE QUESTIONS

31. Matches:

- (A) Ductile → (q) A substance can be drawn into wires.
- (B) Malleable → (r) Ability to withstand longitudinal pull *(Note: Though malleability deals with sheets, contextually in this specific text structure matrix, metals possess high tensile strength/malleability compared to non-metals).*
- (C) Sonority → (p) The property of making a resonating sound.
- (D) Stainless steel → (s) Not ability to withstand longitudinal pull *(Note: Technically an error in the question options, as steel has high tensile strength; however, by structural elimination matching the given keys).*

COMPREHENSION TYPE

32. Answer: A) Zinc is less active than aluminium

- *Explanation:* Since no reaction occurred when Zinc was put into Aluminium Sulphate [$Al_2(SO_4)_3$], it proves Zinc is less reactive than Aluminium and cannot displace it.

33. Answer: C) Metal oxide

- *Explanation:* Metal + Oxygen → Metal Oxide (e.g., $2Mg + O_2 \rightarrow 2MgO$).

34. Answer: B) Copper sulphate

- *Explanation:* $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$.

INTEGER TYPE QUESTIONS

35. Answer: 1

- *Explanation:* Mercury (Hg) is the only metal that is liquid at standard room temperature (25°C).

36. Answer: 1

- *Explanation:* Diamond (an allotrope of Carbon) is the only naturally occurring non-metal framework structure that is exceptionally hard.

37. Answer: 2 *(Generic standard textbook answer: Gallium and Caesium)*

- *Explanation:* Gallium (Ga) and Caesium (Cs) have such low melting points that they melt right on your palm.

38. Answer: 1 *(Standard context: Zinc or Liquid Mercury)*

- *Explanation:* Mercury (Hg) cannot be beaten into sheets because it is liquid. Zinc (Zn) is also highly brittle at room temperature.

39. Answer: 1

- *Explanation:* Graphite (an allotrope of carbon) is the primary non-metal that conducts electricity efficiently.

Uses of Metals and Non-Metals

LEVEL-I:

1. Answer: B) O

- Explanation: Oxygen (O_2) is a nonmetal gas that is absolutely vital for the process of respiration in living organisms to produce energy.

2. Answer: D) Fertilisers

- Explanation: Nonmetals like Nitrogen (N), Phosphorus (P), and Potassium (K) are crucial components used to manufacture chemical fertilizers to boost plant growth.

3. Answer: D) Haematite

- Explanation: Haematite (Fe_2O_3) is a major iron ore. Bauxite is an ore of aluminium, while Malachite and Cuprite are copper ores.

4. Answer: B) Nonmetals

- Explanation: Nonmetals like Iodine are used to make "Tincture of Iodine," which is a highly effective antiseptic applied to wounds.

5. Answer: C) Mg

- Explanation: Magnesium (Mg) is the central metal atom present in the structure of the chlorophyll molecule, which plants use for photosynthesis.

6. Answer: B) Blue colour

- Explanation: Hydrated copper sulphate ($CuSO_4 \cdot 5H_2O$) is characteristically bright blue, often referred to as "blue vitriol."

7. Answer: D) Phosphorous

- Explanation: Phosphorus (specifically white phosphorus) is highly reactive and catches fire spontaneously in open air, which is why it is stored submerged under water. Sodium and Potassium are highly reactive *metals*, not nonmetals.

8. Answer: A) Iron

- Explanation: Iron is a transition metal with strong metallic bonding, giving it a dense, hard structure. Materials like coal, wood, and oxygen gas lack this structural property.

9. Answer: B) Mercury

- Explanation: Mercury (Hg) is the only metal that remains liquid at room temperature. It expands uniformly with temperature changes, making it ideal for thermometers.

10. Answer: C) Sulphide ores

- Explanation: Roasting involves heating the ore strongly in the presence of excess air to convert sulphide ores into their respective metal oxides.

11. Answer: B) 22 carat gold

- Explanation: Pure gold (24 carat) is too soft to maintain its shape in jewelry. Therefore, 22 carat gold (which contains 22 parts gold and 2 parts of copper or silver) is preferred for durability.

12. Answer: D) Duralumin

- Explanation: Duralumin is an alloy of aluminium, copper, manganese, and magnesium. It is favored in aerospace engineering because it is lightweight yet exceptionally strong.

13. Answer: A) in absence of air

- Explanation: Calcination is the thermal treatment used primarily for carbonate and hydrated ores where the material is heated strongly in the absence or a highly limited supply of air.

14. Answer: A) CO_2

- Explanation: Charcoal is carbon. When it burns completely in the presence of atmospheric oxygen, it forms Carbon Dioxide (CO_2).

15. Answer: A) Iron rod

- Explanation: Iron is a metal containing free, mobile electrons that easily conduct an electric current. Sulphur, coal, and wood are non-conductors.

16. Answer: D) Coal

- Explanation: Coal (carbon) is a nonmetal and is brittle. It lacks the property of ductility and will break into pieces if stretched, unlike metals (Fe, Al, Cu).

17. Answer: D) Carbon

- Explanation: Fossil fuels like wood, coal, petroleum, and natural gas are comprised primarily of carbon compounds that release energy upon combustion.

18. Answer: A) Food

- Explanation: Due to its high malleability and non-toxic nature, aluminium can be rolled into thin sheets (foils) to keep food fresh and safe from contamination.

19. Answer: C) Silver foil

- Explanation: Thin, edible silver foils (known as *Vark*) are traditionally applied as decoration on Indian sweets.

20. Answer: B) Fertilisers

- Explanation: Fertilisers supply essential mineral nutrients like Nitrogen, Phosphorus, and Potassium that plants require to thrive.

21. Answer: C) Non-metal

- Explanation: Crackers depend heavily on nonmetals like Sulphur and Phosphorus mixed with Charcoal to create gunpowders and ignition mixtures.

22. Answer: A) $H_2O + CO_2 + O_2$

- Explanation: Moist air contains water vapor (H_2O), carbon dioxide (CO_2), and oxygen (O_2), which collectively drive the atmospheric corrosion/rusting of metals.

23. Answer: A) Froth Floatation

- Explanation: The froth floatation process selectively separates hydrophobic sulphide ore particles from hydrophilic gangue particles using pine oil and water.

LEVEL-II:

Single Answer Questions

1. Answer: B) Aluminium

- Explanation: Bauxite ($Al_2O_3 \cdot 2H_2O$) is the primary commercial resource used for extracting aluminium.

2. Answer: B) Electroplating

- Explanation: Deposition of a thin metal layer (like tin or chromium) onto another metal surface using an electric current is called electroplating. *(Note: Coating specifically with Zinc is called Galvanisation).*

3. Answer: C) Magnesite

- Explanation: Magnesite is Magnesium Carbonate ($MgCO_3$). Haematite is an oxide ore, and Argentite is a sulphide ore.

4. Answer: D) Ti

- Explanation: Titanium (Ti) forms a highly stable, passive protective oxide layer on its surface, giving it exceptional resistance to highly corrosive seawater environment.

5. Answer: C) 93%

- Explanation: Pig iron is the raw product obtained from a blast furnace. It contains around 93--95% pure iron, alongside roughly 4--5% carbon and trace impurities.

6. Answer: A) All ores are minerals

- Explanation: Minerals are naturally occurring substances in the earth's crust. Ores are specific minerals from which a metal can be extracted profitably. Therefore, every ore is fundamentally a mineral, but not all minerals contain enough metal to be classified as ores.

7. Answer: C) are not reactive

- Explanation: Noble metals like Gold (Au) and Platinum (Pt) have exceptionally low reactivity, allowing them to remain uncombined (in their elemental, native state) over geological time.

8. Answer: D) Flux

- Explanation: A flux is added during smelting to react chemically with the infusible earthy impurities (gangue) to yield an easily melting waste liquid called slag.

9. Answer: A) The metal cannot be extracted economically from all the minerals

- Explanation: A mineral is only upgraded to the status of an "ore" if the target metal can be extracted from it easily, efficiently, and with economic viability.

10. Answer: B) Reduction with carbon

- Explanation: Pyrometallurgical reduction using carbon (smelting in a furnace) is the most standard, commercial extraction choice for mid-reactivity metal oxides (like iron, zinc, and tin).

11. Answer: C) copper sulphate

- Explanation: An acidified aqueous solution of copper sulphate ($CuSO_4 + H_2SO_4$) is used to facilitate smooth migration of Cu^{2+} ions from the impure anode to the pure cathode.

12. Answer: D) all five options

- Explanation: Heating an ore during calcination removes moisture (i), makes the structure porous (ii), expels volatile components (iii), breaks down carbonates to oxides (iv), and removes water of crystallization (v).

13. Answer: A) conversion of ore to oxide

- Explanation: The primary goal of roasting is to heat sulphide components in excess oxygen to convert them into solid metal oxides, venting off sulphur as gaseous SO_2.

14. Answer: C) sulphide ores

- Explanation: This technique exploits differences in surface wetting properties, which specifically allows sulphide minerals to float with the generated foam layer.

15. Answer: A) increase of metal ore concentration

- Explanation: Concentration/dressing removes large amounts of unwanted rocky matrix (gangue), systematically increasing the relative percentage weight of the valuable metal compound in the remaining portion.

16. Answer: D) all four options

- Explanation: Depending on where a metal sits in the reactivity series, extraction utilizes electrolytic processes (high reactivity), carbon/CO reduction (mid reactivity), or aluminothermic powder reactions (like chromium/manganese).

17. Answer: A) all except (i)

- Explanation: Highly electropositive metals (e.g., Sodium, Aluminum) have a stronger affinity for oxygen than carbon, carbon monoxide, or aluminum powder do. Consequently, they cannot be reduced by options ii, iii, or iv—they can *only* be isolated using electrolytic processes (i).

18. Answer: D) refines the metal

- Explanation: Flux handles initial waste management by reacting with gangue to make slag and lowering melting temperatures. It does not carry out the subsequent chemical refining or purification of the extracted raw metal.

One or More Than One Correct Answer Questions

19. Answers: A, B, D

- *A is correct:* Reduction chemically pulls oxygen away from a metal oxide.
- *B is correct:* Active elements (like Na, K) require heavy electrical input (electrolysis) to separate.
- *D is correct:* Moderate elements (Zn, Fe) readily give up oxygen using carbon.
- *(C is incorrect because* CO *and* H_2 *are extensively used commercial reducing agents).*

20. Answers: A, C, D

- *A is correct:* Bauxite holds Aluminium.
- *C is correct:* Zinc blende holds Zinc (ZnS).
- *D is correct:* Cinnabar holds Mercury (HgS).

21. Answers: A, B, C

- *A is correct:* Chlorine functions well as an industrial bleaching agent.
- *B is correct:* It kills microbes in public water systems.
- *C is correct:* It serves as a base reactant to synthesis hydrochloric acid (HCl).

22. Answers: A, B, C

- *A, B, and C are correct:* Copper features exceptional structural ductility and outstanding electric conductivity, making it the global standard for wiring and electronics.
- *(D is incorrect because Aluminium is cheaper and lighter than copper).*

Assertion and Reason Type

23. Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

- Explanation: Graphite contains delocalized electrons across its layers, rendering it a highly efficient non-metallic conductor. Because it conducts electrons smoothly and resists high-temperature degradation, it functions perfectly as an anode electrode in various electrolytic setups.

Matrix / Column Matching

24. Answer Matching:

- (A) Brass → (r) Ornaments, bolts, utensils *(Composition:* Cu + Zn*)*
- (B) Bronze → (s) Statues, coins *(Composition:* Cu + Sn*)*
- (C) Duralumin → (p) Aircrafts, pressure cooker *(Lightweight & durable)*
- (D) Stainless steel → (q) Utensils, automobile parts, cutlery *(Corrosion resistant)*

25. Answer Matching:

- (A) Froth floatation process → (p) Zinc sulphide *(Sulphide ores concentrate via froth)*
- (B) Magnetic separation → (q) Ferric Oxide *(Iron oxide holds strong magnetic traits)*
- (C) Thermit process → (r) Chromium oxide *(Reduced effectively using exothermic Aluminium powder)*
- (D) Leaching → (s) Aluminium oxide *(Bayer's chemical dissolution technique)*

Comprehension Type Questions

26. Answer: B) less

- Explanation: Less reactive metals form weaker chemical bonds with oxygen, which means they easily give up oxygen under mild reduction treatments.

27. Answer: A) cathode

- Explanation: The cathode is the negative electrode where positive metallic cations migrate to gain electrons (Reduction: $M^{n+} + ne^- \rightarrow M$).

28. Answer: D) to eliminate H^+ ions.

- Explanation: If an aqueous solution were used, the highly reactive H^+ ions from water would easily discharge at the cathode to produce hydrogen gas, leaving the target high-reactivity metal ion (like Na^+ or Al^{3+}) unreduced in the solution. Using a molten (fused) salt ensures only the metal ions are available to accept electrons.

PREVIOUS YEAR QUESTIONS

1. Answer: B. Carbon, magnesium and gold only

Explanation:

Let's analyze the properties given in the table for each substance to see which ones are incorrect:

- Iron & Aluminium: Both are typical metals. They are malleable (can be beaten into thin sheets), ductile (can be drawn into wires), lustrous (shiny), and hard. Their properties are correctly mentioned.
- Carbon: Carbon is a nonmetal. Nonmetals are brittle and non-ductile (cannot be drawn into wires). The table incorrectly shows Carbon as ductile (ü).
- Magnesium: Magnesium is a metal. Therefore, it is malleable, ductile, and lustrous. The table incorrectly marks these properties with a cross (û).

- Gold: Gold is the most ductile metal known to us. The table incorrectly marks it as non-ductile (û).

Thus, the properties of Carbon, Magnesium, and Gold are incorrectly mentioned.

2. Answer: D. a only

Explanation:

Let's evaluate each characteristic row by row for all the elements: Potassium (K), Zinc (Zn), Carbon (C), Iron (Fe), Copper (Cu), and Phosphorus (P).

- Characteristic a (Metal): * K, Zn, Fe, Cu are correctly marked as metals (ü).
  - C, P are correctly marked as non-metals (û).
  - *Hence, characteristic a is perfectly matched for all elements.*
- Characteristic b (Hardness):
  - Phosphorus (P) is a soft nonmetal, but it is incorrectly marked as hard (ü).
- Characteristic c (Malleability) & d (Ductility):
  - Potassium (K) is a alkali metal and possesses metallic properties, but it is incorrectly marked as non-malleable and non-ductile (û). Similarly, Iron (Fe) is incorrectly marked as non-malleable (û).
- Characteristic e (Conductor of electricity):
  - Potassium (K) is a metal and a good conductor of electricity, but it is incorrectly marked with a cross (û).

Since rows b, c, d, and e contain mistakes, only characteristic 'a' is correctly matched for every single element.

6. Hydrogen and its Compounds

LEVEL – I

1. Correct option: (B, C and D)

When sodium (Na) reacts with water, it forms sodium hydroxide (NaOH) and hydrogen gas (H_2) in the following reaction:

$2Na + 2H_2O \rightarrow 2NaOH + H_2$

Iron react with Sulfuric acid produces hydrogen.

$Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$

When magnesium (Mg) reacts with steam (H_2O), it undergoes a chemical reaction to produce magnesium oxide (MgO) and hydrogen gas (H_2). The balanced chemical equation for this reaction is:

$2\ Mg + 2H_2O \rightarrow 2MgO + H_2$

2. Correct option: (C)

Hydrogen gas (H_2) can be prepared by combining hydrochloric acid (HCl) with a reactive metal, such as magnesium (Mg). The chemical reaction that occurs is as follows:

$2HCl + Mg \rightarrow MgCl_2 + H_2$

3. Correct option: (C)

To determine which of the given elements will not displace hydrogen from an acid, you need to consider the reactivity series of metals. In this series, metals are listed in order of their reactivity, with more reactive metals displacing hydrogen from acids while less reactive metals do not.

Potassium (K) > Sodium (Na) > Calcium (Ca) > Magnesium (Mg) > Aluminum (Al) > Zinc (Zn) > Iron (Fe) > Lead (Pb) > Hydrogen (H_2) > Copper (Cu) > Mercury (Hg) > Silver (Ag) > Gold (Au)

Based on this reactivity series, we can observe that mercury (Hg) is less reactive than hydrogen. So, mercury (Hg) will not displace hydrogen from an acid.

4. Correct option: (A)

Hydrogen gas (H_2) cannot be produced by the action of dilute sulfuric acid (H_2SO_4) on copper (Cu). This is because copper is less reactive than hydrogen in the reactivity series of metals and dilute sulfuric acid is not strong enough to displace hydrogen from metals that are less reactive than hydrogen.

5. Correct option: (A, B, C)

Commercial hydrogen is primarily obtained from various sources, including natural gas, oil, and coal. Among the options you provided: Coal gas, Oil gas, Marsh gas (also known as methane, CH_4)

6. Correct option: (B)

Hydrogen is a light gas, and it is an oxidizing agent because it can support combustion. When hydrogen gas is exposed to air (which contains oxygen), it can readily combust, producing water vapor (H_2O) as a result of the reaction:

$2H_2 + O_2 \rightarrow 2H_2O$

7. Correct option: (B)

Cavendish's work in the late 18th century involved the reaction of zinc (Zn) with sulfuric acid (H_2SO_4), which produce zinc sulfate ($ZnSO_4$) and hydrogen gas (H_2). This experiment allowed him to collect and study the properties of hydrogen, leading to his discovery of the gas's unique characteristics.

8. Correct option: (A)

Carbon dioxide (CO_2) is soluble in water, while hydrogen gas (H_2) is not very soluble in water. Therefore, when the mixture is passed through water, the CO_2 will dissolve into the water, leaving behind the hydrogen gas. This is a common method for separating these gases in various industrial processes, including the water-gas shift reaction.

9. Correct option: (B)

The Bosch process is also known as the water-gas shift reaction, where water gas (a mixture of carbon monoxide and hydrogen) is reacted with steam to produce hydrogen gas and carbon dioxide.

10. Correct option: (C)

In an electric arc, high-energy conditions can dissociate diatomic hydrogen molecules (H_2) into atomic hydrogen (H). This dissociation occurs because of the high temperatures and energy levels generated in the arc. Atomic hydrogen is highly reactive and is used in various industrial processes, such as welding and surface cleaning.

11. Correct option: (A)

$NaandC_2H_5OH$ Both Sodium and ethyl alcohol combine together gives nascent hydrogen.

$Na + C_2H_5OH \rightarrow C_2H_5ONa + [H]$

12. Correct option: (A)

Nascent hydrogen is formed when hydrogen atoms (H) are in a highly reactive state due to excess energy. These highly reactive hydrogen atoms can readily participate in various chemical reactions, making them important in several chemical processes and reactions.

13. Correct option: (B)

Para hydrogen is more stable than ortho hydrogen. Ortho hydrogen and Para hydrogen are two different spin isomers of molecular hydrogen (H_2), and their stability depends on the relative alignment of the spins of the two hydrogen nuclei within the molecule.

14. Correct option: (D)

In Para-hydrogen, the spins of the electrons are in opposite directions (anti parallel), and as a result, the spins of the protons are also in opposite directions (anti parallel).

15. Correct option: (B)

The nucleus of deuterium, also known as heavy hydrogen, consists of one proton and one neutron. It is represented as D, and it is an isotope of hydrogen. It also has one electron orbiting the nucleus, giving it a single positive charge.

16. Correct option: (A)

Tritium is an isotope of Hydrogen.

17. Correct option: (A)

The most reactive state of hydrogen is Atomic hydrogen.

18. Correct option: (D)

H_2 (Two protium atoms)

HD (One protium and one deuterium)

HT (One protium and one tritium)

D_2 (Two deuterium atoms)

T_2 (Two tritium atoms)

HT (One tritium and one deuterium)

So, there are a total of 6 possible diatomic molecules involving these hydrogen isotopes.

19. Correct option: (B)

Hydrogen, like the halogens, has one electron in its outermost energy level and can achieve a stable electron configuration by gaining a single electron to fill its valence shell, similar to how halogens readily gain one electron to complete their valence shells and achieve a stable noble gas configuration.

20. Correct option: (B)

Ionization enthalpy is the energy required to remove an electron from an atom or ion. For hydrogen, this energy is relatively low, making it relatively easy to ionize and form a hydrogen cation (H^+).

LEVEL – II

1. Correct option: (C)

Tritium (symbol T or 3H) is a radioactive isotope of hydrogen, and it contains 1 proton, 1 electron, 2 neutrons.

2. Correct option: (C)

Finely divided nickel is commonly used as a catalyst in the hydrogenation of oils. This process is used in the food industry to convert unsaturated fats and oils into saturated fats, which increases their stability and shelf life.

3. Correct option: (C)

In ortho-hydrogen, the spins of the two protons are parallel (aligned in the same direction), while in Para-hydrogen, the spins of the two protons are anti parallel (aligned in opposite directions). This difference in the spin states of the protons leads to distinct nuclear spin states and affects the physical properties of these two forms of hydrogen.

4. Correct option: (B)

Palladium (Pd) is known to absorb hydrogen gas, forming a solid solution or alloy with hydrogen. This property of palladium is used in various applications, including hydrogen storage and purification processes. The absorption of hydrogen by palladium is reversible, making it a valuable material for storing and releasing hydrogen gas.

5. Correct option: (A)

Hydrogen gas is not obtained when zinc reacts with cold water. Zinc reacts with dilute hydrochloric acid (HCl) and dilute sulfuric acid (H_2SO_4) to produce hydrogen gas, but it does not readily react with cold water to release hydrogen gas.

6. Correct option: (A, B, C)

Zn, Sn, Al are amphoteric metals react with boiling alkali, in this case NaOH to liberate H_2 gas.

$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$
$Sn + 2NaOH + H_2O \rightarrow Na_2SnO_3 + 2H_2$
$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$

7. Correct option: (A, B, C)

Hydrogen is a reducing agent and can reduce metal oxides to their metals.
Hydrogen will reduce heated cupric oxide (CuO) to produce copper (Cu).
Hydrogen will reduce heated ferric oxide (Fe_2O_3) to produce iron (Fe).
Hydrogen will reduce heated zinc oxide (ZnO) to produce zinc (Zn).

8. Correct option: (C)

When sodium (Na) reacts with ethyl alcohol (C_2H_5OH), it can produce hydrogen gas (H_2) along with sodium ethoxide:

$2Na + 2C_2H_5OH \rightarrow 2C_2H_5ONa + H_2$

9. Correct option: (D)

Hydrogen forms covalent compounds by sharing electrons with other nonmetals, and it does not form a large number of ionic compounds by losing an electron to form a cation. Instead, hydrogen usually forms hydride ions when it reacts with metals, which are considered as compounds with covalent character.

10. Correct option: (A)

The mixture of carbon monoxide (CO) and hydrogen (H_2) formed by the action of steam on hydrocarbons is known as "water gas" or "synthesis gas."

11. Correct option: (B)

Hydrogen typically does not act as a cation in ionic salts. In most ionic compounds, other metals or elements take on the role of cations, while hydrogen more commonly participates in covalent compounds.

12. Correct option: (A, B, C)

It is pollution-free and thus is a green fuel.
Hydrogen, when burned or used in fuel cells, produces water vapor as its only byproduct, making it a clean and environmentally friendly fuel.

It releases greater energy per unit weight compared to gasoline and other fuels.
Hydrogen has a high energy-to-weight ratio, making it an efficient fuel in terms of energy output for its weight.

Hydrogen can be transported in the form of a solid as metal hydride.
Hydrogen can indeed be stored and transported in the form of metal hydrides, which can be more convenient and safer than handling gaseous hydrogen in certain applications.

13. Correct option: (B, C)

Its reaction with oxygen is highly exothermic: The combustion of liquid hydrogen with oxygen results in a highly exothermic reaction, producing a large amount of thrust, which is crucial for rocket propulsion.

Small space it occupies: Liquid hydrogen has a high energy density, which means it can provide a significant amount of energy for propulsion while occupying

relatively little space in the rocket's fuel tanks. This is essential for space travel, where minimizing weight and maximizing energy content are critical factors.

14. Correct option: (B)

Hydrogen is often considered one of the cleanest fuels in terms of environmental pollution because when it is burned or used in fuel cells, it produces water vapor (H_2O) as the only byproduct. This means that hydrogen combustion does not release harmful greenhouse gases or air pollutants that contribute to environmental pollution, making it an environmentally friendly fuel.

15. Correct option: (D)

Hydrogen is the lightest of the gases given. It has the lowest molecular mass among common elements and gases, with a molecular weight close to 2 g/mol.

16. Correct option: (A)

The abundance of hydrogen (H_2) in Earth's atmosphere is very small because Earth's gravity is relatively weak when it comes to holding onto extremely light molecules like hydrogen. Hydrogen gas has a very low molecular weight, and its lightness allows it to escape Earth's gravitational pull more easily compared to heavier gases like nitrogen and oxygen.

17. Correct option: (C)

Pure hydrogen can be obtained by carrying out electrolysis of a solution of barium hydroxide ($Ba(OH)_2$). This process involves the use of an electrolytic cell where the electrolyte is a solution of $Ba(OH)_2$, and the electrolysis of water takes place to produce hydrogen gas (H_2) at the cathode. The reaction at the cathode is as follows: $2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-$

18. Correct option: (B)

Iron (Fe) reacts with steam (water vapor, H_2O) to produce hydrogen gas ($H_2(g)$) and Fe_3O_4

$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

19. Correct option: (A)

Sodium (Na) is the metal that reacts with water at room temperature to produce hydrogen gas ($H_2(g)$). This reaction is highly exothermic and can be quite vigorous, often resulting in the evolution of hydrogen gas and the formation of sodium hydroxide (NaOH) as a byproduct.

$2Na + 2H_2O \rightarrow 2NaOH + H_2$

20. Answer

(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Explanation

- **Assertion (A) is True:** Atomic hydrogen torches are routinely used for welding high-melting-point metals (like tungsten or tantalum).
- **Reason (R) is True:** When atomic hydrogen is generated, the recombination of these individual hydrogen atoms into stable molecular hydrogen (H_2) is an extremely exothermic reaction:

  $2H(g) \rightarrow H_2(g) + \Delta H$ (approx. 435.88 kJ/mol)

This process releases immense thermal energy, generating temperatures upwards of **3700°C to 4000°C**, which easily melts high-melting-point metals.

- **Link:** Because this massive release of heat (Reason) is the exact mechanism that enables the welding of these tough metals, the Reason directly explains the Assertion.

21. Answer

(B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

Explanation

- **Assertion (A) is True:** Hydrogen is indeed the lightest known gas in the universe, possessing the lowest molecular mass ($M \approx 2$ g/mol).
- **Reason (R) is True:** Hydrogen gas is non-polar, whereas water is a highly polar solvent. Because of this lack of strong solute-solvent interactions, hydrogen is sparingly (very poorly) soluble in water.
- **Link:** While both statements are scientifically correct facts, the reason *why* hydrogen is the lightest gas is due to its atomic structure (having only one proton and one electron), not because of how well it dissolves in water. Therefore, the Reason does not explain the Assertion.

22. Correct Matching is: a-S; b-P; c-Q; d-R

a) Protium is the most common isotope of hydrogen, and its nucleus contains:
1 proton, 0 neutrons, 1 electron.
Correct match for the option is S.

b) Deuterium is an isotope of hydrogen, and it has one electron, one proton, and one neutron.
Correct match for the option is P.

c) Tritium has 1 proton, 1 electron, and 2 neutrons.
Correct match for the option is Q.

d) Di hydrogen each hydrogen atom still has one proton and one electron.
Correct match for the option is R.

WATER

LEVEL-I

1. Correct option: (D)

Pure water, in its ideal form, has very low conductance of electricity. This is because pure water contains very few ions, and its electrical conductivity primarily depends on the presence of ions or charged particles in the solution. In the absence of impurities or dissolved ions, water does not conduct electricity to a significant degree.

2. Correct option: (C)

Rainwater is considered one of the purest forms of natural water. It starts as water vapor in the atmosphere, condenses to form droplets, and then falls to the ground as rain. During this process, many impurities and contaminants are left behind, resulting in relatively pure water.

3. Correct option: (D)

The shape of a water molecule is V-shape. It has a bent molecular geometry due to the arrangement of its two hydrogen atoms around the central oxygen atom, resulting in a bond angle of approximately 104.5 degrees between the hydrogen-oxygen-hydrogen atoms in a water molecule.

4. Correct option: (B)
The HOH bond angle in a water molecule is 104.5 degrees (104°5').

5. Correct option: (A)
The dipole moment of a water molecule is approximately 1.84 Debye. This dipole moment arises due to the unequal sharing of electrons between the oxygen and hydrogen atoms in the molecule, resulting in a partial positive charge on the hydrogen atoms and a partial negative charge on the oxygen atom.

6. Correct option: (A)
The two hydrogen atoms in a water molecule are joined to an oxygen atom by covalent bonds. Covalent bonds involve the sharing of electrons between atoms, and in the case of water, each hydrogen atom shares an electron with the oxygen atom to form a covalent bond. This sharing of electrons allows the atoms to achieve a stable electron configuration.

7. Correct option: (D)
In liquid water, the aggregates of varying numbers of water molecules are held together primarily by hydrogen bonds. Hydrogen bonds are a type of intermolecular force that occurs between the positively charged hydrogen atoms of one water molecule and the negatively charged oxygen atoms of neighboring water molecules.

8. Correct option: (D)
A water molecule (H_2O) can form a maximum of four hydrogen bonds.

9. Correct option: (B)
At its melting point, ice is lighter than water because of the unique structure of ice crystals. In the solid state (ice), water molecules arrange themselves into a hexagonal lattice structure with open spaces. This arrangement creates a less dense structure compared to liquid water, where the water molecules are more closely packed.

10. Correct option: (C)
The process that will produce hard water is the saturation of water with calcium sulfate (CaSO4). When water is saturated with calcium sulfate, it can lead to the formation of calcium ions (Ca^{+2})and sulfate ions (SO_4^{-2}) in the water, which are responsible for water hardness. These ions can react with soap, forming insoluble precipitates, and create scale deposits in pipes and appliances.

11. Correct option: (C)

12. Correct option: (B)

13. Correct option: (D)
When water is boiled for some time, it becomes free from temporary hardness but not from permanent hardness. Temporary hardness is caused by the presence of bicarbonate ions (HCO_3 in the water, which can be removed by boiling because the heat causes the bicarbonate ions to decompose into carbonate ions (HCO_3^{-2}) and water (H_2O). The carbonate ions can then precipitate as calcium carbonate ($CaCO_3$) or magnesium carbonate ($MgCO_3$), which can be removed by filtration.

14. Correct option: (B)
When Zeolite, which is a hydrated sodium aluminum silicate, is treated with hard water, the sodium ions in the Zeolite are exchanged with calcium ions (Ca^2+ ions).

15. Correct option: (D)

The structure of H_2O_2 (hydrogen peroxide) is non-planar. The molecule has a bent or V-shaped structure with an angle of approximately 104.5 degrees between the two O-H bonds.

16. Correct option: (A)

H_2O_2 (Hydrogen peroxide) is commonly prepared in the laboratory by the action of dilute H_2SO_4 (sulfuric acid) on BaO_2 (barium peroxide).

17. Correct option: (C)

Moist hydrogen peroxide cannot be dried over concentrated H_2SO_4 (sulfuric acid) because it can be oxidized by H_2SO_4. Concentrated sulfuric acid is a strong oxidizing agent, and it can react with and oxidize hydrogen peroxide, potentially leading to the release of oxygen gas and other reactions.

18. Correct option: (D)

An aqueous solution of H_2O_2 (hydrogen peroxide) is weakly acidic. It has a pH slightly below 7, around 6.

19. Correct option: (B)

The reaction $2H_2O_2 \rightarrow 2H_2O + O_2$ shows that hydrogen peroxide (H_2O_2) acts as an oxidizing agent. In this reaction, it is decomposed to produce water (H_2O) and oxygen (O_2).

20. Correct option: (C)

This statement is explaining about the fact that 30 volumes of hydrogen peroxide (H_2O_2) solution will release 30 volumes of oxygen (O_2) gas when decomposed at standard temperature and pressure (STP). This property is commonly used in titration and analytical chemistry to determine the concentration of hydrogen peroxide solutions.

LEVEL-II

1. Correct option: (B)

The maximum density of water is reached at a temperature of approximately 277 K.

2. Correct option: (A)

Potash alum, also known as potassium alum ($KAl(SO_4)_2 \cdot 12H_2O$), is commonly used in water treatment to remove impurities and clarify water. It helps in flocculation and coagulation of impurities, making them easier to settle and filter out of the water.

3. Correct option: (C)

Zeolites are crystalline, hydrated aluminosilicate minerals with a three-dimensional structure that contains pores and channels. They have a high ion exchange capacity, making them useful in various applications, including water softening and catalysis. Sodium aluminum silicate Zeolites are commonly used in ion exchange processes.

4. Correct option: (A)

The true structure of hydrogen peroxide (H_2O_2) is more accurately represented as a "bent" or "V" shape. This molecule has two oxygen atoms bonded to each other, and

each oxygen atom is also bonded to a hydrogen atom. The molecular formula is H-O-O-H.

5. Correct option: (B)

Volume strength of H_2O_2 =5.6× normality

Volume strength of $H_2O_2 = 5.6 \times 15$

Volume strength of $H_2O_2 = = 8.4L$

6. Correct option: (A)

Temporary hardness in water is primarily caused by the presence of bicarbonates of calcium (Ca) and magnesium (Mg).

7. Correct option: (B)

The dihedral angle of H_2O_2 in gaseous phase is 90.2° and in solid phase is 111.5°.

8. Correct option: (C)

Hydrogen peroxide (H_2O_2) is diamagnetic. Diamagnetic substances do not have unpaired electrons, means they have no net magnetic moment and are not attracted to a magnetic field.

9. Correct option: (A)

Hydrogen peroxide (H_2O_2) can readily donate oxygen atoms (in the form of hydroxyl radicals, OH) to other substances, oxidizing them in the process.

10. Correct option: (A, B)

Hydrogen peroxide (H_2O_2) is used for various purposes, but among the given options provided, it is primarily used as:
A bleaching agent, an oxidizing agent.

H_2O_2 Is commonly used for bleaching purposes (such as bleaching hair or fabrics) and as an oxidizing agent in various chemical reactions.

11. Correct option: (C)

Hydrogen peroxide can act as both an oxidizing agent and a reducing agent depending on the reaction conditions and the substances involved. It can oxidize certain substances by donating oxygen atoms, while in other reactions; it can be reduced by accepting electrons.

12. Correct option: (D)

Permanent hardness in water, which is primarily caused by the presence of calcium and magnesium ions in the form of sulfates and chlorides, can be removed or reduced by using the following methods:

Treatment with washing soda (sodium carbonate, Na_2CO_3): Washing soda reacts with calcium and magnesium ions in water to form insoluble calcium carbonate ($CaCO_3$) and magnesium carbonate ($MgCO_3$), which can be removed by precipitation. This method is often used to soften hard water.

13. Correct option: (A, B, C and D)

H_2O_2 Is used for bleaching delicate articles like wool, hair, feathers and ivory.

14. Correct option: (D)

15. Correct option: (B)

The volume strength of a solution is defined as the number of liters of oxygen gas that can be obtained from 1 liter of the solution at STP.

The balanced chemical equation for the decomposition of hydrogen peroxide is:

$2H_2O_2 \rightarrow 2H_2O + O_2$

From the above equation, we observe that 2 moles of hydrogen peroxide decompose to give 1 mole of oxygen gas. Therefore, for a 1M solution of hydrogen peroxide, the volume strength is: $11.2\ L\frac{O_2}{mol} \times \frac{1\ mol}{L} = 11.2\ vol$

16. Correct option: (A)

$$Volume of H_2O_2 = \left(\frac{Volume\ of\ O_2}{22400}\right) \times \left(\frac{Molar\ volume\ at\ STP}{Volume\ strength}\right)$$

Here, the volume of O_2 is given as 0.80 L and the molar volume at STP is 22.4 L/mol. The volume strength of 40 volumes H_2O_2 is 40 mL/100 mL.

Substituting these values in the formula, we get:

$$Volume\ of\ H_2O_2 = \left(\frac{0.80}{22400}\right) \times \left(\frac{22.4}{40}\right)$$

$$Volume\ of\ H_2O_2 = 0.0008\ L$$

We know that the volume strength of 40 volumes H_2O_2 is 40 mL/100 mL. Therefore, we can convert the volume of H_2O_2 from liters to milliliters as follows:

$$Volume\ of\ H_2O_2 = \frac{0.0008 \times 1000}{1000}$$

$$Volume\ of\ H_2O_2 = 0.8\ mL$$

17. Correct option: (A, B)

Temporary hardness of water is caused by the presence of dissolved bicarbonates of calcium and magnesium. It can be removed by adding a base such as slaked lime $Ca(OH)_2$ or sodium carbonate Na_2CO_3

18. Correct option: (A)

Calgon is a trade name for a complex salt, sodium hexametaphosphate $(NaPO_3)_6$, which is used for softening hard water.

19. Correct option: (B)

The reagent commonly used to determine the hardness of water titrimetrically is di sodium salt of EDTA.

20. Answer(D) Assertion is false, but Reason is true.

Explanation

- **Assertion (A) is False:** Hydrogen peroxide (H_2O_2) is **not** a stable liquid at room temperature. It is thermodynamically unstable and slowly undergoes auto-oxidation (decomposition) even when kept in a sealed container at room temperature.
- **Reason (R) is True:** On standing or upon heating, hydrogen peroxide readily decomposes into water and oxygen gas. The reaction is highly exothermic:

  $2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$

  This decomposition is further accelerated by light, rough surfaces, or traces of metals, which is why it is typically stored in dark, wax-lined plastic bottles.

21. Answer(D) Assertion is false, but Reason is true.

Explanation

- **Assertion (A) is False:** Permanent hardness of water **cannot** be removed by simple boiling. Boiling only removes *temporary* hardness, which is caused by soluble calcium and magnesium hydrogen carbonates (bicarbonates).
- **Reason (R) is True:** Permanent hardness is indeed caused by the presence of soluble salts of calcium and magnesium in the form of **chlorides** ($CaCl_2, MgCl_2$) and **sulphates** ($CaSO_4, MgSO_4$). These salts do not precipitate out upon heating and require chemical treatments (like the addition of washing soda or ion-exchange resins) to be removed.

22. Correct option: a-R, b-P, c-S, d-Q

a)Hard water is due to presence of Calcium ions in water.

Correct match for the option is R.

b) Temporary hardness is due to the presence of $Ca(HCO_3)_2$.

Correct match for the option is P.

c) Sodium ions in Zeolite get exchanged with Mg^{+2} ions.

Correct match for the option is S.

d) Paramagnet hardness is due to presence of $MgSO_4$.

Correct match for the option is Q.

Previous year questions

1. Answer: D) $Mg(OH)_2$, $CaCO_3$

Explanation**:** When water containing bicarbonates is boiled, they decompose. Calcium bicarbonate forms insoluble calcium carbonate. However, magnesium bicarbonate forms magnesium hydroxide because the solubility product (K_{sp}) of $Mg(OH)_2$ is much lower than that of $MgCO_3$, making the hydroxide precipitate more readily.

2. Answer: A) placing in wax lined plastic bottle and kept in dark

Explanation**:** H_2O_2 is highly unstable and decomposes when exposed to light or rough surfaces like glass.

- Light**:** Photochemical decomposition occurs, so it must be kept in the dark.
- Container**:** Glass containers have alkali traces that catalyze decomposition; therefore, wax-lined plastic or urea-stabilized bottles are used.

3. Answer: A) Reducing agent, Oxidizing agent

Explanation:

- With KIO4**:** H_2O_2 reduces Periodate (IO_4^-) to Iodate (IO_3^-). In this reaction, H_2O_2 is a reducing agent.
- With NH_2OH: H_2O_2 oxidizes Hydroxylamine (NH_2OH) to compounds like nitrous acid or nitrogen. Here, H_2O_2 acts as an oxidizing agent.

4. Answer: B) Calgon's method

Explanation: Sodium hexametaphosphate ($Na_6P_6O_{18}$) is commercially known as Calgon. It works by sequestering Ca^{2+} and Mg^{2+} ions into a soluble complex, preventing them from reacting with soap to form scum.

5. Answer: A) 111.5° **,** 90.2°

Explanation: H_2O_2 has a non-planar "open book" structure.

- In the gas phase, the dihedral angle is larger (111.5°) due to the absence of intermolecular forces.
- In the solid phase, strong hydrogen bonding between molecules compresses the structure, reducing the angle to 90.2° .

6. Answer: A) 3.8 K

Explanation: * The temperature of maximum density for H_2O is $3.98^\circ C$ (277.13 K).

- The temperature of maximum density for D_2O (Heavy Water) is $11.2^\circ C$ (284.35 K).
- The difference is approximately $7.2^\circ C$. Note: While the arithmetic difference between the provided options often points to the physical properties of the isotopes,

7. Answer: B) $Mg(HCO_3)_2$ $Mg(OH)_2$

Explanation**:** while calcium precipitates as a carbonate, magnesium uniquely precipitates as a hydroxide because $Mg(OH)_2$ is more insoluble than $MgCO_3$ at boiling temperatures.

8. Answer:D

Hydrogen peroxide (H2O2) has a non-planar, open-book structure. The dimensions change depending on whether it is in the gas phase or the solid (crystalline) phase due to intermolecular forces like hydrogen bonding.

- Gas Phase**:** Intermolecular forces are negligible. The dihedral angle is larger to minimize lone pair-lone pair repulsions. The angle is $11.5^\circ C$
- Solid Phase**:** Hydrogen bonding between molecules causes the structure to contract. The dihedral angle decreases significantly to $90.2^\circ C$

9. Answer:C

Calgon" is a trade name for Sodium hexametaphosphate ($Na_6P_6O_{18}$).

- The Process**:** When Calgon is added to hard water, it reacts with the Ca^{2+}} and Mg^{2+} ions (which cause hardness).
- Mechanism: It does not precipitate these ions. Instead, it "sequesters" them by forming a soluble complex. This prevents the calcium and magnesium ions from reacting with soap

10. Answer:C

NH_3 (Ammonia): Electron-rich. Nitrogen (Group 15) has 5 valence electrons. It uses 3 for bonding with Hydrogen and has one lone pair remaining.